

Kronecker-FRI: A Hash-Based Multilinear Polynomial Commitment from a Coefficient-Extraction Identity

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Abstract

We present Kronecker-FRI, a transparent, hash-based polynomial commitment scheme for multilinear polynomials whose evaluation proofs need no sumcheck. The scheme rests on the extraction identity $f(z) = [X^{N-1}] U_f(X) K_z(X)$, where U_f is the Kronecker encoding of f and $K_z(X) = \prod_{k=1}^n (X^{2^{k-1}} + z_k)$: a multilinear evaluation is one coefficient of a univariate product. The prover commits to U_f as a Reed–Solomon codeword. To open at z , it commits to a single opening polynomial; the verifier derives a second one pointwise as a virtual word, batches the words with one challenge, and runs an FRI-style folding test, so that one proximity test enforces both low degree and the claimed value. We prove completeness, soundness, round-by-round knowledge soundness and evaluation binding, for any folding arity 2^k , with correlated agreement for Reed–Solomon codes as the only external result; we derive post-quantum straight-line knowledge soundness of the compiled scheme in the quantum random oracle model; and we extend the scheme to batched openings of several polynomials. All interactive results are machine-checked in Lean 4, with correlated agreement as the single axiom. A Rust implementation over the Goldilocks field commits and verifies as fast as a sumcheck-based Reed–Solomon scheme at identical engineering; with arity 8 and Merkle caps it opens a polynomial with 2^{20} coefficients at a 100-bit interactive soundness level with a 299 KiB proof, verified in 3.4 ms on a single core.

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1 Introduction

Many succinct arguments reduce the correctness of a computation to claims about multilinear polynomials evaluated at a random point. This is the case for sumcheck-based systems such as Spartan [Set20] and HyperPlonk [CBBZ23]. Such systems need a *multilinear polynomial commitment scheme*: the prover commits to a multilinear polynomial f in n variables and later proves that $f(z) = v$ for a point z chosen by the verifier. When the scheme is built from hash functions and error-correcting codes, the resulting arguments are transparent and rely on no number-theoretic assumption, which makes them candidates for post-quantum security.

Hash-based multilinear commitments over Reed–Solomon codes follow, for the most part, the BaseFold paradigm [ZCF24]: the prover commits to a codeword, and proves an evaluation by running a sumcheck whose challenges also drive an FRI-like folding [BSBHR18] of the codeword. The sumcheck is what connects the evaluation claim to the proximity test, and it shapes both the protocol and its soundness analysis, which must track the sumcheck claims and the folded codewords together [ZCF24, GLH⁺25, Mkh26].

This paper takes a different route. It writes the evaluation $f(z)$ as a single coefficient of a product of two univariate polynomials, checks that coefficient through a polynomial identity of bounded degree, and tests the identity with plain FRI. The result is a multilinear commitment

whose evaluation protocol has no sumcheck, and whose verifier is an FRI verifier with one extra local computation per query.

1.1 The idea

Let $N = 2^n$ and write $f = \sum_{a < N} \alpha_a x^a$ in coefficient form, with $x^a = \prod_{k: a_k=1} x_k$. The *Kronecker encoding* of f is the univariate polynomial $U_f(X) = \sum_a \alpha_a X^a = f(X, X^2, X^4, \dots, X^{2^{n-1}})$, and the *kernel polynomial* of a point $z \in \mathbb{F}^n$ is

$$K_z(X) = \prod_{k=1}^n (X^{2^{k-1}} + z_k).$$

Its coefficient of X^{N-1-a} is $z^a = \prod_{k: a_k=1} z_k$, so that

$$f(z) = [X^{N-1}] U_f(X) K_z(X)$$

(Theorem 4.5). To prove $v = f(z)$, the prover splits $X U_f K_z$ around the degree N :

$$X U_f(X) K_z(X) = A(X) + v X^N + X^{N+1} H(X), \quad A, H \in \mathbb{F}[X]_{<N}$$

(Lemma 4.10). Conversely, if polynomials U, A, H of degree less than N satisfy this identity, then $v = [X^{N-1}] U K_z$; the factor X puts the checked coefficient at the degree N , so that the natural degree bound N of the code is all that must be enforced (Lemma 4.12 and Remark 4.13).

The scheme commits to $\text{Enc}(f) = \text{ev}_L(U_f)$ on a smooth multiplicative domain L of order $M = N/\rho$. To open, the prover sends the word $w_A = \text{ev}_L(A)$ only. The verifier does not need H as an oracle: the identity determines $H(\xi)$ from $U_f(\xi)$ and $A(\xi)$ at every point, so the verifier computes a *virtual word* h from the commitment and w_A (Definition 6.5). It then batches $w + \beta w_A + \beta^2 h$ with a random β and runs an FRI-style folding test on the result. A wrong value v turns h into a word far from the code, so the evaluation claim is enforced by the proximity test alone.

1.2 Contributions

The extraction identity and the opening identity (Section 4). We prove the extraction identity, the split of $X U_f K_z$ into opening polynomials, and the identity lemma, together with algorithms and their exact costs: $K_z(\xi)$ with $n-1$ squarings, $n-1$ multiplications and n additions, the product $U_f K_z$ with at most $2nN$ multiplications, and K_z on all of L with fewer than $2M$ (Lemmas 4.9 and 6.7). The identity is the classical expression of an inner product as one coefficient of a product, applied to the coefficient vector of f and the vector $(z^a)_a$, whose generating polynomial has the product form K_z up to reversal (Lemma 4.14). The same tensor structure is used, in the Lagrange basis and with KZG commitments, by Mercury [EG25]. Our contribution is its use in a transparent, hash-based scheme over Reed–Solomon codes, together with the analysis below.

A folding proximity test for arbitrary words (Section 5). We give a self-contained analysis of the FRI-style test with the kernel fold of [Mkh26], for an arbitrary input word and without sumcheck: fibre distance, unique fibre decoding, good challenges, witness sets and the codeword chain, and a soundness theorem (Theorem 5.24). The witness sets are stated for an arbitrary prover, so that they can serve directly as the round-by-round state of the scheme. The analysis covers higher folding arity exactly: committing only to a subset of the folded words leaves every error term unchanged (Proposition 5.28).

The scheme and its soundness (Sections 6 and 7). The evaluation protocol Π_{KF} (Definition 6.8) has $\ell + 2$ rounds: one opening oracle, then the folding test on the batched word. We prove completeness, exact operation counts (Proposition 6.11), soundness with error $\varepsilon_{\text{KF}} < 3M/|\mathbb{F}| + (1 - \delta)^\kappa$ for every $\delta \leq (1 - \rho)/2$ (Theorem 7.8), round-by-round knowledge soundness with an explicit decoding extractor (Theorem 7.11), and evaluation binding (Corollary 7.12). The only new algebraic step is a batching lemma (Lemma 7.4), which combines correlated agreement for curves of degree 2 [BSCI⁺23] with the identity lemma through the virtual word; from the second round on, the analysis is that of the folding test. The only external result about codes is the correlated-agreement theorem of Ben-Sasson, Carmon, Ishai, Kopparty and Saraf [BSCI⁺23] in the unique-decoding regime.

The non-interactive scheme (Section 8). We derive relaxed round-by-round knowledge soundness, with fibres as oracle symbols and partially defined strings (Theorem 8.5), and apply the BCS theorem for interactive oracle reductions of Chiesa, Di, Hu and Zheng [CDHZ25]. The compiled scheme is straightline knowledge sound, for one opening, against quantum adversaries in the quantum random oracle model (Corollary 8.6); with the parameters of Remark 8.8, the extraction error against 2^{64} oracle queries is below $2^{-31.8}$.

Formal verification (Appendix B). The results of Sections 4 to 7, including the arity result, round-by-round knowledge soundness and the batched openings of Section 11.2, are formalised in Lean 4 with Mathlib, without `sorry`; the only axiom is the correlated-agreement theorem for curves (Theorem 2.16).

Implementation, experiments and practical use (Sections 9 to 11). We give a Rust implementation over the Goldilocks field, with quadratic and quartic extensions and salted Merkle trees, whose test suite checks the algebraic statements of Sections 4 to 6 listed in Appendix B against direct computations. We compare it at identical engineering with the sumcheck-based scheme of [Mkh26] and a coefficient-form (BaseFold-style) baseline: Kronecker-FRI commits and verifies at the same cost, with proofs within 5% of theirs (Table 3). With arity 8 and Merkle caps, proofs at $n = 20$ shrink from 1117 to 315 KiB at the same ℓ , and to 299 KiB with $\ell = n - 8$, with verification in 3.4 ms (unsalted, interactive soundness error below 2^{-99} ; Tables 4 and 5). We extend the scheme to the opening of several polynomials at one point with a proved soundness bound (Theorem 11.2), recommend parameter sets, and compare the scheme with existing multilinear commitments (Section 11).

1.3 Related work

Proximity tests for Reed–Solomon codes. FRI [BSBHR18] tests proximity to Reed–Solomon codes by repeated even–odd folding. Its soundness in the unique-decoding and list-decoding regimes rests on proximity gaps for Reed–Solomon codes [BSCI⁺23]. WHIR [ACFY24] tests proximity to constrained Reed–Solomon codes with very fast verification and supports multilinear evaluation claims. We use FRI with the kernel fold of [Mkh26], which is the classical fold up to a scalar and a reparametrisation of the challenge (Remark 5.6), and only the unique-decoding correlated-agreement theorem of [BSCI⁺23].

Multilinear commitments from folding. BaseFold [ZCF24] introduced foldable codes and the interleaving of sumcheck rounds with folding rounds under shared challenges. DeepFold [GLH⁺25]

instantiates the paradigm with Reed–Solomon codes in the list-decoding regime and improves proof size. Haböck [Hab24] analyses BaseFold over Reed–Solomon codes up to the Johnson bound, and Zeilberger [Zei26] proves proximity gaps for multilinear evaluation for all linear codes. FRI-Binius [DP24] adapts BaseFold to binary towers using the novel polynomial basis of Lin, Chung and Han [LCH14]. Our companion paper [Mkh26] introduced the Boolean-kernel basis $\{K_b\}_{b \in \{0,1\}^n}$, in which the fold of a Reed–Solomon codeword acts as restriction of a table, and built a BaseFold-type scheme on it; the kernel polynomial K_z of the present paper is the same product evaluated at an arbitrary point z instead of a Boolean one. Kronecker-FRI keeps the Reed–Solomon commitment and the folding test of these schemes, and replaces the sumcheck by the extraction identity.

Multilinear-to-univariate reductions. A different line of work proves multilinear evaluations with a univariate polynomial commitment, typically KZG. Gemini [BCHO22] reduces a coefficient-form claim to univariate claims through recursive even–odd splitting. Zeromorph [KT24] works in table form and uses quotients and degree checks. Mercury [EG25] reaches constant proof size and expresses the evaluation as a coefficient of a product with a tensor-structured polynomial; Samaritan [GPS25] also reaches constant proof size with a linear-time prover. Mohamed [Moh25] relates univariate and multilinear sumchecks through the inverse Kronecker substitution. Our scheme shares the univariate viewpoint of these works, but commits through a Reed–Solomon code and a hash function, and checks the identity by a proximity test instead of a pairing.

Virtual oracles. Computing a word from other oracles instead of committing to it is standard in hash-based arguments; Aurora [BSCR⁺19] calls such words virtual oracles. The virtual word h of Definition 6.5 is of this kind.

Other codes and domains. Brakedown [GLS⁺23] avoids FFTs by using linear-time encodable codes. Circle STARKs [HLP24] obtain smooth folding domains over the Mersenne prime $2^{31} - 1$. Our scheme needs a smooth multiplicative subgroup in odd characteristic (Section 11.1).

1.4 Organisation

Section 2 fixes notation and collects the facts on multilinear polynomials and Reed–Solomon codes that we use, including the correlated-agreement theorem. Section 3 recalls interactive oracle proofs, round-by-round (knowledge) soundness, polynomial commitments, and the definitions and the theorem of [CDHZ25]. Section 4 proves the extraction identity and the opening identity, and Section 5 analyses the folding proximity test. Section 6 defines the scheme, Section 7 proves its soundness, and Section 8 treats the non-interactive scheme. Sections 9 and 10 describe the implementation and the measurements, Section 11 discusses practical use, parameters and related schemes, and Section 12 concludes. Appendix A collects the notation, and Appendix B records how each result is verified.

2 Preliminaries

This section fixes notation and collects the algebraic and coding-theoretic facts used in the paper. Every result specific to this paper is proved. Standard facts from algebra and coding theory are quoted with a reference and labelled *Fact*. The only external result about codes used in the soundness analysis is the correlated-agreement theorem of Section 2.7; Fact 2.13 is used only for the running time of the extractor. Several statements coincide with results of the companion

paper [Mkh26], whose proofs are formalised in Lean 4; we reproduce them with their proofs so that the present paper is self-contained, and we indicate the corresponding item of [Mkh26].

2.1 Integers and bits

Integers. $\mathbb{N} = \{0, 1, 2, \dots\}$. For integers a and c we write $\llbracket a, c \rrbracket = \{i \in \mathbb{Z} : a \leq i \leq c\}$, which is empty if $a > c$.

Number of variables. The letter $n \in \mathbb{N}$ denotes a number of variables, and $N = 2^n$. Unless a hypothesis states otherwise, the statements of Section 2 hold for every $n \geq 0$.

Binary decomposition. Every $a \in \llbracket 0, N - 1 \rrbracket$ has a unique binary decomposition

$$a = \sum_{k=1}^n a_k 2^{k-1}, \quad a_k \in \{0, 1\},$$

and $(a_1, \dots, a_n)$ is its *bit vector*; a_1 is the least significant bit. The map $a \mapsto (a_1, \dots, a_n)$ is a bijection from $\llbracket 0, N - 1 \rrbracket$ onto $B_n = \{0, 1\}^n$, the *Boolean hypercube*, and we identify a with its bit vector. When a letter is declared to denote an index, its subscripts denote its bits. For $a, b \in \llbracket 0, N - 1 \rrbracket$:

- $\text{wt}(a) = \sum_{k=1}^n a_k$ is the *Hamming weight* of a ;
- $\bar{a} = (N - 1) - a$ is the *complement* of a ;
- $b \preceq a$ means $b_k \leq a_k$ for all $k \in \llbracket 1, n \rrbracket$ (b is a *submask* of a).

Lemma 2.1 (Complement). *For $a \in \llbracket 0, N - 1 \rrbracket$, the bits of $\bar{a}$ are $\bar{a}_k = 1 - a_k$ for $k \in \llbracket 1, n \rrbracket$.*

Proof. $\sum_k (1 - a_k) 2^{k-1} = \sum_k 2^{k-1} - \sum_k a_k 2^{k-1} = (N - 1) - a$, and $1 - a_k \in \{0, 1\}$; conclude by uniqueness of the binary decomposition. $\square$

2.2 Fields and univariate polynomials

Throughout the paper, $\mathbb{F}_q$ is a finite field with q elements and *odd characteristic*, so that 2 is invertible in $\mathbb{F}_q$, and $\mathbb{F} \supseteq \mathbb{F}_q$ is a finite extension of $\mathbb{F}_q$, the *challenge field* (possibly $\mathbb{F} = \mathbb{F}_q$). For a field $\mathbb{F}$, $\mathbb{F}[X]$ is the ring of univariate polynomials over $\mathbb{F}$; for $d \in \mathbb{N}$, $\mathbb{F}[X]_{<d}$ is the $\mathbb{F}$ -vector space of polynomials of degree less than d , of dimension d (with $\deg 0 = -\infty$). For $P \in \mathbb{F}[X]$ and $j \in \mathbb{N}$, $[X^j]P$ is the coefficient of X^j in P .

Lemma 2.2 (Root counting; [Mkh26, Lemma 2.1]). *Let $\mathbb{F}$ be a field. A nonzero polynomial $P \in \mathbb{F}[X]$ of degree d has at most d roots in $\mathbb{F}$. Consequently, two polynomials of $\mathbb{F}[X]_{<d}$ that agree on d distinct points of $\mathbb{F}$ are equal.*

Proof. If $P(a) = 0$, Euclidean division by $X - a$ gives $P = (X - a)Q$ with $\deg Q = d - 1$, and every root of P other than a is a root of Q , because $\mathbb{F}$ has no zero divisors. Induction on d gives the first statement. For the second, apply it to the difference, which has degree $< d$. $\square$

Fact 2.3 ([LN97, Theorem 2.8]). *The multiplicative group $\mathbb{F}^\times$ of a finite field $\mathbb{F}$ is cyclic.*

Lemma 2.4 (Even–odd split of polynomials; [Mkh26, Lemma 2.3(1)]). *Let $\mathbb{F}$ be a field and $d \geq 1$. Every $P \in \mathbb{F}[X]_{<2d}$ can be written uniquely as*

$$P(X) = P_e(X^2) + X P_o(X^2), \quad P_e, P_o \in \mathbb{F}[X]_{<d},$$

namely with $[X^i] P_e = [X^{2i}] P$ and $[X^i] P_o = [X^{2i+1}] P$ for $i \in \llbracket 0, d-1 \rrbracket$. The map $P \mapsto (P_e, P_o)$ is $\mathbb{F}$ -linear.

Proof. Each monomial X^j with $j < 2d$ appears on the right-hand side exactly once: as $(X^2)^i$ in $P_e(X^2)$ if $j = 2i$, and as $X(X^2)^i$ in $X P_o(X^2)$ if $j = 2i + 1$. Comparing coefficients gives existence and uniqueness; linearity is clear from the coefficient formulas. $\square$

2.3 Multilinear polynomials

Let $\mathbb{A}$ be a commutative ring. A polynomial in $\mathbb{A}[x_1, \dots, x_n]$ is *multilinear* if its degree in each variable is at most 1. The multilinear polynomials form an $\mathbb{A}$ -module $\mathcal{L}_n(\mathbb{A})$. For $a \in \llbracket 0, N-1 \rrbracket$, the *monomial* of index a is

$$x^a = \prod_{k=1}^n x_k^{a_k} = \prod_{k: a_k=1} x_k,$$

and the monomials x^a , $a \in \llbracket 0, N-1 \rrbracket$, form a basis of $\mathcal{L}_n(\mathbb{A})$, since the monomials form a basis of $\mathbb{A}[x_1, \dots, x_n]$.

Definition 2.5 (Coefficient form and table form). Every $f \in \mathcal{L}_n(\mathbb{A})$ is written uniquely as

$$f = \sum_{a=0}^{N-1} \alpha_a x^a, \quad \alpha_a \in \mathbb{A};$$

the family $\alpha = (\alpha_a)_{a \in \llbracket 0, N-1 \rrbracket}$ is the *coefficient form* of f . The *table form* of f is the family $(f(b))_{b \in \llbracket 0, N-1 \rrbracket}$ of its values on $B_n \subseteq \mathbb{A}^n$. A polynomial with coefficients in $\mathbb{A}$ is evaluated at points of $\mathbb{A}^n$ for every commutative ring $\mathbb{A}' \supseteq \mathbb{A}$; for $z \in \mathbb{A}^n$ we write $z^a = \prod_{k: a_k=1} z_k$, so that $f(z) = \sum_a \alpha_a z^a$.

Lemma 2.6 (Table form $\leftrightarrow$ coefficient form; [Mkh26, Lemma 2.9]). *With the notation of Definition 2.5, for all $a, b \in \llbracket 0, N-1 \rrbracket$,*

$$f(b) = \sum_{c \preceq b} \alpha_c \quad \text{and} \quad \alpha_a = \sum_{c \preceq a} (-1)^{\text{wt}(a) - \text{wt}(c)} f(c).$$

In particular, a multilinear polynomial is determined by its table form, and the map from table form to coefficient form (the Möbius transform) is computed with $\frac{n}{2}N$ subtractions.

Proof. For $b \in B_n$ we have $b^c = \prod_{k: c_k=1} b_k$, which equals 1 if $c \preceq b$ and 0 otherwise; evaluating $f = \sum_c \alpha_c x^c$ at b gives the first identity. For the second, substitute the first into its right-hand side:

$$\sum_{b \preceq a} (-1)^{\text{wt}(a) - \text{wt}(b)} \sum_{c \preceq b} \alpha_c = \sum_{c \preceq a} \alpha_c \sum_{b \preceq c} (-1)^{\text{wt}(a) - \text{wt}(b)}.$$

The inner sum runs over the subsets of the $\text{wt}(a) - \text{wt}(c)$ positions where a has a 1 and c a 0; it equals $(1-1)^{\text{wt}(a) - \text{wt}(c)}$, which is 1 if $c = a$ and 0 otherwise.

For the cost, the second identity factors over the bits: define $g^{(0)} = f|_{B_n}$ and, for $k \in \llbracket 1, n \rrbracket$, $g^{(k)}(a) = g^{(k-1)}(a)$ if $a_k = 0$ and $g^{(k)}(a) = g^{(k-1)}(a) - g^{(k-1)}(a - 2^{k-1})$ if $a_k = 1$. By induction on k , $g^{(k)}(a) = \sum (-1)^{\sum_{i \leq k} (a_i - c_i)} f(c)$ over the c with $c \preceq a$ and $c_i = a_i$ for $i > k$; for $k = n$ this is α_a . Each step performs $N/2$ subtractions. $\square$

2.4 Smooth domains

Definition 2.7 (Smooth domain; [Mkh26, Def. 2.14]). A *smooth domain* is a subgroup $L \leq \mathbb{F}_q^\times$ of order $M = 2^\mu$ with $\mu \in \mathbb{N}$. For $j \in \llbracket 0, \mu \rrbracket$, $L^{2^j} = \{\xi^{2^j} : \xi \in L\}$ is the image of the 2^j -th power map (not a Cartesian power).

Lemma 2.8 (Power maps on smooth domains; [Mkh26, Lemma 2.15]). *Let $\mu \in \mathbb{N}$.*

1. $\mathbb{F}_q^\times$ has a subgroup of order 2^μ if and only if $2^\mu \mid q-1$; it is then unique and cyclic.
2. Let L be a smooth domain of order $M = 2^\mu$ and $j \in \llbracket 0, \mu \rrbracket$. The map $\xi \mapsto \xi^{2^j}$ is a group homomorphism from L onto L^{2^j} , which is the smooth domain of order $M/2^j$, and every element of L^{2^j} has exactly 2^j preimages in L .
3. $(L^{2^j})^2 = L^{2^{j+1}}$ for $j \in \llbracket 0, \mu-1 \rrbracket$.
4. If $\mu \geq 1$, then $-1 \in L$, $-1 \neq 1$, and the preimages of $\eta = \xi^2 \in L^2$ under squaring are exactly ξ and $-\xi$, which are distinct.

Proof. (1) A subgroup of order e of a group of order $q-1$ exists only if $e \mid q-1$ (Lagrange's theorem). Conversely, let $e \mid q-1$. By Fact 2.3, $\mathbb{F}_q^\times$ is generated by some ω of order $q-1$, and $\omega^{(q-1)/e}$ generates a cyclic subgroup of order e . Every subgroup of order e is contained in the set of roots of $X^e - 1$, which has at most e elements by Lemma 2.2; so there is exactly one such subgroup. (2) The power map is a homomorphism because L is commutative. If ω generates L , then ω^{2^j} generates the image and has order $2^{\mu-j}$; by (1) the image is the smooth domain of that order. The preimages of an element form a coset of the kernel, which has order $|L|/|L^{2^j}| = 2^j$. (3) $(\xi^{2^j})^2 = \xi^{2^{j+1}}$. (4) The kernel of squaring has order 2 by (2) and consists of roots of $X^2 - 1 = (X-1)(X+1)$; since the characteristic is odd, $-1 \neq 1$, so the kernel is $\{1, -1\}$. The preimages of ξ^2 form the coset $\{\xi, -\xi\}$, and $\xi \neq -\xi$ because $2\xi \neq 0$. $\square$

Definition 2.9 (Fibres; [Mkh26, Def. 2.16]). Let L be a smooth domain of order $M \geq 2$. The *fibre* over $\eta \in L^2$ is the set $\{\xi, -\xi\}$ of its two square roots in L (Lemma 2.8(4)). The $M/2$ fibres partition L .

2.5 Reed–Solomon codes

Definition 2.10 (Reed–Solomon code; [Mkh26, Def. 2.17]). Let L be a smooth domain of order M and $1 \leq d \leq M$. The *evaluation map* is $\text{ev}_L : \mathbb{F}[X] \rightarrow \mathbb{F}^L$, $P \mapsto (P(\xi))_{\xi \in L}$, and the *Reed–Solomon code* of degree bound d on L over $\mathbb{F}$ is

$$\text{RS}_{\mathbb{F}}[L, d] = \text{ev}_L(\mathbb{F}[X]_{<d}) \subseteq \mathbb{F}^L,$$

of rate $\rho = d/M$. Elements of $\mathbb{F}^L$ are *words*, and elements of $\text{RS}_{\mathbb{F}}[L, d]$ are *codewords*. For $w, w' \in \mathbb{F}^L$, the *relative Hamming distance* is $\Delta(w, w') = |\{\xi \in L : w(\xi) \neq w'(\xi)\}|/M$, and $\Delta(w, \mathcal{C}) = \min_{c \in \mathcal{C}} \Delta(w, c)$ for a nonempty $\mathcal{C} \subseteq \mathbb{F}^L$. Two words *agree* on $S \subseteq L$ if they are equal at every point of S .

Lemma 2.11 (Hamming distance; [Mkh26, Lemma 2.18]). Δ is a metric on $\mathbb{F}^L$. In particular, if w agrees with w' on at least αM points and w' agrees with w'' on at least βM points, then w agrees with w'' on at least $(\alpha + \beta - 1)M$ points.

Proof. Symmetry, and $\Delta(w, w') = 0$ if and only if $w = w'$, are clear. If $w(\xi) \neq w''(\xi)$, then $w(\xi) \neq w'(\xi)$ or $w'(\xi) \neq w''(\xi)$; counting gives the triangle inequality, and the second statement is the triangle inequality written with agreements. $\square$

Lemma 2.12 (Parameters of Reed–Solomon codes; [Mkh26, Lemma 2.19]). *Let $\mathcal{C} = \text{RS}_{\mathbb{F}}[L, d]$ with $|L| = M$ and rate $\rho = d/M$.*

1. $\text{ev}_L : \mathbb{F}[X]_{<d} \rightarrow \mathcal{C}$ is an $\mathbb{F}$ -linear bijection; for $c \in \mathcal{C}$ we write P_c for the unique polynomial of $\mathbb{F}[X]_{<d}$ with $\text{ev}_L(P_c) = c$. Since ev_L is injective on $\mathbb{F}[X]_{<M}$, P_c depends only on the word c .
2. Two distinct codewords agree on at most $d - 1$ points of L .
3. If $2\delta < 1 - (d - 1)/M$, in particular if $\delta \leq (1 - \rho)/2$, every word is within distance δ of at most one codeword.

Proof. Items 1 and 2 follow from Lemma 2.2, as $d \leq M$ and the points of L are distinct. For item 3, two codewords within distance δ of w would be within distance $2\delta < 1 - (d - 1)/M$ of each other (Lemma 2.11), so they would agree on more than $d - 1$ points, contradicting item 2. Finally $(1 - \rho)/2 < (1 - (d - 1)/M)/2$. $\square$

Fact 2.13 (Efficient unique decoding [WB86, Gao03]). *There is an algorithm that, given L , d and $w \in \mathbb{F}^L$, outputs the unique $c \in \text{RS}_{\mathbb{F}}[L, d]$ with $\Delta(w, c) < \frac{1}{2}(1 - (d - 1)/M)$ whenever it exists, using a number of operations in $\mathbb{F}$ polynomial in M . Checking the distance of the output to w decides whether such a codeword exists, with $O(M)$ further comparisons.*

Lemma 2.14 (Decoding over an extension field; [Mkh26, Lemma 2.21]). *Let L be a smooth domain of order M , $1 \leq d \leq M$, and $w \in \mathbb{F}_q^L$. If $c \in \text{RS}_{\mathbb{F}}[L, d]$ satisfies $\Delta(w, c) < \frac{1}{2}(1 - (d - 1)/M)$, then $P_c \in \mathbb{F}_q[X]_{<d}$; in particular $c \in \mathbb{F}_q^L$.*

Proof. Let $\text{Frob} : \mathbb{F} \rightarrow \mathbb{F}$, $x \mapsto x^q$; it is an injective ring homomorphism, since q is a power of the characteristic p and $(x + y)^p = x^p + y^p$. Its fixed points are roots of $X^q - X$, of which there are at most q (Lemma 2.2), and every element of $\mathbb{F}_q$ is one; so they are exactly the elements of $\mathbb{F}_q$. Apply Frob entrywise to words and coefficientwise to polynomials. Since $L \subseteq \mathbb{F}_q$, $\text{Frob}(P(\xi)) = \text{Frob}(P)(\xi)$ for $\xi \in L$, so $\text{Frob}(c) = \text{ev}_L(\text{Frob}(P_c)) \in \text{RS}_{\mathbb{F}}[L, d]$. As Frob is injective and $\text{Frob}(w) = w$, $\Delta(w, \text{Frob}(c)) = \Delta(w, c)$; by Lemma 2.12(3), $\text{Frob}(c) = c$. Then $\text{Frob}(P_c)$ and P_c lie in $\mathbb{F}[X]_{<d}$ and evaluate to c , so they are equal (Lemma 2.12(1)), and the coefficients of P_c lie in $\mathbb{F}_q$. $\square$

2.6 Even–odd split of words

Let L be a smooth domain of order $M \geq 2$. For a word $w \in \mathbb{F}^L$, define $w_e, w_o \in \mathbb{F}^{L^2}$ by

$$w_e(\xi^2) = \frac{w(\xi) + w(-\xi)}{2}, \quad w_o(\xi^2) = \frac{w(\xi) - w(-\xi)}{2\xi} \quad (\xi \in L). \quad (1)$$

Lemma 2.15 (Even–odd split of words; [Mkh26, Lemma 2.22]). *Let $w \in \mathbb{F}^L$ with $|L| = M \geq 2$.*

1. w_e and w_o are well defined, and $w \mapsto (w_e, w_o)$ is $\mathbb{F}$ -linear.
2. (Locality.) For $\eta \in L^2$, $w_e(\eta)$ and $w_o(\eta)$ depend only on the values of w on the fibre over η .
3. For every $\xi \in L$,

$$w(\xi) = w_e(\xi^2) + \xi w_o(\xi^2), \quad w(-\xi) = w_e(\xi^2) - \xi w_o(\xi^2). \quad (2)$$

Consequently, w vanishes on the fibre over η if and only if $w_e(\eta) = w_o(\eta) = 0$.

4. If $1 \leq d \leq M/2$, $P \in \mathbb{F}[X]_{<2d}$ and $w = \text{ev}_L(P)$, then $w_e = \text{ev}_{L^2}(P_e)$ and $w_o = \text{ev}_{L^2}(P_o)$ (Lemma 2.4); in particular $w_e, w_o \in \text{RS}_{\mathbb{F}}[L^2, d]$.

Proof. (1) The denominators are nonzero because the characteristic is odd and $\xi \neq 0$. Replacing ξ by $-\xi$ leaves both right-hand sides of (1) unchanged, and $\pm\xi$ are the only square roots of ξ^2 in L (Lemma 2.8(4)), so the words are well defined. Linearity is clear. (2) is visible in (1). (3) Add and subtract the two formulas of (1). If $w_e(\eta) = w_o(\eta) = 0$, (2) gives $w(\pm\xi) = 0$; the converse follows from (1). (4) For $\xi \in L$, $P(\pm\xi) = P_e(\xi^2) \pm \xi P_o(\xi^2)$; substituting into (1) gives $w_e(\xi^2) = P_e(\xi^2)$ and $w_o(\xi^2) = P_o(\xi^2)$. Moreover $P_e, P_o \in \mathbb{F}[X]_{<d}$ and $d \leq M/2 = |L^2|$ (Lemma 2.8(2)). $\square$

2.7 Correlated agreement

The external result used in the soundness analysis is the correlated-agreement theorem of Ben-Sasson, Carmon, Ishai, Kopparty and Saraf [BSCI⁺23], for parameterised curves, in the unique-decoding regime. For $t \geq 1$ and $u_0, \dots, u_t \in \mathbb{F}^L$, the *curve* generated by $(u_0, \dots, u_t)$ is the family $(u_\beta)_{\beta \in \mathbb{F}}$ with $u_\beta = \sum_{i=0}^t \beta^i u_i$; for $t = 1$ it is the line $u_0 + \beta u_1$.

Theorem 2.16 ([BSCI⁺23, Theorem 1.5 with Theorem 1.2]). *Let $\mathcal{C} = \text{RS}_{\mathbb{F}}[L, d]$ with $|L| = M$ and rate $\rho = d/M$, let $0 < \delta \leq (1 - \rho)/2$, and let $t \geq 1$ and $u_0, \dots, u_t \in \mathbb{F}^L$. Let $Z = \{\beta \in \mathbb{F} : \Delta(u_\beta, \mathcal{C}) \leq \delta\}$. If $|Z| > tM$, then there are a set $L' \subseteq L$ with $|L'| \geq (1 - \delta)M$ and codewords $\hat{c}_0, \dots, \hat{c}_t \in \mathcal{C}$ such that u_i agrees with $\hat{c}_i$ on L' for every $i \in \llbracket 0, t \rrbracket$.*

We use the numbering of the full version (IACR ePrint 2020/654). There, $\text{RS}[\mathbb{F}, D, k]$ is the code of polynomials of degree at most k on a set $D \subseteq \mathbb{F}$, of blocklength $|D|$ and rate $(k + 1)/|D|$; we apply it with $D = L$ and $k = d - 1$. Theorem 1.2 of [BSCI⁺23] sets the error to $\epsilon = |D|/|\mathbb{F}|$ for $\delta \in (0, (1 - \rho)/2]$, and Theorem 1.5 concludes correlated agreement when $\Pr_\beta[\Delta(u_\beta, \mathcal{C}) \leq \delta] > t\epsilon$ for β uniform in $\mathbb{F}$, that is, when $|Z| > tM$. The theorem is stated over any finite field; we apply it with $\mathbb{F} \supseteq \mathbb{F}_q$ and $L \subseteq \mathbb{F}_q^\times \subseteq \mathbb{F}$. For $t = 1$ it is the statement used in [Mkh26, Theorem 2.23]. Theorem 2.16 is the only axiom of the Lean formalisation of this paper (Appendix B).

3 Interactive Oracle Proofs and Polynomial Commitments

This section recalls the proof-system notions used in the rest of the paper: interactive oracle proofs, their probability space and soundness notions, polynomial commitment schemes, Merkle trees, and the compilation of interactive oracle reductions into non-interactive arguments. Everything in Sections 3.1 to 3.3 is proved; this material is [Mkh26, §3.1–3.4], reproduced with its proofs. Section 3.4 concerns the compiled, non-interactive scheme; it restates the definitions of Chiesa, Di, Hu and Zheng [CDHZ25] that we need and quotes their theorem (Theorem 3.15), as in [Mkh26, §3.5]. It is used only in Section 8.

3.1 Interactive oracle proofs

A *relation* $\mathcal{R}$ is a set of pairs $(\mathbf{x}, \mathbf{w})$ of an *instance* $\mathbf{x}$ and a *witness* $\mathbf{w}$; its *language* is $L_{\mathcal{R}} = \{\mathbf{x} : \exists \mathbf{w}, (\mathbf{x}, \mathbf{w}) \in \mathcal{R}\}$. An instance may have an *oracle part*: a word to which the verifier has query access only.

Definition 3.1 (Interactive oracle proof). A *public-coin interactive oracle proof* (IOP) [BSCS16] with $\nu \geq 1$ rounds and *challenge sets* $\text{Ch}_1, \dots, \text{Ch}_\nu$ (finite and nonempty) is a pair (P, V) of a prover and a verifier. On a common instance $\mathbf{x}$ (and a witness given to P only), the interaction proceeds as follows. For $i = 1, \dots, \nu$:

1. P sends a message $\mathbf{m}_i$. Parts of $\mathbf{m}_i$ may be designated as *oracles*: V does not read them, but may later query individual entries.
2. V replies with a *challenge* $\mathbf{c}_i \in \text{Ch}_i$.

After the last round, V reads the non-oracle parts of the instance and of the messages, queries entries of the oracles (including the oracle part of the instance) at positions that are determined by these non-oracle parts and by the challenges, and outputs *accept* or *reject* as a deterministic function of everything it has read.

A *partial transcript* of i rounds is $\tau = (\mathbf{x}, \mathbf{m}_1, \mathbf{c}_1, \dots, \mathbf{m}_i, \mathbf{c}_i)$; for $i = 0$ it is the *empty transcript* $(\mathbf{x})$, and for $i = \nu$ it is a *full transcript*. Since the verifier's decision is a deterministic function of the full transcript (with oracles), we say that V *accepts* or *rejects* a full transcript. When the verifier makes checks “during” a round, as in the protocols below, these checks are predicates on the transcript that are evaluated at the end; the interaction always runs all ν rounds.

Provers and the probability space. A *deterministic prover* P^* is a function that maps each partial transcript of $i - 1$ rounds to a message $\mathbf{m}_i$. Its interaction with V on an instance $\mathbf{x}$ is determined by the challenges. All probabilities are over the *challenge space* $\Omega = \text{Ch}_1 \times \dots \times \text{Ch}_\nu$ with the uniform distribution, that is, with independent uniform challenges. We write τ_i for the partial transcript after i rounds, which is a function of $(\mathbf{c}_1, \dots, \mathbf{c}_i)$, and $\mathbf{m}_i$ for the i -th message, which is a function of $(\mathbf{c}_1, \dots, \mathbf{c}_{i-1})$. A *randomised prover* is a finitely supported probability distribution over deterministic provers, independent of the challenges.

Lemma 3.2 (Sequential challenges). *For $i \in \llbracket 1, \nu \rrbracket$, let $E_i \subseteq \text{Ch}_1 \times \dots \times \text{Ch}_i$, and suppose that for every $(\mathbf{c}_1, \dots, \mathbf{c}_{i-1})$ there are at most k_i values $\mathbf{c}_i \in \text{Ch}_i$ with $(\mathbf{c}_1, \dots, \mathbf{c}_i) \in E_i$. Then*

$$\Pr_\Omega [\exists i \in \llbracket 1, \nu \rrbracket : (\mathbf{c}_1, \dots, \mathbf{c}_i) \in E_i] \leq \sum_{i=1}^{\nu} \frac{k_i}{|\text{Ch}_i|}.$$

Proof. Under the uniform distribution on Ω , the prefix $(\mathbf{c}_1, \dots, \mathbf{c}_i)$ is uniform on $\text{Ch}_1 \times \dots \times \text{Ch}_i$. Counting the elements of E_i prefix by prefix gives $|E_i| \leq k_i \prod_{i' < i} |\text{Ch}_{i'}|$, so the probability of the i -th event is at most $k_i/|\text{Ch}_i|$. Conclude by the union bound. $\square$

Lemma 3.3 (Independent queries). *Let Ω' be a finite nonempty set with the uniform distribution, L a finite nonempty set, $\kappa \geq 1$, and let $\varpi \mapsto \mathcal{S}(\varpi) \subseteq L$ and $\mathcal{A} \subseteq \Omega'$ be given. On $\Omega' \times L^{\times \kappa}$ with the uniform distribution, where $L^{\times \kappa}$ denotes the Cartesian power (the set of κ -tuples of elements of L),*

$$\Pr[\varpi \in \mathcal{A} \text{ and } \xi^{(1)}, \dots, \xi^{(\kappa)} \in \mathcal{S}(\varpi)] = \mathbb{E}_\varpi \left[\mathbf{1}_{\mathcal{A}}(\varpi) \cdot \left(\frac{|\mathcal{S}(\varpi)|}{|L|} \right)^\kappa \right],$$

where $\mathbf{1}_{\mathcal{A}}$ is the indicator function of $\mathcal{A}$. In particular, if $\pi \geq 0$ and $|\mathcal{S}(\varpi)| \leq \pi|L|$ for all $\varpi \in \mathcal{A}$, the probability is at most $\Pr[\mathcal{A}] \pi^\kappa \leq \pi^\kappa$.

Proof. For fixed $\varpi \in \mathcal{A}$, the number of κ -tuples of elements of $\mathcal{S}(\varpi)$ is $|\mathcal{S}(\varpi)|^\kappa$; divide by $|\Omega'| |L|^\kappa$ and sum over ϖ . $\square$

Lemma 3.4 (Randomised provers). *Fix an instance $\mathbf{x}$, and let E be a set of full transcripts of $\mathbf{x}$ (for instance, the accepted ones). If $\Pr_\Omega[\tau_\nu \in E] \leq \varepsilon$ for every deterministic prover, then the same bound holds for every randomised prover. Equivalently, if a randomised prover achieves $\Pr[\tau_\nu \in E] > \varepsilon$, so does some deterministic prover in its support.*

Proof. Since the prover's randomness is independent of the challenges, the probability for a randomised prover is the average, over its distribution, of the probabilities for deterministic provers. $\square$

Definition 3.5 (Completeness and soundness). The IOP is *complete* for $\mathcal{R}$ if, for every $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$, the honest prover makes $\mathcal{V}$ accept with probability 1. It has *soundness error* ε if, for every $\mathbf{x} \notin L_{\mathcal{R}}$ and every deterministic prover P^* , $\mathcal{V}$ accepts with probability at most ε .

By Lemma 3.4, soundness also holds against randomised provers.

3.2 Round-by-round soundness

For the Fiat–Shamir compilation, the relevant notions are round by round: every partial transcript is classified as *doomed* or not, and no round can turn a doomed transcript into a non-doomed one except with small probability. The following definitions are those of [BGTZ24, Definitions 3.11 and 3.12]; the soundness notion was introduced in [CCH⁺19], where it is phrased with a *state function*, which amounts to specifying a set $\mathcal{D}$ of doomed transcripts, and its knowledge analogue in [CMS19]. We place the verifier's query positions in the last challenge c_ν .

Definition 3.6 (Round-by-round soundness). An IOP with ν rounds has *round-by-round soundness error* $(\varepsilon_1, \dots, \varepsilon_\nu) \in [0, 1]^\nu$ for $\mathcal{R}$ if there is a set $\mathcal{D}$ of partial transcripts, the *doomed* ones, such that:

1. if $\mathbf{x} \notin L_{\mathcal{R}}$, then the empty transcript $(\mathbf{x})$ is doomed;
2. for every $i \in \llbracket 1, \nu \rrbracket$, every doomed partial transcript τ of $i - 1$ rounds and every message $\mathbf{m}_i$,

$$\Pr_{c_i \leftarrow \text{Ch}_i} [(\tau, \mathbf{m}_i, c_i) \notin \mathcal{D}] \leq \varepsilon_i;$$

3. $\mathcal{V}$ rejects every doomed full transcript.

We write $\varepsilon_{\text{rbr}} = \max_i \varepsilon_i$.

Definition 3.7 (Round-by-round knowledge soundness). An IOP with ν rounds is *round-by-round knowledge sound* for $\mathcal{R}$ with error $(\varepsilon_1, \dots, \varepsilon_\nu) \in [0, 1]^\nu$ if there are a set $\mathcal{D}$ of partial transcripts and an algorithm Ext , the *extractor*, running in time polynomial in the size of the instance (including its oracle part), defined on all pairs of a partial transcript and a next message, such that:

1. the empty transcript $(\mathbf{x})$ is doomed for every instance $\mathbf{x}$;
2. for every $i \in \llbracket 1, \nu \rrbracket$, every doomed partial transcript τ of $i - 1$ rounds and every message $\mathbf{m}_i$,
if

$$\Pr_{c_i \leftarrow \text{Ch}_i} [(\tau, \mathbf{m}_i, c_i) \notin \mathcal{D}] > \varepsilon_i,$$

then $(\mathbf{x}, \text{Ext}(\tau, \mathbf{m}_i)) \in \mathcal{R}$, where $\mathbf{x}$ is the instance of τ ;

3. $\mathcal{V}$ rejects every doomed full transcript.

We write $\varepsilon_{\text{rbr}} = \max_i \varepsilon_i$.

The polynomial-time requirement on Ext matters only for the compiled scheme of Section 3.4; the results of Sections 3.2 and 3.3 use conditions 1–3 but not the running time of Ext .

Lemma 3.8 (Round-by-round implies overall). *Let P^* be a deterministic prover and $\mathbf{x}$ an instance, and let τ_i and $\mathbf{m}_i$ be as in Section 3.1.*

1. *If the IOP has round-by-round soundness error $(\varepsilon_i)_i$ and $\mathbf{x} \notin L_{\mathcal{R}}$, then V accepts with probability at most $\sum_{i=1}^{\nu} \varepsilon_i$.*
2. *If the IOP is round-by-round knowledge sound with error $(\varepsilon_i)_i$, then*

$$\Pr[V \text{ accepts}] \leq \sum_{i=1}^{\nu} \varepsilon_i + \Pr[\exists i \in \llbracket 1, \nu \rrbracket : \tau_{i-1} \in \mathcal{D} \text{ and } (\mathbf{x}, \text{Ext}(\tau_{i-1}, \mathbf{m}_i)) \in \mathcal{R}].$$

In particular, if V accepts with probability greater than $\sum_i \varepsilon_i$, then there exist $i \in \llbracket 1, \nu \rrbracket$ and challenges $\mathbf{c}_1, \dots, \mathbf{c}_{i-1}$ such that $\tau_{i-1} \in \mathcal{D}$ and $(\mathbf{x}, \text{Ext}(\tau_{i-1}, \mathbf{m}_i)) \in \mathcal{R}$.

Proof. In both cases τ_0 is doomed, and an accepted full transcript τ_{ν} is not doomed (condition 3). So if V accepts, there is a first round i with $\tau_{i-1} \in \mathcal{D}$ and $\tau_i \notin \mathcal{D}$. For (2), let E_i be the set of $(\mathbf{c}_1, \dots, \mathbf{c}_i)$ such that $\tau_{i-1} \in \mathcal{D}$, $(\mathbf{x}, \text{Ext}(\tau_{i-1}, \mathbf{m}_i)) \notin \mathcal{R}$ and $\tau_i \notin \mathcal{D}$. For fixed $(\mathbf{c}_1, \dots, \mathbf{c}_{i-1})$, the transcript τ_{i-1} and the message $\mathbf{m}_i$ are fixed; if the first two conditions hold, condition 2 of Definition 3.7 (in contrapositive form) shows that at most $\varepsilon_i |\text{Ch}_i|$ values of $\mathbf{c}_i$ satisfy the third. Acceptance implies that some E_i occurs or that the event in the displayed probability occurs, and Lemma 3.2 bounds the probability of the former by $\sum_i \varepsilon_i$. For (1), the same argument applies with E_i defined without the condition on Ext , using condition 2 of Definition 3.6. $\square$

3.3 Polynomial commitment schemes

A *multilinear polynomial commitment scheme* lets a prover commit to a multilinear polynomial $f \in \mathcal{L}_n(\mathbb{F})$ (Section 2.3) and later prove claims of the form $f(z) = v$ with $z \in \mathbb{F}^n$ and $v \in \mathbb{F}$. We describe it in the IOP model, where the commitment is an oracle.

Definition 3.9 (Polynomial commitment in the IOP model). A multilinear polynomial commitment scheme in the IOP model consists of:

- an *encoding* Enc that maps each $f \in \mathcal{L}_n(\mathbb{F})$ to a word $\text{Enc}(f)$, the *commitment*;
- an *evaluation protocol*: a public-coin IOP whose instance is $(w; z, v)$, with a word w (the commitment) as oracle part, a point $z \in \mathbb{F}^n$ and a value $v \in \mathbb{F}$.

It is *complete* if the evaluation protocol accepts with probability 1 when the prover is honest, $w = \text{Enc}(f)$ and $f(z) = v$.

Every claim (z, v) is true for *some* multilinear polynomial (for instance the constant polynomial v), so the relation $\{((z, v), f) : f(z) = v\}$ says nothing about the commitment. The meaningful requirement is that the commitment determines the polynomial. To express it, fix a function dist from pairs of words to real numbers, and a threshold δ , such that

$$\text{for every word } w, \text{ there is at most one } f \in \mathcal{L}_n(\mathbb{F}) \text{ with } \text{dist}(w, \text{Enc}(f)) \leq \delta, \quad (3)$$

and consider the relation

$$\mathcal{R}_{\text{eval}}^{\delta} = \{((w; z, v), f) : \text{dist}(w, \text{Enc}(f)) \leq \delta \text{ and } f(z) = v\}.$$

Definition 3.10 (Evaluation binding). The evaluation protocol is ε -*evaluation binding* if for every word w , point z and values $v \neq v'$, it is not the case that some deterministic prover makes the verifier accept $(w; z, v)$ with probability greater than ε and some deterministic prover makes it accept $(w; z, v')$ with probability greater than ε .

By Lemma 3.4, ε -evaluation binding also holds when randomised provers are allowed.

Lemma 3.11 (Knowledge soundness implies binding). *If the evaluation protocol is round-by-round knowledge sound for $\mathcal{R}_{\text{eval}}^\delta$ with error $(\varepsilon_i)_i$, and (3) holds, then it is ε -evaluation binding with $\varepsilon = \sum_i \varepsilon_i$.*

Proof. Suppose that provers are accepted on $(w; z, v)$ and on $(w; z, v')$, each with probability greater than ε . By Lemma 3.8(2), the extractor outputs, on some partial transcript of each interaction, polynomials f and f' with $((w; z, v), f) \in \mathcal{R}_{\text{eval}}^\delta$ and $((w; z, v'), f') \in \mathcal{R}_{\text{eval}}^\delta$. Both satisfy $\text{dist}(w, \text{Enc}(\cdot)) \leq \delta$, so $f = f'$ by (3), and $v = f(z) = f'(z) = v'$. $\square$

3.4 Non-interactive arguments

This subsection concerns only the non-interactive scheme (Section 8).

Merkle trees. Let $H : \{0, 1\}^* \rightarrow \{0, 1\}^{\lambda_H}$ be a hash function with output length λ_H , and let $\parallel$ denote concatenation of bit strings. The *Merkle tree* [Mer89] over a vector $\mathbf{d} = (\mathbf{d}_0, \dots, \mathbf{d}_{T-1})$ of bit strings, with T a power of 2, is the complete binary tree whose leaves are the hashes $H(0 \parallel \mathbf{d}_i)$ and whose internal nodes are $H(1 \parallel \text{left child} \parallel \text{right child})$; its *root* $\text{MT}(\mathbf{d})$ is a commitment to the vector. An *opening* of position i consists of $\mathbf{d}_i$ and its *authentication path*, the $\log_2 T$ siblings on the path from the leaf to the root; it is checked by recomputing the root. The entries $\mathbf{d}_i$ may be encodings of several field elements, for instance the two values of a word on one fibre.

Lemma 3.12 (Merkle binding). *1. Given two accepting openings of the same position to different values under the same root, one can compute a collision of H , that is, two distinct inputs with the same hash, with $O(\log_2 T)$ hash evaluations.*

2. Given two vectors $\mathbf{d} \neq \mathbf{d}'$ of length T with $\text{MT}(\mathbf{d}) = \text{MT}(\mathbf{d}')$, one can compute a collision of H with $O(T)$ hash evaluations.

Proof. (1) Number the levels of the tree from 0 (leaf hashes) to $\log_2 T$ (root). Node values are strings of the fixed length λ_H , and the position i fixes, at every level, on which side the path node lies; so the input $1 \parallel \text{left} \parallel \text{right}$ hashed at a level determines the path node below it and its sibling. Recompute both paths from the leaves to the root, and let t be the first level at which the two recomputed path nodes are equal; it exists because both paths end at the same root. If $t = 0$, the hashed inputs $0 \parallel \mathbf{d}_i$ and $0 \parallel \mathbf{d}'_i$ are distinct. If $t > 0$, the inputs hashed at level t are distinct, because they contain the path nodes of level $t - 1$, which differ by the choice of t . In both cases these two inputs have the same hash. (2) Compute both trees, and apply (1) at a position where $\mathbf{d}$ and $\mathbf{d}'$ differ, taking the authentication paths from the two trees. $\square$

Random oracles. In the *random oracle model*, H is modelled as a uniformly random function to which all parties have oracle access; in the *quantum random oracle model* (QROM) [BDF⁺11], the adversary is a quantum algorithm that may query H in superposition. The *BCS transformation* of Ben-Sasson, Chiesa and Spooner [BSCS16] compiles a public-coin IOP into a non-interactive argument: each oracle message is replaced by a Merkle root, each challenge is derived by hashing

the instance and the transcript so far (the Fiat–Shamir paradigm), and the proof contains the non-oracle messages, the roots, and an opening of every queried oracle entry. The BCS theorem in the QROM of Chiesa, Manohar and Spooner [CMS19, Theorem 8.6] covers IOPs whose instance has no oracle part. An evaluation protocol has an oracle part, the commitment, which in the non-interactive scheme is replaced by its Merkle root and is chosen by the adversary. This setting is covered by the extension of the BCS transformation to *interactive oracle reductions* (IORs) of Chiesa, Di, Hu and Zheng [CDHZ25], which we now recall in the special case that we need. All definitions below are those of [CDHZ25]; we restate them to fix notation, and we indicate the specialisation. We write ε for the error terms that [CDHZ25] denotes by κ , since κ is our number of queries.

Relations with an implicit instance [CDHZ25, §3.3]. Fix a finite alphabet Σ and a length $l_y \geq 1$. A *partial string* is an element of $(\Sigma \sqcup \{\perp\})^{l_y}$, the symbol $\perp$ marking an undefined entry; $\text{Dom}(y)$ is the set of positions where y is defined, and y is *full* if $\text{Dom}(y)$ is everything. The fractional Hamming distance $\Delta(y, y')$ of two partial strings is the fraction of positions at which they differ, $\perp$ being different from every symbol of Σ . A *relation with implicit instance* is a set $\mathcal{R}$ of triples $(\mathbf{x}, y, \mathbf{w})$ of an *explicit instance* $\mathbf{x}$, a full string $y \in \Sigma^{l_y}$, the *implicit instance*, to which the verifier has oracle access only, and a witness $\mathbf{w}$. For $\delta \in [0, 1]$, the *relaxed relation* is

$$\mathcal{R}_{\leq \delta} = \{ (\mathbf{x}, y, \mathbf{w}) : y \in (\Sigma \sqcup \{\perp\})^{l_y}, \exists y^* \in \Sigma^{l_y} \text{ with } (\mathbf{x}, y^*, \mathbf{w}) \in \mathcal{R} \text{ and } \Delta(y, y^*) \leq \delta \}.$$

IOPs as interactive oracle reductions [CDHZ25, §3.4]. Consider an IOP (Definition 3.1) with ν rounds whose instance is $(\mathbf{x}; y)$ with oracle part y , in which every prover message is a string Π_i over Σ , whose challenges are uniformly random bit strings $\text{vr}_i \in \{0, 1\}^{\lambda_i}$, and whose verifier reads (by queries) positions of y and of the Π_i that are determined by $\mathbf{x}$ and the challenges only; a message part that the verifier reads entirely is a string all of whose positions are read. Let $\mathcal{R}_{\text{acc}} = \{(\top, (), \mathbf{w})\}$, where $\top$ is a fixed symbol, $()$ is the empty implicit instance, and $\mathbf{w}$ is arbitrary; then $(\mathcal{R}_{\text{acc}})_{\leq \delta} = \mathcal{R}_{\text{acc}}$ for every δ . If the verifier outputs the instance $(\top, ())$ when it accepts and rejects otherwise, the IOP is an IOR from $\mathcal{R}$ to $\mathcal{R}_{\text{acc}}$ in the sense of [CDHZ25, §3.4], with $n_y = 1$ input implicit instance and $n'_y = 0$ output implicit instances. $\mathcal{R}_{\text{acc}}$ is *monotone* in the sense of [CDHZ25, Definition 11.1], because it has no implicit instance. A *transcript* is a sequence $\text{tr} = (\Pi_1, \text{vr}_1, \Pi_2, \dots)$; the empty transcript is written $\text{tr}_\emptyset$.

In the proof of the theorem below, the verifier and the knowledge state function are evaluated on strings produced by an extractor, whose entries may be undefined [CDHZ25, Constructions 11.8 and 11.9]. We therefore require the verifier to be defined when y and the Π_i are partial strings, and we take the statements $(\mathbf{x}, y)$ and the messages Π_i in Definition 3.13 to range over partial strings. For the protocols of this paper, the verifier rejects as soon as one of the entries it reads is $\perp$.

Definition 3.13 (Relaxed round-by-round knowledge soundness [CDHZ25]). This is [CDHZ25, Definitions 3.5 and 3.6] with output relation $\mathcal{R}_{\text{acc}}$. A *knowledge state function* for an IOR from $\mathcal{R}$ to $\mathcal{R}_{\text{acc}}$ is a deterministic function KState that, on input a proximity bound $\delta \in [0, 1]$, a statement $(\mathbf{x}, y)$, a transcript tr and a witness $\mathbf{w}$, outputs a bit $\text{KState}_\delta(\mathbf{x}, y, \text{tr}, \mathbf{w})$, such that:

1. (empty transcript) $\text{KState}_\delta(\mathbf{x}, y, \text{tr}_\emptyset, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\leq \delta}$;
2. (prover moves) if the prover is about to move after tr and $\text{KState}_\delta(\mathbf{x}, y, \text{tr}, \mathbf{w}) = 0$, then $\text{KState}_\delta(\mathbf{x}, y, \text{tr} \parallel \Pi, \mathbf{w}) = 0$ for every message Π ;
3. (full transcript) for a full transcript, $\text{KState}_\delta(\mathbf{x}, y, \text{tr}, \mathbf{w}) = 1$ if and only if the verifier accepts tr .

The IOR has *relaxed round-by-round knowledge soundness errors* $(\varepsilon_1^{\text{rr}}, \dots, \varepsilon_\nu^{\text{rr}})$, functions of $(\mathbf{x}, y, \delta)$, if there are a knowledge state function KState and a deterministic polynomial-time algorithm E , which does not depend on δ , such that for every $\delta \in [0, 1]$, every statement $(\mathbf{x}, y)$, every $i \in [1, \nu]$ and every transcript $\text{tr} = (\Pi_1, \text{vr}_1, \dots, \Pi_{i-1}, \text{vr}_{i-1}, \Pi_i)$,

$$\Pr_{\text{vr}_i \leftarrow \{0,1\}^{\lambda_i}} \left[\exists \mathbf{w} : \text{KState}_\delta(\mathbf{x}, y, \text{tr}, \text{E}(\mathbf{x}, y, \text{tr} \parallel \text{vr}_i, \mathbf{w})) = 0 \right. \\ \left. \text{and } \text{KState}_\delta(\mathbf{x}, y, \text{tr} \parallel \text{vr}_i, \mathbf{w}) = 1 \right] \leq \varepsilon_i^{\text{rr}}(\mathbf{x}, y, \delta).$$

For $\mathcal{R}_{\text{acc}}$, condition 3 is the specialisation of [CDHZ25, Definition 3.5], which asks that the verifier output an instance $(\mathbf{x}', y')$ with $(\mathbf{x}', y', \mathbf{w}) \in (\mathcal{R}_{\text{acc}})_{\leq \delta}$, that is, that it accept.

Salted Merkle commitments [CDHZ25, Definition 8.2]. The compiler of [CDHZ25] uses a salted variant of the Merkle tree: to commit to $\mathbf{a} = (a_1, \dots, a_l) \in \Sigma^l$ (l a power of 2, after padding), it samples uniformly random salts $\gamma_1, \dots, \gamma_l$ of a fixed length, sets the leaves to $\text{H}(a_h, \gamma_h)$ and the internal nodes to $\text{H}(\text{left child}, \text{right child})$, where H is a random oracle with output length λ_{H} ; the root is the commitment cm , and the salts and the tree form the trapdoor td . An opening of a set I of positions consists of (a_h, γ_h) and the authentication path of every $h \in I$, and is checked by recomputing the root. The proof of Lemma 3.12(1) applies verbatim, with the input (a_h, γ_h) in place of $0 \parallel \mathbf{d}_i$ and $\text{left} \parallel \text{right}$ in place of $1 \parallel \text{left} \parallel \text{right}$: *two accepting openings of the same position to different values under the same root and the same oracle yield a collision of that oracle.*

Definition 3.14 (Committed relations [CDHZ25, Definition 11.2]). Let $\mathcal{R}$ be a relation with implicit instance and $\delta \in [0, 1]$. A *committed instance* is $\text{cx} = (\mathbf{x}, \text{cm})$, where cm is a Merkle root, and a *committed witness* is $\text{cw} = (y, \mathbf{w}, \text{td})$.

- $(\text{cx}, \text{cw}) \in \text{CR}[\mathcal{R}]$ if $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}$ and cm and td are the root and the trapdoor of the salted Merkle tree of y with the salts recorded in td ;
- $(\text{cx}, \text{cw}) \in \widehat{\text{CR}}[\mathcal{R}, \delta]$ (the *relaxed committed relation*) if y is a partial string with $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\leq \delta}$, and the opening of all positions of $\text{Dom}(y)$ computed from td is accepted for y under the root cm .

In [CDHZ25] the committed instance also carries an index that selects, by domain separation, the random oracle used by the commitment. We fix this index in the scheme (the verifier rejects any other value) and omit it from the notation. This extra check only removes accepted proofs and makes no oracle query, so the extraction event of [CDHZ25, Definition 10.3] (acceptance and failed extraction) can only become smaller, and Theorem 3.15 still applies.

The compiler. $\text{BCS}[\Pi]$ denotes the non-interactive reduction of [CDHZ25, Construction 11.7] applied to an IOR Π , with salted Merkle commitments of output length λ_{H} . Its prover receives $(\text{cx}, \text{cw}) \in \text{CR}[\mathcal{R}]$; in round i it commits to Π_i by a salted Merkle tree, samples a fresh random salt, and derives vr_i by querying a random oracle on $\mathbf{x}$, the root cm of y , the roots of $\Pi_1, \dots, \Pi_i$ and the salts of rounds $1, \dots, i$. The proof consists of the roots, the salts, and the openings of every position of y and of the Π_i that the IOR verifier reads; the verifier recomputes the challenges, checks the openings, and runs the IOR verifier on the opened values. All random oracles are derived from one random oracle by domain separation.

Knowledge soundness. The security notion is post-quantum straightline knowledge soundness [CDHZ25, Definition 10.3]. It is stated for an *emulated oracle*, which answers the adversary's queries and gives the extractor an additional extraction interface. It has three error terms: a simulation error ε_{sim} , bounding the distinguishing advantage between the emulated oracle and a true random oracle; a commutativity error ε_{com} ; and the extraction error ε_{ext} , which bounds, for a Q -query quantum adversary $\tilde{P}$ that outputs a committed instance $\mathbf{cx}$ of its choice, a proof, and a witness for the output relation, the probability that

the verifier accepts, and the output $\mathbf{cw}$ of the extractor satisfies $(\mathbf{cx}, \mathbf{cw}) \notin \widehat{\text{CR}}[\mathcal{R}, \delta]$.

(For $\mathcal{R}_{\text{acc}}$ the output relation holds exactly when the verifier accepts, and its decider makes no oracle queries.) The extractor is a polynomial-time quantum algorithm and is *straightline*: it does not rewind the adversary.

Theorem 3.15 ([CDHZ25, Theorems 6.10, 8.3 and 11.3], specialised). *Let Π be an IOR from $\mathcal{R}$ to $\mathcal{R}_{\text{acc}}$ as above, with ν rounds, challenge lengths $\lambda_1, \dots, \lambda_\nu$ and $\lambda_{\min} = \min_i \lambda_i$, relaxed round-by-round knowledge soundness errors $(\varepsilon_i^{\text{rr}})_i$ (Definition 3.13), and let $\varepsilon_{\text{rr}}(\delta) = \max_{i, (\mathbf{x}, y)} \varepsilon_i^{\text{rr}}(\mathbf{x}, y, \delta)$, the maximum being over the statements of bounded size under consideration. Let $l_{\max}$ be the largest length of y and of the strings Π_i , q_V the number of random-oracle queries of the verifier of $\text{BCS}[\Pi]$, $q_{\widehat{\text{CR}}}$ that of the decider of $\widehat{\text{CR}}[\mathcal{R}, \delta]$, q_s the number of positions of y and of the Π_i that the IOR verifier reads, and $T_1 = Q + 1 + q_V + \nu + q_{\widehat{\text{CR}}}$. Then there is a polynomial-time quantum extractor such that, for every $\delta \in [0, 1]$ and every query bound Q , $\text{BCS}[\Pi]$ is a non-interactive reduction from $(\text{CR}[\mathcal{R}], \widehat{\text{CR}}[\mathcal{R}, \delta])$ to $(\text{CR}[\mathcal{R}_{\text{acc}}], \widehat{\text{CR}}[\mathcal{R}_{\text{acc}}, \delta])$ with post-quantum straightline knowledge soundness errors $(0, \varepsilon_{\text{com}}, \varepsilon_{\text{ext}})$, where $\varepsilon_{\text{com}}(q) \leq 240(q + 1)/2^{\lambda_H}$ and*

$$\begin{aligned} \varepsilon_{\text{ext}}(Q) \leq & 320(Q + \nu + 1)(Q + \nu) \varepsilon_{\text{rr}}(\delta) + \frac{4(Q + \nu + 1)\nu}{2^{\lambda_{\min}}} \\ & + \frac{32 l_{\max} + 1280 Q (2Q + 1)^2 + 128 q_s \lceil \log_2 l_{\max} \rceil}{2^{\lambda_H}} \\ & + \frac{960(Q + 1) T_1^2 + 480 q_V^2 (Q + \nu + q_V + 2)}{2^{\lambda_H}}. \end{aligned}$$

Derivation from [CDHZ25]. By [CDHZ25, Theorem 6.10], Π has post-quantum state-restoration straightline knowledge soundness error $\varepsilon^{\text{sr}} \leq 80(Q + \nu + 1)(\mathbf{m}_{\text{FS}} + \nu) \varepsilon_{\text{rr}}(\delta) + (Q + \nu + 1)\nu/2^{\lambda_{\min}}$ for a Q -query prover with total query mass $\mathbf{m}_{\text{FS}}$ to the challenge oracle. By [CDHZ25, Theorem 8.3], the salted Merkle commitments are post-quantum multi-extractable with errors, at query bound Q' ,

$$\begin{aligned} \varepsilon_{\text{com}}(Q') &\leq \frac{240(Q' + 1)}{2^{\lambda_H}}, & \varepsilon_{\text{extract}} &\leq \frac{8 n_y l_{\max}}{2^{\lambda_H}}, \\ \varepsilon_{\text{offline}} &\leq \frac{160 \mathbf{m} Q'^2 + 16 q_s \log_2 l_{\max}}{2^{\lambda_H}}, & \varepsilon_{\text{online}} &\leq \frac{240 \mathbf{m}' Q'^2}{2^{\lambda_H}}, \end{aligned}$$

where $\mathbf{m}, \mathbf{m}'$ are the total query masses in the third argument of these functions in [CDHZ25]. Substitute these bounds into [CDHZ25, Theorem 11.3] with $n_y = 1$, $n'_y = 0$ and $q_{\widehat{\text{CR}}} = 0$. Every total query mass is at most the number of queries [CDHZ25, Definition 4.3 and Remark 4.2], so the masses $\mathbf{m}_{\text{FS}}, \mathbf{m}_{\text{VC}}, \mathbf{m}_{\text{MultiExtract}}$ over which the maximum in [CDHZ25, Theorem 11.3] is taken are at most Q . This gives the displayed terms in order: $4\varepsilon^{\text{sr}}$; $4\varepsilon_{\text{extract}}$ with $n_y = 1$; $8\varepsilon_{\text{offline}}$ at query bound $2Q + 1$ with mass $\mathbf{m}_{\text{VC}} \leq Q$ (by [CDHZ25, Remark 11.5] this term vanishes when $n'_y = 0$, and we keep it as an upper bound); $4\varepsilon_{\text{online}}$ at query bound T_1 with mass $\mathbf{m}_{\text{MultiExtract}} + 1 \leq Q + 1$; and $2q_V^2 \varepsilon_{\text{com}}$ at query bound $Q + 1 + \nu + q_V$. $\square$

We use the constants of [CDHZ25, Theorem 8.3] as stated there (version of 2 March 2026); they follow from [CDHZ25, Theorem 8.4, Lemmas 8.7 and 8.8] with the query counts of [CDHZ25, §8.3]. A classical adversary is a special case of a quantum one, so Theorem 3.15 also applies to classical adversaries in the random oracle model, with the same quantum extractor. For classical adversaries, Bünz, Mishra, Nguyen and Wang [BMNW24, Theorem 5.9] give an asymptotic statement of the same shape for their notion of round-by-round knowledge [BMNW24, Definition 5.5]; we do not use it.

4 The Kernel Polynomial and the Extraction Identity

This section introduces the two univariate polynomials of the scheme, the Kronecker encoding U_f of a multilinear polynomial and the kernel polynomial K_z of a point, and proves the identity $f(z) = [X^{N-1}] U_f K_z$ on which the commitment scheme rests. It then gives the algorithms used by the prover and the verifier, the algebraic form of an opening (Section 4.5), and the relation with other forms of the same identity (Section 4.6). Throughout, $n \geq 1$, $N = 2^n$, and $\mathbb{F}$ is a field; the results of this section hold over any field (and, except where a division is performed, over any commutative ring).

4.1 The Kronecker encoding and the kernel polynomial

Definition 4.1 (Kronecker encoding). For $f = \sum_{a=0}^{N-1} \alpha_a x^a \in \mathcal{L}_n(\mathbb{F})$ (Definition 2.5), the *Kronecker encoding* of f is

$$U_f(X) = \sum_{a=0}^{N-1} \alpha_a X^a \in \mathbb{F}[X]_{<N}.$$

Lemma 4.2 (Kronecker substitution). 1. $U_f(X) = f(X, X^2, X^4, \dots, X^{2^{n-1}})$, that is, U_f is obtained from f by the substitution $x_k \mapsto X^{2^{k-1}}$.

2. The map $f \mapsto U_f$ is an $\mathbb{F}$ -linear bijection from $\mathcal{L}_n(\mathbb{F})$ onto $\mathbb{F}[X]_{<N}$.

Proof. (1) Under the substitution, $x^a = \prod_{k:a_k=1} x_k$ becomes $\prod_{k:a_k=1} X^{2^{k-1}} = X^{\sum_k a_k 2^{k-1}} = X^a$. (2) The map sends the coefficient vector $(\alpha_a)_a$ in the basis $(x^a)_a$ to the same vector in the basis $(X^a)_{a < N}$ of $\mathbb{F}[X]_{<N}$. $\square$

Definition 4.3 (Kernel polynomial). For $z = (z_1, \dots, z_n) \in \mathbb{F}^n$, the *kernel polynomial* of z is

$$K_z(X) = \prod_{k=1}^n (X^{2^{k-1}} + z_k) \in \mathbb{F}[X].$$

For $z \in B_n$, K_z is the Boolean kernel polynomial of [Mkh26, Def. 4.1], whose family $(K_b)_{b \in B_n}$ is a basis of $\mathbb{F}[X]_{<N}$; here z is an arbitrary point.

Lemma 4.4 (Basic properties). Let $z \in \mathbb{F}^n$.

1. K_z is monic of degree $\sum_{k=1}^n 2^{k-1} = N - 1$.
2. (Recursion.) Writing $z = (z_1, z')$ with $z' = (z_2, \dots, z_n)$, $K_z(X) = (X + z_1) K_{z'}(X^2)$, where $K_{z'}$ has $n - 1$ variables.

3. (Coefficient rule; [Mkh26, Lemma 4.3].) For $j \in \llbracket 0, N-1 \rrbracket$,

$$[X^j] K_z = \prod_{k: j_k=0} z_k = z^{\bar{j}}.$$

Proof. (1) Each factor $X^{2^{k-1}} + z_k$ is monic of degree 2^{k-1} . (2) $\prod_{k=2}^n (X^{2^{k-1}} + z_k) = \prod_{k'=1}^{n-1} ((X^2)^{2^{k'-1}} + z_{k'+1}) = K_{z'}(X^2)$. (3) Expanding the product, each term chooses, for each k , either $X^{2^{k-1}}$ or z_k . The term that chooses $X^{2^{k-1}}$ exactly for k in a set $T \subseteq \llbracket 1, n \rrbracket$ is $X^{\sum_{k \in T} 2^{k-1}} \prod_{k \notin T} z_k$. By uniqueness of the binary decomposition, the exponent j arises from exactly one set, $T = \{k : j_k = 1\}$, so $[X^j] K_z = \prod_{k: j_k=0} z_k$. By Lemma 2.1, $j_k = 0$ if and only if $\bar{j}_k = 1$, so this product is $z^{\bar{j}}$. $\square$

4.2 The extraction identity

Theorem 4.5 (Extraction identity). *For every $f \in \mathcal{L}_n(\mathbb{F})$ and every $z \in \mathbb{F}^n$,*

$$f(z) = [X^{N-1}] U_f(X) K_z(X).$$

Proof. Since $U_f = \sum_a \alpha_a X^a$ and $\deg K_z = N-1$,

$$[X^{N-1}] U_f K_z = \sum_{a=0}^{N-1} \alpha_a [X^{N-1-a}] K_z = \sum_{a=0}^{N-1} \alpha_a z^{\bar{a}} = \sum_a \alpha_a z^a = f(z),$$

by Lemma 4.4(3) with $j = N-1-a = \bar{a}$, and $\bar{\bar{a}} = a$. $\square$

Corollary 4.6 (Linear combinations of points). *For $z^{(1)}, \dots, z^{(s)} \in \mathbb{F}^n$ and $\gamma_1, \dots, \gamma_s \in \mathbb{F}$,*

$$\sum_{i=1}^s \gamma_i f(z^{(i)}) = [X^{N-1}] U_f \cdot \sum_i \gamma_i K_{z^{(i)}}.$$

Proof. The map $K \mapsto [X^{N-1}] U_f K$ is linear; apply Theorem 4.5 to each point. $\square$

Corollary 4.6 is the starting point for openings at several points (Section 12).

Corollary 4.7 (Sums over the hypercube). *For every $f \in \mathcal{L}_n(\mathbb{F})$,*

$$\sum_{b \in B_n} f(b) = [X^{N-1}] U_f(X) \prod_{k=1}^n (1 + 2X^{2^{k-1}}).$$

If the characteristic of $\mathbb{F}$ is not 2, both sides equal $2^n f(\frac{1}{2}, \dots, \frac{1}{2})$.

Proof. $\sum_{b \in B_n} b^a = \prod_{k: a_k=0} 2 \cdot \prod_{k: a_k=1} 1 = 2^{n-\text{wt}(a)}$, since the factor b_k ranges over $\{0, 1\}$ for $a_k = 1$ and each k with $a_k = 0$ doubles the count. Hence $\sum_b f(b) = \sum_a \alpha_a 2^{n-\text{wt}(a)}$. Expanding $\prod_k (1 + 2X^{2^{k-1}})$ as in Lemma 4.4(3), $[X^j] \prod_k (1 + 2X^{2^{k-1}}) = 2^{\text{wt}(j)}$, and $[X^{N-1-a}] \cdot = 2^{\text{wt}(\bar{a})} = 2^{n-\text{wt}(a)}$; summing against α_a gives the identity. Finally $\prod_k (1 + 2X^{2^{k-1}}) = 2^n K_{(1/2, \dots, 1/2)}(X)$, so by Theorem 4.5 the right-hand side is $2^n f(\frac{1}{2}, \dots, \frac{1}{2})$. $\square$

4.3 Further identities

The following identities are not used by the scheme; they record the structure of the family $(K_z)_{z \in \mathbb{F}^n}$.

Proposition 4.8 (Bilinear and squaring identities). *For $y, z \in \mathbb{F}^n$:*

1. $[X^{N-1}] K_y K_z = \prod_{k=1}^n (y_k + z_k)$;
2. $K_y(X) K_{-y}(X) = \prod_{k=1}^n (X^{2^k} - y_k^2)$.

Proof. (1) By Lemma 4.4(3), $K_y = U_g$ for the multilinear $g = \sum_a y^{\bar{a}} x^a$, which equals $\prod_{k=1}^n (y_k + x_k)$ (expand the product: the term choosing x_k exactly for k with $a_k = 1$ is $y^{\bar{a}} x^a$). By Theorem 4.5, $[X^{N-1}] K_y K_z = g(z) = \prod_k (y_k + z_k)$. (2) Multiply the factors pairwise: $(X^{2^{k-1}} + y_k)(X^{2^{k-1}} - y_k) = X^{2^k} - y_k^2$. $\square$

4.4 Algorithms

We count operations in $\mathbb{F}$.

Lemma 4.9 (Costs). *Let $z \in \mathbb{F}^n$, $U \in \mathbb{F}[X]_{<N}$ and $\xi \in \mathbb{F}$.*

1. $K_z(\xi)$ is computed with $n - 1$ squarings, $n - 1$ multiplications and n additions, and ξ^N with one further squaring.
2. The vector $(z^a)_{a \in \llbracket 0, N-1 \rrbracket}$ is computed with $N - n - 1$ multiplications, and $[X^{N-1}] U K_z$ with N further multiplications and $N - 1$ additions.
3. The product $U K_z$ is computed with at most $2nN$ multiplications and $2nN$ additions.

Proof. (1) Compute $\xi^{2^{k-1}}$ for $k \in \llbracket 1, n \rrbracket$ by $n - 1$ successive squarings, add z_k to each, and multiply the n factors; one more squaring gives $\xi^{2^n} = \xi^N$. (2) The vector is built bit by bit: after processing bits $1, \dots, k$ it holds $(z^a)_{a < 2^k}$, and bit $k+1$ appends $z_{k+1} z^a$ for $a < 2^k$; this costs $\sum_{k=1}^n 2^{k-1} = N - 1$ multiplications, of which the n multiplications of z_k by $z^0 = 1$ are free. Then $[X^{N-1}] U K_z = \sum_a [X^a] U z^a$ by Theorem 4.5. (3) Put $G_0 = U$ and $G_k = G_{k-1} \cdot (X^{2^{k-1}} + z_k)$ for $k \in \llbracket 1, n \rrbracket$, so that $G_n = U K_z$. Coefficientwise, $[X^j] G_k = z_k [X^j] G_{k-1} + [X^{j-2^{k-1}}] G_{k-1}$. The polynomial G_{k-1} has at most $N + 2^{k-1} - 1 < 2N$ coefficients, so step k costs fewer than $2N$ multiplications and $2N$ additions. $\square$

4.5 The opening polynomials

The scheme checks the extraction identity through a polynomial identity in which the coefficient to be checked is moved to the degree N by a factor X .

Lemma 4.10 (Split). *Let $U \in \mathbb{F}[X]_{<N}$, $z \in \mathbb{F}^n$ and $v \in \mathbb{F}$. There exist $A, H \in \mathbb{F}[X]_{<N}$ with*

$$X U(X) K_z(X) = A(X) + v X^N + X^{N+1} H(X) \quad (4)$$

if and only if $v = [X^{N-1}] U K_z$. In that case A and H are unique, $A(0) = 0$, and $\deg H \leq N - 2$.

Proof. Let $G = U K_z$; $\deg G \leq (N - 1) + (N - 1) = 2N - 2$, so XG has no constant term and degree $\leq 2N - 1$. Put $A = \sum_{j=0}^{N-1} [X^j] XG X^j$, $a_N = [X^N] XG = [X^{N-1}] G$ and $H = \sum_{j \geq 0} [X^{N+1+j}] XG X^j$; then $XG = A + a_N X^N + X^{N+1} H$, with $\deg A < N$, $A(0) = [X^0] XG = 0$ and $\deg H \leq 2N - 1 - (N + 1) = N - 2$. If (4) holds with $A', H' \in \mathbb{F}[X]_{<N}$ and some v , then $A' + v X^N + X^{N+1} H'$ and $A + a_N X^N + X^{N+1} H$ are two expressions of XG in which the three parts occupy the disjoint degree ranges $\llbracket 0, N - 1 \rrbracket$, $\{N\}$ and $\llbracket N + 1, 2N \rrbracket$; comparing coefficients gives $A' = A$, $v = a_N$ and $H' = H$. Conversely, $v = a_N$ gives (4) with (A, H) . $\square$

Definition 4.11 (Opening polynomials). For $U \in \mathbb{F}[X]_{<N}$ and $z \in \mathbb{F}^n$, the *opening polynomials* $A_{U,z}, H_{U,z} \in \mathbb{F}[X]_{<N}$ are the polynomials A, H of Lemma 4.10 for $v = [X^{N-1}]UK_z$.

By Theorem 4.5, for $U = U_f$ the value is $v = f(z)$, and the opening polynomials are computed from $U_f K_z$ at the cost of Lemma 4.9(3).

Lemma 4.12 (Identity lemma). *Let $U, A, H \in \mathbb{F}[X]_{<N}$, $z \in \mathbb{F}^n$, $v \in \mathbb{F}$, and*

$$\Phi(X) = XU(X)K_z(X) - A(X) - vX^N - X^{N+1}H(X).$$

Then $\deg \Phi \leq 2N$, and if $\Phi = 0$ then $v = [X^{N-1}]UK_z$. In particular, if $U = U_f$ and $\Phi = 0$, then $v = f(z)$.

Proof. The four terms have degrees at most $2N - 1$, $N - 1$, N and $2N$. If $\Phi = 0$, compare the coefficients of X^N : on the left it is $[X^N]XUK_z = [X^{N-1}]UK_z$; on the right it is v , since $\deg A < N$ and $X^{N+1}H$ has no monomial of degree N . The last statement is Theorem 4.5. $\square$

Remark 4.13 (Why the factor X). Without the factor X , the identity $UK_z = A_0 + vX^{N-1} + X^N H_0$ requires $\deg A_0 < N - 1$: a polynomial A_0 of degree $N - 1$ can absorb any error in v . The degree bound $N - 1$ is not a power of two, so a folding proximity test for the bound N does not enforce it, and a separate degree-correction argument would be needed. Multiplying by X moves the checked coefficient to X^N , so that the natural bound $\deg A < N$ suffices (Lemma 4.12).

4.6 Related forms of the identity

Lemma 4.14 (Reversal). *For $z \in \mathbb{F}^n$, let $K_z^*(X) = X^{N-1}K_z(1/X)$ be the reversal of K_z . Then*

$$K_z^*(X) = \prod_{k=1}^n (1 + z_k X^{2^{k-1}}) = \sum_{a=0}^{N-1} z^a X^a.$$

Proof. $X^{N-1} = \prod_k X^{2^{k-1}}$, so $X^{N-1}K_z(1/X) = \prod_k X^{2^{k-1}}(X^{-2^{k-1}} + z_k) = \prod_k (1 + z_k X^{2^{k-1}})$. Expanding the product as in Lemma 4.4(3), the coefficient of X^a is $\prod_{k:a_k=1} z_k = z^a$. $\square$

For vectors $u = (u_a)_{a < N}$ and $u' = (u'_a)_{a < N}$ over $\mathbb{F}$, the classical identity $\sum_a u_a u'_a = [X^{N-1}](\sum_a u_a X^a)(\sum_a u'_a X^{N-1-a})$ writes an inner product as one coefficient of a product. By Lemma 4.14, Theorem 4.5 is this identity for the coefficient vector $(\alpha_a)_a$ of f and the vector $(z^a)_a$ of monomial values, whose generating polynomial is K_z^* and whose reversed generating polynomial is K_z . The same tensor structure appears in the Lagrange basis in pairing-based schemes: Mercury [EG25] writes $f(u) = \langle \hat{P}_u, \hat{g} \rangle$, with $\hat{P}_u(X) = \prod_{k=0}^{n-1} ((1 - u_k) + u_k X^{2^k})$ and $\hat{g}$ the table form of f , and proves the inner product with KZG commitments. Kronecker-FRI turns this identity into a transparent, hash-based scheme over Reed–Solomon codes, through (4) and one folding proximity test (Section 6), with the analysis of Sections 7 and 8.

5 The Folding Proximity Test

The scheme of Section 6 reduces an evaluation claim to a single proximity claim: a word w_0 on L , available as an oracle, must be close to a Reed–Solomon codeword of degree bound N . This section gives the test used for that claim, a FRI-style folding test [BSBHR18] with the kernel fold of [Mkh26], and proves its soundness for an *arbitrary* word w_0 . It is self-contained: the only external result it uses is the correlated-agreement theorem for lines (Theorem 2.16 with $t = 1$).

The results of Sections 5.1, 5.3 and 5.4 are those of [Mkh26, §5.2 and §7.1–7.3], reproduced with their proofs, except Lemma 5.5, which is new; Sections 5.2 and 5.5 adapt [Mkh26, §7.4 and §7.6] to a test without sumcheck and with a final polynomial instead of a final table.

Setting. Throughout this section, $\mathbb{F} \supseteq \mathbb{F}_q$ is the challenge field, $n \geq 1$ and $N = 2^n$, and $L \subseteq \mathbb{F}_q^\times$ is a smooth domain of order $M = 2^\mu = 2^R N$ with $R \geq 1$, so that the rate $\rho = N/M = 2^{-R}$ satisfies $\rho \leq 1/2$. We fix a number of folding rounds $\ell \in \llbracket 1, n \rrbracket$ and put, for $j \in \llbracket 0, \ell \rrbracket$,

$$L_j = L^{2^j}, \quad M_j = |L_j| = M/2^j, \quad N_j = N/2^j, \quad \mathcal{C}_j = \text{RS}_{\mathbb{F}}[L_j, N_j].$$

By Lemma 2.8, L_j is the smooth domain of order M_j and $L_{j+1} = (L_j)^2$; every $\mathcal{C}_j$ has rate $N_j/M_j = \rho$, and $M_j \geq M_\ell = 2^R N_\ell \geq 2$, so fibres (Definition 2.9) exist at every level $j \leq \ell - 1$. We fix a proximity parameter $0 < \delta \leq (1 - \rho)/2$.

5.1 The kernel fold

Let L' be any smooth domain of order at least 2; the definitions below apply to every L_j with $j \leq \ell - 1$. For a word $w \in \mathbb{F}^{L'}$, let $w_e, w_o \in \mathbb{F}^{L'^2}$ be its even and odd parts (1).

Definition 5.1 (Kernel fold). For $r \in \mathbb{F}$, the *kernel fold* of a word $w \in \mathbb{F}^{L'}$ is the word $\text{fold}_r(w) \in \mathbb{F}^{L'^2}$,

$$\text{fold}_r(w) = (2r - 1)w_e + (1 - r)w_o,$$

and the kernel fold of a polynomial $P \in \mathbb{F}[X]$, written $P(X) = P_e(X^2) + XP_o(X^2)$, is $\text{pfold}_r(P) = (2r - 1)P_e + (1 - r)P_o$.

For $P \in \mathbb{F}[X]_{<2d}$ we have $\text{pfold}_r(P) \in \mathbb{F}[X]_{<d}$ (Lemma 2.4). The name comes from [Mkh26], where pfold_r acts on the coordinates of P in a basis of kernel polynomials as a partial evaluation at r .

Lemma 5.2 (Properties of the word fold). *Let $w \in \mathbb{F}^{L'}$ and $r \in \mathbb{F}$.*

1. (Explicit formula.) *For every $\xi \in L'$,*

$$\text{fold}_r(w)(\xi^2) = \frac{((2r - 1)\xi + (1 - r))w(\xi) + ((2r - 1)\xi - (1 - r))w(-\xi)}{2\xi}. \quad (5)$$

2. (Linearity.) *$w \mapsto \text{fold}_r(w)$ is $\mathbb{F}$ -linear.*

3. (Locality.) *For $\eta \in L'^2$, $\text{fold}_r(w)(\eta)$ depends only on the values of w on the fibre over η ; it is computed from these two values with $O(1)$ operations in $\mathbb{F}$, given $(2\xi)^{-1}$.*

Proof. (1) By (1), $(2r - 1)w_e(\xi^2) = \frac{(2r-1)\xi(w(\xi)+w(-\xi))}{2\xi}$ and $(1 - r)w_o(\xi^2) = \frac{(1-r)(w(\xi)-w(-\xi))}{2\xi}$; add and collect the coefficients of $w(\xi)$ and $w(-\xi)$. (2) and (3) follow from Lemma 2.15(1,2) and from (1). $\square$

Lemma 5.3 (Line form). *For $w \in \mathbb{F}^{L'}$, put $u^0 = w_o - w_e$ and $u^1 = 2w_e - w_o$, both in $\mathbb{F}^{L'^2}$. Then $\text{fold}_r(w) = u^0 + r u^1$ for every $r \in \mathbb{F}$. Conversely, $w_e = u^0 + u^1$ and $w_o = 2u^0 + u^1$.*

Proof. $(2r - 1)w_e + (1 - r)w_o = (w_o - w_e) + r(2w_e - w_o)$; and $u^0 + u^1 = w_e$, $2u^0 + u^1 = w_o$. $\square$

Proposition 5.4 (Consistency with the polynomial fold). *Let $|L'| = M'$, $1 \leq d \leq M'/2$, $r \in \mathbb{F}$ and $P \in \mathbb{F}[X]_{<2d}$. Then*

$$\text{fold}_r(\text{ev}_{L'}(P)) = \text{ev}_{L'^2}(\text{pfold}_r(P)).$$

In particular, fold_r maps $\text{RS}_{\mathbb{F}}[L', 2d]$ into $\text{RS}_{\mathbb{F}}[L'^2, d]$, and $P_{\text{fold}_r(c)} = \text{pfold}_r(P_c)$ for every $c \in \text{RS}_{\mathbb{F}}[L', 2d]$.

Proof. By Lemma 2.15(4), $\text{ev}_{L'}(P)_e = \text{ev}_{L'^2}(P_e)$ and $\text{ev}_{L'}(P)_o = \text{ev}_{L'^2}(P_o)$; since $\text{ev}_{L'^2}$ is linear, $\text{fold}_r(\text{ev}_{L'}(P)) = \text{ev}_{L'^2}((2r-1)P_e + (1-r)P_o)$. Since $\text{pfold}_r(P) \in \mathbb{F}[X]_{<d}$ and $d \leq M'/2 = |L'^2|$, the last statement follows from Lemma 2.12(1). $\square$

For $r \in \mathbb{F}^\ell$ and $j \in \llbracket 0, \ell \rrbracket$, write $\text{pfold}_r^{(j)} = \text{pfold}_{r_j} \circ \dots \circ \text{pfold}_{r_1}$ (the identity for $j = 0$). For $t \in \llbracket 0, 2^j - 1 \rrbracket$ with bits $t_1, \dots, t_j$ (Section 2.1), put $\chi_t(r) = \prod_{k=1}^j \chi_{t_k}(r_k)$, where $\chi_0(x) = 2x - 1$ and $\chi_1(x) = 1 - x$.

Lemma 5.5 (Iterated fold). *Let $r \in \mathbb{F}^\ell$, $j \in \llbracket 0, \ell \rrbracket$ and $P \in \mathbb{F}[X]_{<N}$. Then $\text{pfold}_r^{(j)}(P) \in \mathbb{F}[X]_{<N_j}$, and for $b \in \llbracket 0, N_j - 1 \rrbracket$,*

$$[X^b] \text{pfold}_r^{(j)}(P) = \sum_{t=0}^{2^j-1} \chi_t(r) [X^{2^j b+t}] P.$$

If $w_0 = \text{ev}_L(P)$ and $w_j = \text{fold}_{r_j}(w_{j-1})$ for $j \in \llbracket 1, \ell \rrbracket$, then $w_j = \text{ev}_{L_j}(\text{pfold}_r^{(j)}(P))$ for every $j \in \llbracket 0, \ell \rrbracket$.

Proof. Induction on j ; the case $j = 0$ is trivial. Let $Q = \text{pfold}_r^{(j-1)}(P) \in \mathbb{F}[X]_{<N_{j-1}}$. By Lemma 2.4, $[X^b] \text{pfold}_{r_j}(Q) = \chi_0(r_j)[X^{2b}]Q + \chi_1(r_j)[X^{2b+1}]Q$, and $\text{pfold}_{r_j}(Q) \in \mathbb{F}[X]_{<N_j}$. By the induction hypothesis, for $s \in \{0, 1\}$, $[X^{2b+s}]Q = \sum_{t' < 2^{j-1}} \chi_{t'}(r)[X^{2^{j-1}(2b+s)+t'}]P$. Put $t = t' + 2^{j-1}s$: then t ranges over $\llbracket 0, 2^j - 1 \rrbracket$, its bits are those of t' followed by $t_j = s$, so $\chi_s(r_j)\chi_{t'}(r) = \chi_t(r)$, and $2^{j-1}(2b+s) + t' = 2^j b + t$. This gives the formula. For the words, apply Proposition 5.4 on L_{j-1} with $2d = N_{j-1} \leq M_{j-1}$, so $d = N_j \leq M_{j-1}/2$, and use induction. $\square$

Remark 5.6 (Relation to the classical FRI fold). The classical fold of FRI [BSBHR18] is $\text{cfold}_\theta(w) = w_e + \theta w_o$. If $r \neq 1/2$, then $\text{fold}_r = (2r-1)\text{cfold}_\theta$ with $\theta = (1-r)/(2r-1)$, and conversely $\text{cfold}_\theta = (1+2\theta)\text{fold}_r$ with $r = (1+\theta)/(1+2\theta)$ when $\theta \neq -1/2$ [Mkh26, Prop. 5.11]. The proofs below use only three properties of the fold: linearity and locality (Lemma 5.2), consistency (Proposition 5.4), and the line form with an invertible change of coordinates (Lemma 5.3). The classical fold has them too (with $u^0 = w_e$, $u^1 = w_o$), and all results of this section hold for it with the same error terms, except that the coefficients $\chi_t(r)$ of Lemma 5.5 change. We use the kernel fold so that the implementation (Section 9) shares its folding code with the one of [Mkh26]; the only requirement is that prover and verifier use the same fold [Mkh26, Rem. 5.12].

5.2 The test

Definition 5.7 (Folding proximity test). The test $\Pi_{\text{FT}}[\ell, \kappa]$, with $\kappa \geq 1$ queries, has as instance a word $w_0 \in \mathbb{F}^L$, given as an oracle. It has $\ell + 1$ rounds, numbered as in Definition 3.1:

1. Round 1: the prover sends nothing; the verifier sends $r_1 \in \mathbb{F}$.
2. Round $j \in \llbracket 2, \ell \rrbracket$: the prover sends an oracle $w_{j-1} \in \mathbb{F}^{L_{j-1}}$; the verifier sends $r_j \in \mathbb{F}$.
3. Round $\ell + 1$: the prover sends a polynomial $P \in \mathbb{F}[X]_{<N_\ell}$ in the clear, as its N_ℓ coefficients; the verifier sends κ points $\xi^{(1)}, \dots, \xi^{(\kappa)} \in L$, independent and uniform.

Put $w_\ell = \text{ev}_{L_\ell}(P) \in \mathcal{C}_\ell$ and $\hat{w}_j = \text{fold}_{r_j}(w_{j-1}) \in \mathbb{F}^{L_j}$ for $j \in \llbracket 1, \ell \rrbracket$. The verifier accepts if and only if, for every $i \in \llbracket 1, \kappa \rrbracket$ and every $j \in \llbracket 1, \ell \rrbracket$, the *fold check*

$$\hat{w}_j(\xi^{2^j}) = w_j(\xi^{2^j}), \quad \xi = \xi^{(i)}, \quad (6)$$

holds. By locality, it computes the left-hand side from $w_{j-1}(\pm \xi^{2^{j-1}})$ with (5); for $j < \ell$ the right-hand side is one of the two values of w_j read for the check of level $j+1$, and for $j = \ell$ it is $P(\xi^{2^\ell})$.

Remark 5.8 (Challenge space). The challenge space is $\mathbb{F}^\ell \times L^{\times \kappa}$. All messages are determined by the challenges of the earlier rounds: w_j ($j \in \llbracket 1, \ell-1 \rrbracket$) by $r_{\leq j}$, and P by $r_{\leq \ell}$.

Lemma 5.9 (Completeness and costs). 1. If $w_0 = \text{ev}_L(U)$ with $U \in \mathbb{F}[X]_{<N}$, the honest prover, which sends $w_j = \text{ev}_{L_j}(\text{pfold}_r^{(j)}(U))$ for $j \in \llbracket 1, \ell-1 \rrbracket$ and $P = \text{pfold}_r^{(\ell)}(U)$, makes the verifier accept with probability 1.

2. The honest prover computes all its messages with $O(M)$ operations in $\mathbb{F}$ from w_0 and the coefficients of U : each w_j with at most $7M_j$ operations from w_{j-1} given the inverses $(2\xi)^{-1}$, $\xi \in L_{j-1}$, and P with $3N$ operations by Lemma 5.5.

3. The verifier makes $2\kappa\ell$ queries (two per fibre and level) and $O(\kappa(\ell + N_\ell))$ operations in $\mathbb{F}$, besides computing the points ξ^{2^j} and the inverses $(2\xi^{2^{j-1}})^{-1}$, which lie in $\mathbb{F}_q$. The non-oracle proof consists of the N_ℓ coefficients of P .

Proof. (1) By Lemma 5.5, $\text{fold}_{r_j}(w_{j-1}) = w_j$ for $j \in \llbracket 1, \ell \rrbracket$, including $j = \ell$ where $w_\ell = \text{ev}_{L_\ell}(P)$. So every fold check holds at every point. (2) By (5), each value of w_j costs a constant number of operations once $(2r_j - 1)$, $(1 - r_j)$ and $(2\xi)^{-1}$ are known; there are M_j values, and $\sum_{j \geq 1} M_j < M$. Each fold of coefficients costs 3 operations per output coefficient, and $\sum_j N_j < N$. (3) For each point and level $j \in \llbracket 1, \ell \rrbracket$ the verifier reads w_{j-1} at $\pm \xi^{2^{j-1}}$ and evaluates (5); at level ℓ it evaluates P by Horner's rule with $2N_\ell$ operations. $\square$

5.3 Fibre distance and decoding

Definition 5.10 (Fibre distance). Let $j \in \llbracket 0, \ell-1 \rrbracket$. For $w, c \in \mathbb{F}^{L_j}$,

$$\Delta_j^{\text{fib}}(w, c) = \frac{|\{\eta \in L_{j+1} : w \text{ and } c \text{ differ somewhere on the fibre over } \eta\}|}{M_{j+1}},$$

and $\Delta_j^{\text{fib}}(w, \mathcal{C}_j) = \min_{c \in \mathcal{C}_j} \Delta_j^{\text{fib}}(w, c)$.

Lemma 5.11 (Fibre distance). Let $j \in \llbracket 0, \ell-1 \rrbracket$ and $w, c \in \mathbb{F}^{L_j}$.

1. $\Delta(w, c) \leq \Delta_j^{\text{fib}}(w, c)$; in particular $\Delta(w, \mathcal{C}_j) \leq \Delta_j^{\text{fib}}(w, \mathcal{C}_j)$.
2. $\Delta_j^{\text{fib}}(w, c) \leq \delta$ if and only if w and c agree on every fibre over some set of at least $(1 - \delta)M_{j+1}$ points of L_{j+1} .

Proof. (1) Each point of L_j at which w and c differ lies in a fibre over which they differ; there are $\Delta_j^{\text{fib}}(w, c)M_{j+1}$ such fibres, each with two points, so $\Delta(w, c)M_j \leq 2\Delta_j^{\text{fib}}(w, c)M_{j+1} = \Delta_j^{\text{fib}}(w, c)M_j$. Minimise over c . (2) The set of points of L_{j+1} over whose fibre w and c agree has $(1 - \Delta_j^{\text{fib}}(w, c))M_{j+1}$ elements. $\square$

Lemma 5.12 (Unique fibre decoding). *Let $j \in \llbracket 0, \ell - 1 \rrbracket$. For every $w \in \mathbb{F}^{L_j}$ there is at most one $c \in \mathcal{C}_j$ with $\Delta_j^{\text{fib}}(w, c) \leq \delta$.*

Proof. Such a c satisfies $\Delta(w, c) \leq \delta \leq (1 - \rho)/2$ by Lemma 5.11(1); apply Lemma 2.12(3). $\square$

Definition 5.13 (Decoded codeword). Let $j \in \llbracket 0, \ell - 1 \rrbracket$ and $w \in \mathbb{F}^{L_j}$ with $\Delta_j^{\text{fib}}(w, \mathcal{C}_j) \leq \delta$. The *decoded codeword* $\text{dec}_j(w)$ is the unique $c \in \mathcal{C}_j$ of Lemma 5.12.

Remark 5.14 (Computing the decoded codeword). Since $\Delta(w, \text{dec}_j(w)) \leq \delta < \frac{1}{2}(1 - (N_j - 1)/M_j)$, the algorithm of Fact 2.13 outputs $\text{dec}_j(w)$ in time polynomial in M_j , and checking $\Delta_j^{\text{fib}}(w, \text{dec}_j(w)) \leq \delta$ costs $O(M_j)$ comparisons. Conversely, if the decoder fails or its output fails this check, no codeword is within fibre distance δ of w . So decoding followed by the check decides whether $\Delta_j^{\text{fib}}(w, \mathcal{C}_j) \leq \delta$, and computes $\text{dec}_j(w)$ and $P_{\text{dec}_j(w)}$ (by an inverse NTT) when it is. If $w \in \mathbb{F}_q^{L_j}$, then $P_{\text{dec}_j(w)} \in \mathbb{F}_q[X]$ by Lemma 2.14.

5.4 Folding far words and good challenges

Lemma 5.15 (Folding far words). *Let $j \in \llbracket 0, \ell - 1 \rrbracket$ and $w \in \mathbb{F}^{L_j}$ with $\Delta_j^{\text{fib}}(w, \mathcal{C}_j) > \delta$. Then*

$$|\{r \in \mathbb{F} : \Delta(\text{fold}_r(w), \mathcal{C}_{j+1}) \leq \delta\}| \leq M_{j+1}.$$

Proof. Suppose the set has more than M_{j+1} elements. By Lemma 5.3, $\text{fold}_r(w) = u^0 + ru^1$ with $u^0 = w_o - w_e$ and $u^1 = 2w_e - w_o$. Apply Theorem 2.16 with $t = 1$ to the code $\mathcal{C}_{j+1}$, of length M_{j+1} and rate ρ , and to $\delta \leq (1 - \rho)/2$: there are $L' \subseteq L_{j+1}$ with $|L'| \geq (1 - \delta)M_{j+1}$ and codewords $\hat{c}^0, \hat{c}^1 \in \mathcal{C}_{j+1}$ with $u^0 = \hat{c}^0$ and $u^1 = \hat{c}^1$ on L' . Let $T_e = P_{\hat{c}^0} + P_{\hat{c}^1}$ and $T_o = 2P_{\hat{c}^0} + P_{\hat{c}^1}$, both in $\mathbb{F}[X]_{<N_{j+1}}$, and $T(X) = T_e(X^2) + XT_o(X^2) \in \mathbb{F}[X]_{<N_j}$. By Lemma 5.3, $w_e = T_e$ and $w_o = T_o$ on L' . So for every $\eta \in L'$ and $\xi \in L_j$ with $\xi^2 = \eta$, (2) gives $w(\pm\xi) = T_e(\eta) \pm \xi T_o(\eta) = T(\pm\xi)$. Thus w agrees with the codeword $\text{ev}_{L_j}(T) \in \mathcal{C}_j$ on every fibre over L' , and $\Delta_j^{\text{fib}}(w, \mathcal{C}_j) \leq \delta$ by Lemma 5.11(2), a contradiction. $\square$

Lemma 5.16 (Fibre errors survive folding). *Let $j \in \llbracket 0, \ell - 1 \rrbracket$, $w, c \in \mathbb{F}^{L_j}$, and $\eta \in L_{j+1}$ such that w and c differ somewhere on the fibre over η . Then there is at most one $r \in \mathbb{F}$ with $\text{fold}_r(w)(\eta) = \text{fold}_r(c)(\eta)$.*

Proof. Let $y = w - c$. By linearity and Lemma 5.3, $\text{fold}_r(w)(\eta) - \text{fold}_r(c)(\eta) = \text{fold}_r(y)(\eta) = (y_o(\eta) - y_e(\eta)) + r(2y_e(\eta) - y_o(\eta))$, a polynomial of degree at most 1 in r . If it were identically zero, the inverse formulas of Lemma 5.3 would give $y_e(\eta) = y_o(\eta) = 0$, so y would vanish on the fibre over η (Lemma 2.15(3)), a contradiction. Hence it has at most one root. $\square$

Definition 5.17 (Good challenge). Let $j \in \llbracket 1, \ell \rrbracket$, $w \in \mathbb{F}^{L_{j-1}}$ and $r \in \mathbb{F}$. We say that r is *good* for w at level j if:

1. in case $\Delta_{j-1}^{\text{fib}}(w, \mathcal{C}_{j-1}) > \delta$: $\Delta(\text{fold}_r(w), \mathcal{C}_j) > \delta$;
2. in case $\Delta_{j-1}^{\text{fib}}(w, \mathcal{C}_{j-1}) \leq \delta$, with $c = \text{dec}_{j-1}(w)$: $\text{fold}_r(w)(\eta) \neq \text{fold}_r(c)(\eta)$ for every $\eta \in L_j$ such that w and c differ somewhere on the fibre over η .

Lemma 5.18 (Few bad challenges). *For every $j \in \llbracket 1, \ell \rrbracket$ and $w \in \mathbb{F}^{L_{j-1}}$, at most M_j values $r \in \mathbb{F}$ are not good for w at level j .*

Proof. In case 1 of Definition 5.17, this is Lemma 5.15 (at level $j - 1$). In case 2, by Lemma 5.16, each of the at most M_j points η over whose fibre w and c differ excludes at most one r . $\square$

Lemma 5.19 (One folding step). *Let $j \in \llbracket 1, \ell \rrbracket$, $w \in \mathbb{F}^{L_{j-1}}$, and $r \in \mathbb{F}$ good for w at level j . Let $c' \in \mathcal{C}_j$ and $Y = \{\eta \in L_j : \text{fold}_r(w)(\eta) = c'(\eta)\}$, and suppose $|Y| \geq (1 - \delta)M_j$. Then $\Delta_{j-1}^{\text{fib}}(w, \mathcal{C}_{j-1}) \leq \delta$, the codeword $c = \text{dec}_{j-1}(w)$ satisfies $\text{fold}_r(c) = c'$, and $w = c$ on the fibre over every $\eta \in Y$.*

Proof. We have $\Delta(\text{fold}_r(w), c') \leq \delta$, so case 1 of Definition 5.17 is excluded, and $\Delta_{j-1}^{\text{fib}}(w, \mathcal{C}_{j-1}) \leq \delta$. By Proposition 5.4 (on L_{j-1} , with $2d = N_{j-1}$ and $d = N_j \leq M_{j-1}/2$), $\text{fold}_r(c) \in \mathcal{C}_j$. By locality (Lemma 5.2(3)), $\text{fold}_r(c)$ agrees with $\text{fold}_r(w)$ at every η over whose fibre $w = c$, hence on at least $(1 - \delta)M_j$ points (Lemma 5.11(2)); and $\text{fold}_r(w)$ agrees with c' on Y . By Lemma 2.11, $\text{fold}_r(c)$ and c' agree on at least $(1 - 2\delta)M_j \geq \rho M_j = N_j$ points, so they are equal (Lemma 2.12(2)). Finally, for $\eta \in Y$, $\text{fold}_r(w)(\eta) = c'(\eta) = \text{fold}_r(c)(\eta)$, so by case 2 of Definition 5.17, $w = c$ on the fibre over η . $\square$

5.5 Witness sets and the codeword chain

Fix a deterministic prover and an instance w_0 , and consider the words w_j , $\hat{w}_j$ and the polynomial P of Definition 5.7 as functions of the challenges $r \in \mathbb{F}^\ell$. For $j \in \llbracket 1, \ell \rrbracket$, let

$$\mathcal{B}_j = \{ \eta \in L_j : \hat{w}_j(\eta) \neq w_j(\eta) \},$$

the set of points where the fold check of level j fails. It is determined by w_{j-1} , r_j and w_j , that is, by $r_{\leq j}$ for $j < \ell$ and by $r_{\leq \ell}$ for $j = \ell$.

Definition 5.20 (Witness sets). For $c \in \mathcal{C}_0$, let $\mathcal{W}_0(c) = \{\xi \in L : w_0(\xi) = c(\xi) \text{ and } w_0(-\xi) = c(-\xi)\}$. For $j \in \llbracket 1, \ell \rrbracket$ and $c \in \mathcal{C}_j$, let

$$\mathcal{W}_j(c) = \{ \xi \in L : \xi^{2^i} \notin \mathcal{B}_i \text{ for all } i \in \llbracket 1, j-1 \rrbracket, \text{ and } \hat{w}_j(\xi^{2^j}) = c(\xi^{2^j}) \}.$$

Let $\text{dens}_j(c) = |\mathcal{W}_j(c)|/M$.

$\mathcal{W}_j(c)$ depends on $w_0, \dots, w_{j-1}$ and $r_1, \dots, r_j$ only; in particular, it is determined before the prover sends w_j (or P , for $j = \ell$). It is the set of points ξ for which the fold checks of levels $1, \dots, j-1$ pass, and the level- j fold of the prover's words at ξ^{2^j} is consistent with c .

Lemma 5.21 (Fold checks and witness sets). *Let $r \in \mathbb{F}^\ell$.*

1. *A point $\xi \in L$ passes the fold checks (6) of all levels $j \in \llbracket 1, \ell \rrbracket$ if and only if $\xi \in \mathcal{W}_\ell(w_\ell)$.*
2. *The verifier accepts if and only if $\xi^{(1)}, \dots, \xi^{(\kappa)} \in \mathcal{W}_\ell(w_\ell)$; for fixed r , this has probability $\text{dens}_\ell(w_\ell)^\kappa$.*

Proof. (1) By definition, $\xi \in \mathcal{W}_\ell(w_\ell)$ if and only if $\xi^{2^i} \notin \mathcal{B}_i$ for $i \in \llbracket 1, \ell-1 \rrbracket$ and $\hat{w}_\ell(\xi^{2^\ell}) = w_\ell(\xi^{2^\ell})$, that is, $\xi^{2^\ell} \notin \mathcal{B}_\ell$; and $\xi^{2^j} \notin \mathcal{B}_j$ is exactly the fold check of level j at ξ . (2) follows from (1), since the points are independent and uniform in L and $\mathcal{W}_\ell(w_\ell)$ is determined by r . $\square$

Lemma 5.22 (One round of the chain). *Let $j \in \llbracket 1, \ell \rrbracket$, fix $r_{\leq j}$ (hence $w_0, \dots, w_{j-1}$), and suppose r_j is good for w_{j-1} at level j . If $c' \in \mathcal{C}_j$ satisfies $\text{dens}_j(c') \geq 1 - \delta$, then $\Delta_{j-1}^{\text{fib}}(w_{j-1}, \mathcal{C}_{j-1}) \leq \delta$, the codeword $c = \text{dec}_{j-1}(w_{j-1})$ satisfies $\text{fold}_{r_j}(c) = c'$, and $\mathcal{W}_j(c') \subseteq \mathcal{W}_{j-1}(c)$; in particular $\text{dens}_{j-1}(c) \geq 1 - \delta$.*

Proof. Let $Y = \{\eta \in L_j : \hat{w}_j(\eta) = c'(\eta)\}$. Every $\xi \in \mathcal{W}_j(c')$ has $\xi^{2^j} \in Y$, and $\xi \mapsto \xi^{2^j}$ is 2^j -to-1 from L onto L_j (Lemma 2.8(2)); so $|Y| \geq |\mathcal{W}_j(c')|/2^j \geq (1 - \delta)M_j$. Lemma 5.19 gives the first two claims, and $w_{j-1} = c$ on the fibre over every $\eta \in Y$. Let $\xi \in \mathcal{W}_j(c')$ and $\eta = \xi^{2^j} \in Y$. The fibre over η is $\{\pm \xi^{2^{j-1}}\}$ (Lemma 2.8(4)), and $w_{j-1} = c$ on it. If $j = 1$, this says $\xi \in \mathcal{W}_0(c)$. If $j \geq 2$, the condition $\xi^{2^{j-1}} \notin \mathcal{B}_{j-1}$, part of the definition of $\mathcal{W}_j(c')$, gives $\hat{w}_{j-1}(\xi^{2^{j-1}}) = w_{j-1}(\xi^{2^{j-1}}) = c(\xi^{2^{j-1}})$; the conditions $\xi^{2^i} \notin \mathcal{B}_i$ for $i \leq j-2$ are common to both sets. Hence $\xi \in \mathcal{W}_{j-1}(c)$. $\square$

Proposition 5.23 (The codeword chain). *Let $r \in \mathbb{F}^\ell$ be such that r_j is good for w_{j-1} at level j for every $j \in \llbracket 1, \ell \rrbracket$, and suppose $\text{dens}_\ell(w_\ell) \geq 1 - \delta$. Put $c_\ell = w_\ell$. Then, for $j = \ell - 1, \dots, 0$, $\Delta_j^{\text{fib}}(w_j, \mathcal{C}_j) \leq \delta$, and the codewords $c_j = \text{dec}_j(w_j)$ satisfy*

$$\text{fold}_{r_{j+1}}(c_j) = c_{j+1} \quad \text{and} \quad \mathcal{W}_\ell(w_\ell) \subseteq \mathcal{W}_j(c_j).$$

In particular:

1. w_0 agrees with c_0 at every point of $\mathcal{W}_\ell(w_\ell)$, and $\Delta_0^{\text{fib}}(w_0, \mathcal{C}_0) \leq \delta$;
2. $P = \text{pfold}_r^{(\ell)}(P_{c_0})$.

Proof. Downward induction on j , applying Lemma 5.22 at level $j+1$ with $c' = c_{j+1}$: its hypothesis $\text{dens}_{j+1}(c_{j+1}) \geq 1 - \delta$ holds for $j+1 = \ell$ by assumption, and for $j+1 < \ell$ by the previous step, since $\mathcal{W}_\ell(w_\ell) \subseteq \mathcal{W}_{j+1}(c_{j+1})$. (1) $\xi \in \mathcal{W}_0(c_0)$ means $w_0 = c_0$ at $\pm \xi$. (2) By Lemma 5.5 applied to $P_{c_0} \in \mathbb{F}[X]_{<N}$, the words c_j are $\text{ev}_{L_j}(\text{pfold}_r^{(j)}(P_{c_0}))$; so $\text{ev}_{L_\ell}(P) = w_\ell = c_\ell = \text{ev}_{L_\ell}(\text{pfold}_r^{(\ell)}(P_{c_0}))$, and both polynomials lie in $\mathbb{F}[X]_{<N_\ell}$ with $N_\ell \leq M_\ell$, so they are equal (Lemma 2.12(1)). $\square$

5.6 Soundness of the test

Put

$$\varepsilon_{\text{fold}} = \sum_{j=1}^{\ell} \frac{M_j}{|\mathbb{F}|} = \frac{M(1 - 2^{-\ell})}{|\mathbb{F}|} < \frac{M}{|\mathbb{F}|}.$$

Theorem 5.24 (Soundness of the folding test). *Fix a deterministic prover for $\Pi_{\text{FT}}[\ell, \kappa]$ and an instance $w_0 \in \mathbb{F}^L$, and let **Good** be the event, over $r \in \mathbb{F}^\ell$, that r_j is good for w_{j-1} at level j for every $j \in \llbracket 1, \ell \rrbracket$. Then:*

1. $\Pr[\neg \text{Good}] \leq \varepsilon_{\text{fold}}$;
2. for every $r \in \text{Good}$, either $\text{dens}_\ell(w_\ell) < 1 - \delta$, or the conclusions of Proposition 5.23 hold;
3. if $\Delta_0^{\text{fib}}(w_0, \mathcal{C}_0) > \delta$, the verifier accepts with probability at most $\varepsilon_{\text{fold}} + (1 - \delta)^\kappa$.

By Lemma 3.4, item 3 also holds for randomised provers.

Proof. (1) The word w_{j-1} is a function of $r_{\leq j-1}$, and for each value of $r_{\leq j-1}$ at most M_j values of r_j are not good for it (Lemma 5.18). Apply Lemma 3.2. (2) is Proposition 5.23. (3) If $r \in \text{Good}$, then $\text{dens}_\ell(w_\ell) < 1 - \delta$, for otherwise Proposition 5.23(1) would give $\Delta_0^{\text{fib}}(w_0, \mathcal{C}_0) \leq \delta$. By Lemma 5.21(2) and Lemma 3.3 (with $\Omega' = \mathbb{F}^\ell$, $\mathcal{A} = \text{Good}$ and $\mathcal{S}(r) = \mathcal{W}_\ell(w_\ell)$), the verifier accepts and $r \in \text{Good}$ with probability at most $(1 - \delta)^\kappa$. Add item 1. $\square$

Remark 5.25 (Fibre distance versus distance). Since $\Delta \leq \Delta_0^{\text{fib}}$ (Lemma 5.11(1)), item 3 applies in particular to every word with $\Delta(w_0, \mathcal{C}_0) > \delta$. The fibre distance is the natural quantity for the analysis because a single fold check reads a whole fibre, and it is at most twice the Hamming distance; Sections 7 and 8 measure the distance of the commitment in it.

Remark 5.26 (Use in the scheme). The scheme of Section 6 runs $\Pi_{\text{FT}}[\ell, \kappa]$ on the batched word $w_0 = w + \beta w_A + \beta^2 h$, whose values the verifier computes from two oracles and the virtual word h (Definition 6.5), and adds no other check. Its analysis (Section 7) needs more than item 3 of Theorem 5.24: it uses item 2, the witness sets of Definition 5.20, and Lemma 5.22 to define the round-by-round state. This is why these statements are given for an arbitrary prover and word, and not only for far words.

5.7 Virtual levels and higher folding arity

The prover of Π_{FT} commits to every folded word. Committing only to some of them reduces the number of Merkle trees and authentication paths, and the analysis above covers this variant without change: an uncommitted word is replaced by the exact fold of the previous one, whose set of failing points is empty.

Definition 5.27 (Folding test with committed levels C). Let $C \subseteq \llbracket 1, \ell - 1 \rrbracket$. The test $\Pi_{\text{FT}}^C[\ell, \kappa]$ is $\Pi_{\text{FT}}[\ell, \kappa]$ (Definition 5.7) in which, in round $j + 1$ for $j \in \llbracket 1, \ell - 1 \rrbracket$, the prover sends the oracle w_j only if $j \in C$; for $j \notin C$ the word w_j is *virtual*, defined as $w_j = \hat{w}_j = \text{fold}_{r_j}(w_{j-1})$. The verifier makes the fold checks (6) of the levels $j \in C \cup \{\ell\}$. For such a level j , let $j' < j$ be the previous level in $C \cup \{0\}$; the verifier computes $\hat{w}_j(\xi^{2^j})$ from the values of $w_{j'}$ on the $2^{j-j'}$ points $\zeta \in L_{j'}$ with $\zeta^{2^{j-j'}} = \xi^{2^j}$, by $j - j'$ successive folds (Lemma 5.2(3)). For $k \geq 1$, the test of *arity* 2^k is $\Pi_{\text{FT}}^{C_k}$ with $C_k = \{k, 2k, 3k, \dots\} \cap \llbracket 1, \ell - 1 \rrbracket$; arity 2 is Π_{FT} .

Proposition 5.28 (Virtual levels). *Let $C \subseteq \llbracket 1, \ell - 1 \rrbracket$.*

1. *Every deterministic prover P^C of $\Pi_{\text{FT}}^C[\ell, \kappa]$ defines a deterministic prover P of $\Pi_{\text{FT}}[\ell, \kappa]$, which sends the words of P^C at the levels of C and the virtual words at the other levels, such that for all challenges the two verifiers accept or reject together.*
2. *Consequently Theorem 5.24 holds for Π_{FT}^C with the same bounds, and Definition 5.20, Lemmas 5.21 and 5.22, and Proposition 5.23 apply to it with $\mathcal{B}_j = \emptyset$ for every virtual level j .*
3. *Per query point, the verifier reads $w_{j'}$ on $2^{j-j'}$ points for every pair of consecutive levels $j' < j$ of $C \cup \{0, \ell\}$, that is, $2^{j-j'-1}$ fibres. For arity 2^k , the oracles have total length $\sum_{j \in C_k} M_j < M/(2^k - 1)$.*

Proof. (1) The virtual word $w_j = \text{fold}_{r_j}(w_{j-1})$ is a function of w_{j-1} and r_j , hence of the earlier messages and challenges, so P is a prover of Π_{FT} . At a virtual level j , $\mathcal{B}_j = \{\eta : \hat{w}_j(\eta) \neq w_j(\eta)\} = \emptyset$, so the fold check of level j holds at every point. At a level $j \in C \cup \{\ell\}$, with j' as in Definition 5.27, the words at the levels between j' and j are the successive folds of $w_{j'}$, so by locality (Lemma 5.2(3)) the value $\hat{w}_j(\xi^{2^j})$ computed by the verifier of Π_{FT} is the one computed by the verifier of Π_{FT}^C . The two verifiers therefore check the same equations. (2) The events of Theorem 5.24 are events about the challenges and the transcript of P , and the acceptance events coincide by (1). The other statements are about arbitrary words and provers. (3) The $2^{j-j'}$ points above a point of L_j form $2^{j-j'-1}$ fibres of $L_{j'}$. The total length is $\sum_{i \geq 1} M_{ik} = M \sum_{i \geq 1} 2^{-ik} < M/(2^k - 1)$. $\square$

Remark 5.29 (Choice of arity). Larger arity removes Merkle trees and authentication paths but increases the number of values read per query, 2^k per committed word; the error terms of Theorem 5.24 do not change. In our implementation, with Merkle caps, arities 8 and 16 give the smallest proofs, within 3% of each other, and arity 8 verifies faster than arity 16 (Section 10.3). The same observation applies to any FRI-type test, and folding several levels between commitments is standard practice in FRI implementations; Proposition 5.28 records that the analysis of this section covers it exactly.

6 The Commitment Scheme

This section defines the scheme in the IOP model (Definition 3.9): the encoding, the evaluation protocol Π_{KF} , its completeness, and its costs. The protocol has three parts: the prover sends the opening polynomial A of Definition 4.11 as an oracle; the verifier defines from the commitment and this oracle a third word, the *virtual word* h , whose values it computes from (4) instead of querying them, and batches the three words into one by a random challenge β ; and the folding proximity test of Section 5 is run on the batched word. Soundness is proved in Section 7, and the compilation into a non-interactive argument is in Section 8.

6.1 Parameters

The parameters are:

- the number of variables $n \geq 1$, and $N = 2^n$;
- a base field $\mathbb{F}_q$ of odd characteristic and a challenge field $\mathbb{F} \supseteq \mathbb{F}_q$, both finite;
- a rate $\rho = 2^{-R}$ with $R \geq 2$, and a smooth domain $L \leq \mathbb{F}_q^\times$ of order $M = N/\rho = 2^{n+R}$, which exists if and only if $2^{n+R} \mid q - 1$ (Lemma 2.8(1));
- a number of folding rounds $\ell \in \llbracket 1, n \rrbracket$, a number of queries $\kappa \geq 1$, and a set $C \subseteq \llbracket 1, \ell - 1 \rrbracket$ of committed levels, by default $C = C_k$ for a folding arity 2^k (Definition 5.27).

The levels L_j , M_j , N_j and codes $\mathcal{C}_j = \text{RS}_{\mathbb{F}}[L_j, N_j]$, $j \in \llbracket 0, \ell \rrbracket$, are those of Section 5; in particular $\mathcal{C}_0 = \text{RS}_{\mathbb{F}}[L, N]$. The setting of Section 5 only requires $R \geq 1$; the stronger condition $R \geq 2$, that is $\rho \leq 1/4$, is used by the batching lemma of Section 7, and explained in Remark 6.14. The polynomials to be committed have coefficients in $\mathbb{F}_q$ or in $\mathbb{F}$, and evaluation points lie in $\mathbb{F}^n$.

Example 6.1 (Goldilocks). For the Goldilocks prime $q = 2^{64} - 2^{32} + 1$, $q - 1 = 2^{32}(2^{32} - 1)$, so smooth domains of order up to 2^{32} exist, and $n + R \leq 32$. With $R = 2$ this allows $n \leq 30$. For $\mathbb{F}$ one takes an extension of degree 2 to 4 of $\mathbb{F}_q$ (Section 8).

6.2 Encoding

Definition 6.2 (Encoding). For $f \in \mathcal{L}_n(\mathbb{F})$, $\text{Enc}(f) = \text{ev}_L(U_f) \in \mathcal{C}_0$, where U_f is the Kronecker encoding (Definition 4.1).

Lemma 6.3 (Properties of the encoding). 1. Enc is an $\mathbb{F}$ -linear bijection from $\mathcal{L}_n(\mathbb{F})$ onto $\mathcal{C}_0$, and $P_{\text{Enc}(f)} = U_f$.

2. For $0 < \delta \leq (1 - \rho)/2$, every word $w \in \mathbb{F}^L$ satisfies $\Delta_0^{\text{fib}}(w, \text{Enc}(f)) \leq \delta$ for at most one $f \in \mathcal{L}_n(\mathbb{F})$, and likewise with Δ in place of Δ_0^{fib} ; that is, (3) holds with $\text{dist} = \Delta_0^{\text{fib}}$ and with $\text{dist} = \Delta$.

3. If $f \in \mathcal{L}_n(\mathbb{F}_q)$, then $\text{Enc}(f) \in \mathbb{F}_q^L$. Conversely, if $w \in \mathbb{F}_q^L$ and $\Delta(w, \text{Enc}(f)) \leq (1 - \rho)/2$ (in particular if $\Delta_0^{\text{fib}}(w, \text{Enc}(f)) \leq (1 - \rho)/2$), then $f \in \mathcal{L}_n(\mathbb{F}_q)$.

Proof. (1) $f \mapsto U_f$ is a linear bijection onto $\mathbb{F}[X]_{<N}$ (Lemma 4.2(2)), and ev_L is a linear bijection from $\mathbb{F}[X]_{<N}$ onto $\mathcal{C}_0$ (Lemma 2.12(1)). (2) Two such polynomials f, f' would give code-words $\text{Enc}(f), \text{Enc}(f')$ within fibre distance (or distance) δ of w , hence equal by Lemma 5.12 (or Lemma 2.12(3)), and $f = f'$ by (1). (3) The coefficients of U_f are those of f , and $L \subseteq \mathbb{F}_q$. Conversely ($\Delta \leq \Delta_0^{\text{fib}}$ by Lemma 5.11(1)), $(1 - \rho)/2 < \frac{1}{2}(1 - (N - 1)/M)$, so Lemma 2.14 gives $U_f = P_{\text{Enc}(f)} \in \mathbb{F}_q[X]$. $\square$

Fact 6.4 (Fast evaluation on a smooth domain [CT65]). *Let L be a smooth domain of order $M = 2^\mu$ with a known generator. For $P \in \mathbb{F}[X]_{<M}$, the word $\text{ev}_L(P)$ is computed with $\frac{M}{2}\mu$ multiplications and $M\mu$ additions in $\mathbb{F}$ (radix-2 number-theoretic transform, NTT), and P is recovered from $\text{ev}_L(P)$ at the same cost plus M multiplications (inverse NTT). If P has coefficients in $\mathbb{F}_q$, all operations are in $\mathbb{F}_q$; if $\mathbb{F}$ has degree e over $\mathbb{F}_q$, an NTT over $\mathbb{F}$ costs e NTTs over $\mathbb{F}_q$, since the multipliers lie in $\mathbb{F}_q$.*

Committing. To commit to f , the prover computes $\text{Enc}(f)$: from the coefficient form, one NTT of size M on the coefficients of U_f padded with zeros; from the table form, first the Möbius transform (Lemma 2.6, $\frac{n}{2}N$ subtractions). In the IOP model the commitment is the oracle $w = \text{Enc}(f)$; in the compiled scheme it is a Merkle root (Section 8). Unlike [Mkh26], whose encoding $\sum_b f(b)K_b$ uses the table form directly, the Kronecker encoding needs the coefficient form; the difference is one Möbius transform, of $\frac{n}{2}N$ subtractions, against $\frac{3}{2}M\mu$ operations for the NTT that follows it, that is, at most a fraction $n/(3 \cdot 2^R\mu) < 1/(3 \cdot 2^R)$ of it.

6.3 The virtual word

The verifier needs the values of the second opening polynomial H only through (4), which determines them from those of U_f and A . We therefore do not send H , and let the verifier compute the word it would be. Words computed by the verifier from other oracles are the *virtual oracles* of [BSCR⁺19].

Definition 6.5 (Virtual word). For words $w, w_A \in \mathbb{F}^L$, a point $z \in \mathbb{F}^n$ and a value $v \in \mathbb{F}$, the *virtual word* $h = h[w, w_A; z, v] \in \mathbb{F}^L$ is

$$h(\xi) = \frac{\xi w(\xi) K_z(\xi) - w_A(\xi) - v \xi^N}{\xi^{N+1}} \quad (\xi \in L).$$

The denominator is nonzero because $0 \notin L$.

Lemma 6.6 (Properties of the virtual word). *Let $w, w_A \in \mathbb{F}^L$, $z \in \mathbb{F}^n$, $v \in \mathbb{F}$ and $h = h[w, w_A; z, v]$.*

1. (Identity.) $\xi w(\xi) K_z(\xi) = w_A(\xi) + v \xi^N + \xi^{N+1} h(\xi)$ for every $\xi \in L$.
2. (Locality.) $h(\xi)$ depends only on $w(\xi)$ and $w_A(\xi)$, and is computed from them with $O(n)$ operations, given ξ^{-1} .
3. (Honest openings.) If $w = \text{ev}_L(U)$ with $U \in \mathbb{F}[X]_{<N}$, $v = [X^{N-1}] U K_z$ and $w_A = \text{ev}_L(A_{U,z})$, then $h = \text{ev}_L(H_{U,z})$.
4. (Linearity.) $h[w, w_A; z, v]$ is linear in (w, w_A, v) : $h[w + w', w_A + w'_A; z, v + v'] = h[w, w_A; z, v] + h[w', w'_A; z, v']$.

Proof. (1) Multiply the definition by ξ^{N+1} . (2) $K_z(\xi)$ costs $2n - 1$ operations besides $n - 1$ squarings, and one more squaring gives ξ^N (Lemma 4.9(1)); $\xi^{-(N+1)} = (\xi^{-1})^{N+1}$ costs $n + 1$ further multiplications. (3) By Lemma 4.10, $XU K_z = A + v X^N + X^{N+1} H$ with $A = A_{U,z}$ and $H = H_{U,z}$. Evaluating at ξ and dividing by ξ^{N+1} gives $H(\xi) = h(\xi)$. (4) The numerator is linear in (w, w_A, v) , and the denominator does not depend on them. $\square$

Lemma 6.7 (The kernel polynomial on the domain). *Given $z \in \mathbb{F}^n$ and a generator of L , the word $\text{ev}_L(K_z)$ is computed with fewer than $2M$ multiplications and $2M$ additions in $\mathbb{F}$, besides the powers of the generator, which lie in $\mathbb{F}_q$.*

Proof. For $k \in \llbracket 0, n \rrbracket$ let $z^{(k)} = (z_{k+1}, \dots, z_n)$, so that $K_{z^{(n)}} = 1$ and $K_{z^{(k-1)}}(x) = (x + z_k) K_{z^{(k)}}(x^2)$ by Lemma 4.4(2). The domains $L_k = L^{2^k}$ satisfy $\{x^2 : x \in L_{k-1}\} = L_k$ (Lemma 2.8). So $\text{ev}_{L_{k-1}}(K_{z^{(k-1)}})$ is obtained from $\text{ev}_{L_k}(K_{z^{(k)}})$ with one addition and one multiplication per point of L_{k-1} , for $k = n, \dots, 1$; and $\sum_{k=1}^n M/2^{k-1} < 2M$. Here L_k is defined for $k \leq n < \mu$, since $M = 2^{n+R}$. $\square$

6.4 The evaluation protocol

Definition 6.8 (The evaluation protocol Π_{KF}). The instance is $(w; z, v)$, with an oracle $w \in \mathbb{F}^L$, a point $z \in \mathbb{F}^n$ and a value $v \in \mathbb{F}$; honestly, $w = \text{Enc}(f)$ and $v = f(z)$, and the prover holds f . The protocol has $\ell + 2$ rounds, numbered as in Definition 3.1.

1. *Round 1 (opening)*. The prover sends an oracle $w_A \in \mathbb{F}^L$; honestly $w_A = \text{ev}_L(A_{U_f, z})$ (Definition 4.11). The verifier sends $\beta \in \mathbb{F}$. Put $h = h[w, w_A; z, v]$ (Definition 6.5) and

$$w_0 = w + \beta w_A + \beta^2 h \in \mathbb{F}^L.$$

2. *Round 2*. The prover sends nothing; the verifier sends $r_1 \in \mathbb{F}$.
3. *Round $j + 1$, $j \in \llbracket 2, \ell \rrbracket$* . If $j - 1 \in C$, the prover sends an oracle $w_{j-1} \in \mathbb{F}^{L_{j-1}}$, honestly $w_{j-1} = \text{fold}_{r_{j-1}}(w_{j-2})$; otherwise it sends nothing, and $w_{j-1} = \text{fold}_{r_{j-1}}(w_{j-2})$ is virtual. The verifier sends $r_j \in \mathbb{F}$.
4. *Round $\ell + 2$ (final)*. The prover sends $P \in \mathbb{F}[X]_{< N_\ell}$ by its N_ℓ coefficients; honestly $\text{ev}_{L_\ell}(P) = \text{fold}_{r_\ell}(w_{\ell-1})$. The verifier sends κ independent uniform points $\xi^{(1)}, \dots, \xi^{(\kappa)} \in L$.

Put $w_\ell = \text{ev}_{L_\ell}(P)$ and $\hat{w}_j = \text{fold}_{r_j}(w_{j-1})$ for $j \in \llbracket 1, \ell \rrbracket$. The verifier accepts if and only if the *fold checks* $\hat{w}_j(\xi^{2^j}) = w_j(\xi^{2^j})$ hold for every $j \in \llbracket 1, \ell \rrbracket$ at every query point $\xi = \xi^{(i)}$, as in (6). The values $w_0(\pm\xi)$ needed by the fold check of level 1 are computed from the four queries $w(\pm\xi)$ and $w_A(\pm\xi)$, through $h(\pm\xi)$ (Lemma 6.6(2)). The challenge sets are $\mathbb{F}$ for rounds 1 to $\ell + 1$ and $L^{\times \kappa}$ for round $\ell + 2$.

Rounds 2 to $\ell + 2$ of Π_{KF} are rounds 1 to $\ell + 1$ of the folding test $\Pi_{\text{FT}}^C[\ell, \kappa]$ (Definition 5.27) on the instance w_0 , which is determined after round 1; the checks of Π_{KF} are exactly those of that test. By Proposition 5.28 we may, and do, analyse it as $\Pi_{\text{FT}}[\ell, \kappa]$ with virtual words at the levels outside C ; for $C = \llbracket 1, \ell - 1 \rrbracket$ (arity 2) the two coincide. The words $w_j, \hat{w}_j$, the sets $\mathcal{B}_j$ and $\mathcal{W}_j(c)$ of Section 5.5, and the good challenges of Definition 5.17 are therefore defined for Π_{KF} , as functions of β and r .

Remark 6.9 (Design). 1. The evaluation claim enters only through the virtual word: by Lemma 6.6(1), the identity (4) holds at every point *by construction*, and a wrong value v shows up as a word h far from $\mathcal{C}_0$. For instance, if w and w_A are honest and the claim is $v' \neq v$, then $h[w, w_A; z, v'] = \text{ev}_L(H) + (v - v') \text{ev}_L(X^{-1})$ by Lemma 6.6(3,4), where X^{-1} is evaluated pointwise; since $\xi^{-1} = \xi^{M-1}$ and $1 - \xi T(\xi)$ has at most N roots for $T \in \mathbb{F}[X]_{< N}$ (it is a nonzero polynomial of degree at most N), the word $\xi \mapsto \xi^{-1}$ agrees with every codeword of $\mathcal{C}_0$ on at most N points.

2. The commitment w is not tested on its own: it enters the proximity test through w_0 , together with w_A and h . The batching lemma (Lemma 7.4) shows that if the batched word agrees with a codeword on a large set of fibres for more than $2M$ values of β , then w, w_A and h are *simultaneously* close to codewords of $\mathcal{C}_0$ on a common set of fibres.

3. All three words are tested for the degree bound N , the natural bound of $\mathcal{C}_0$; the factor X in (4) is what makes this suffice (Remark 4.13). The properties $A(0) = 0$ and $\deg H \leq N - 2$ of honest openings are not checked and not needed.
4. The verifier never computes v from z : it only evaluates K_z at the 2κ points $\pm\xi^{(i)}$, with $O(n)$ operations each.
5. There is no sumcheck and no separate identity check: the evaluation claim is carried by the virtual word, and the folding test is a plain proximity test.

The honest prover. On input f (in coefficient form, with $w = \text{Enc}(f)$ already computed) and z :

1. compute $G = U_f K_z$ (Lemma 4.9(3)); read off $v = [X^{N-1}]G$ and the opening polynomials $A = A_{U_f, z}$ and $H = H_{U_f, z}$, whose coefficients are those of XG in degrees $\llbracket 0, N-1 \rrbracket$ and $\llbracket N+1, 2N \rrbracket$ (proof of Lemma 4.10);
2. compute $w_A = \text{ev}_L(A)$ by one NTT of size M over $\mathbb{F}$, and send it;
3. on receiving β , compute $\text{ev}_L(K_z)$ (Lemma 6.7), the virtual word h , which equals $\text{ev}_L(H)$ (Lemma 6.6(3)), the word $w_0 = w + \beta w_A + \beta^2 h$, and $U_0 = U_f + \beta A + \beta^2 H \in \mathbb{F}[X]_{<N}$;
4. on receiving r_j ($j \in \llbracket 1, \ell-1 \rrbracket$), compute $w_j = \text{fold}_{r_j}(w_{j-1})$ with (5), and send it if $j \in C$;
5. on receiving r_ℓ , send $P = \text{pfold}_r^{(\ell)}(U_0)$, computed on coefficients (Lemma 5.5).

6.5 Completeness

Theorem 6.10 (Completeness). *If $w = \text{Enc}(f)$ and $v = f(z)$, the honest prover makes the verifier of Π_{KF} accept with probability 1.*

Proof. Let $A = A_{U_f, z}$ and $H = H_{U_f, z}$. By Theorem 4.5, $v = f(z) = [X^{N-1}]U_f K_z$, so Lemma 6.6(3) applies: $h = \text{ev}_L(H)$. By linearity of ev_L , $w_0 = \text{ev}_L(U_0)$ with $U_0 = U_f + \beta A + \beta^2 H \in \mathbb{F}[X]_{<N}$. The honest messages of rounds 2 to $\ell+2$ are those of the honest prover of $\Pi_{\text{FT}}^C[\ell, \kappa]$ for U_0 , whose induced prover of $\Pi_{\text{FT}}[\ell, \kappa]$ (Proposition 5.28(1)) is the honest one; so all fold checks hold at every point, by Lemma 5.9(1). $\square$

6.6 Costs

We count operations in $\mathbb{F}$ (and note when they are in $\mathbb{F}_q$), with $\mu = n + R = \log_2 M$; hashing is counted in Section 8.

Proposition 6.11 (Costs of Π_{KF}). *1. Commit. One NTT of size M : $\frac{M}{2}\mu$ multiplications and $M\mu$ additions, in $\mathbb{F}_q$ if f has coefficients in $\mathbb{F}_q$; plus $\frac{n}{2}N$ subtractions if f is given in table form.*

2. Prover, opening. *At most $2nN$ multiplications and $2nN$ additions for $U_f K_z$; one NTT of size M over $\mathbb{F}$; fewer than $4M$ operations for $\text{ev}_L(K_z)$; $6M$ operations for h and $4M$ for w_0 , given the powers ξ^N and $\xi^{-(N+1)}$, $\xi \in L$, which lie in $\mathbb{F}_q$; at most $7M$ operations for the words $w_1, \dots, w_{\ell-1}$ (given the inverses $(2\xi)^{-1}$, which lie in $\mathbb{F}_q$ and are precomputed once per domain); and at most $7N$ operations for U_0 and P . In total $O(M\mu)$, dominated by the NTT.*

3. Verifier, arity 2. *Per query point ξ : n squarings, one multiplication and one inversion in $\mathbb{F}_q$ for $\xi^2, \dots, \xi^{2^n} = \xi^N$ and $\xi^{-(N+1)}$; $2n+1$ operations for $K_z(\xi)$ and $K_z(-\xi)$; 12 operations for $h(\pm\xi)$ and 8 for $w_0(\pm\xi)$; at most 7ℓ operations for the fold checks, besides ℓ exponentiations in $\mathbb{F}_q$ for the inverses $(2\xi^{2^{j-1}})^{-1}$ or, alternatively, ℓ squarings of ξ^{-1} ; and $2N_\ell$ operations for $P(\xi^{2^\ell})$. In total $O(\kappa(n + N_\ell))$.*
4. Communication, IOP model. *Oracles: w_A of length M , and w_j of length M_j for $j \in C$, of total length less than $2M$ (less than $M + M/(2^k - 1)$ for $C = C_k$). Non-oracle messages: the N_ℓ coefficients of P . Queries, per point: for arity 2, 4 to (w, w_A) and 2 to each of $w_1, \dots, w_{\ell-1}$, that is, $\kappa(2\ell + 2)$ in total; for general C , 2^g values of w and of w_A and $2^{g'}$ values of each committed word, where g, g' are the gaps between consecutive levels of $C \cup \{0, \ell\}$ (Proposition 5.28(3)). For arity 2^k , the verifier's field operations are $O(\kappa(2^k(n + \ell) + N_\ell))$.*

Proof. (1) Fact 6.4 and Lemma 2.6. (2) The product is Lemma 4.9(3); the NTT is Fact 6.4, over $\mathbb{F}$ since the coefficients of A lie in $\mathbb{F}$; $\text{ev}_L(K_z)$ is Lemma 6.7. Each value $h(\xi) = ((w(\xi)K_z(\xi))\xi - w_A(\xi) - v\xi^N)\xi^{-(N+1)}$ costs four multiplications and two subtractions, and each entry of w_0 two multiplications and two additions, with β^2 computed once. For the folds, each value $\text{fold}_r(w)(\xi^2) = (2r - 1)w_e(\xi^2) + (1 - r)w_o(\xi^2)$ costs one addition and one multiplication by $\frac{1}{2}$ for w_e , one subtraction and one multiplication by $(2\xi)^{-1}$ for w_o , and two multiplications and one addition for the combination, so 7 operations; there are $\sum_{j=1}^{\ell-1} M_j < M$ values. For U_0 , $2N$ multiplications and $2N$ additions; for P , 3 operations per output coefficient and $\sum_{j=1}^{\ell} N_j < N$ outputs. (3) The powers ξ^{2^k} for $k \in \llbracket 1, n \rrbracket$ take n squarings; $\xi^{-(N+1)} = (\xi^N \cdot \xi)^{-1}$ one multiplication and one inversion, and $(-\xi)^N = \xi^N$, $(-\xi)^{-(N+1)} = -\xi^{-(N+1)}$ since N is even. By Lemma 4.4(2), $K_z(\pm\xi) = (z_1 \pm \xi)K_{z'}(\xi^2)$, where $K_{z'}(\xi^2)$ takes $n-1$ additions and $n-2$ multiplications given the powers (Lemma 4.9(1)); this gives both values with $2n+1$ operations. The values $h(\pm\xi)$ and $w_0(\pm\xi)$ are as in (2). The fold check of level j needs $(2\xi_{j-1})^{-1}$ for the point $\xi_{j-1} = \pm\xi^{2^{j-1}}$ of its fibre, in $\mathbb{F}_q$; each check then costs 7 operations as in (2) and one comparison. P is evaluated by Horner's rule. (4) $M + \sum_{j \in C} M_j < 2M$, and Proposition 5.28(3) for arity 2^k ; the virtual word must be computed at the 2^g points read at level 0, with $O(n)$ operations each. At the query point ξ , the fold check of level 1 reads w and w_A at $\pm\xi$, and the fold check of level $j \geq 2$ reads w_{j-1} at $\pm\xi^{2^{j-1}}$; the right-hand sides $w_j(\xi^{2^j})$ for $j < \ell$ are among the values read at level $j+1$. $\square$

Remark 6.12 (Choice of ℓ). With $\ell = n$, P is a constant and the verifier's work per point is $O(n)$; with $\ell < n$, the prover sends $N_\ell = 2^{n-\ell}$ coefficients instead of the last $n - \ell$ oracles, which in the compiled scheme with arity 2 removes $n - \ell$ Merkle roots and $\kappa(n - \ell)$ authentication paths at the cost of $2^{n-\ell}$ field elements and $O(\kappa 2^{n-\ell})$ verifier operations. The choice is made in Section 10.

Remark 6.13 (Comparison with the kernel-basis scheme). The evaluation protocol of [Mkh26] runs a sumcheck of ℓ rounds alongside the folds, and sends no oracle besides the folded words. Π_{KF} replaces the sumcheck by the virtual word and one oracle w_A (one NTT over $\mathbb{F}$ and one Merkle tree for the prover, 2 more queries per point), computed from the product $U_f K_z$. A quantitative comparison is in Section 10.

Remark 6.14 (Why $R \geq 2$). The batching lemma of Section 7 shows, for a wrong value v , that the polynomial Φ of Lemma 4.12, of degree at most $2N$, would vanish on the agreement set produced by the proximity test, of size at least $(1 - \delta)M$. This needs $2N < (1 - \delta)M$, that is, $2\rho < 1 - \delta$; with $\delta \leq (1 - \rho)/2$ it holds for every admissible δ when $\rho < 1/3$, so for $\rho = 2^{-R}$ with $R \geq 2$, and for $\rho = 1/2$ it fails for every $\delta > 0$ (Remark 7.6).

7 Soundness

We prove that Π_{KF} is sound, round-by-round knowledge sound, and evaluation binding, in the unique-decoding regime. The only new ingredient is the batching lemma (Lemma 7.4), which combines correlated agreement for curves of degree 2 (Theorem 2.16 with $t = 2$) with the identity lemma (Lemma 4.12) through the virtual word; the folding part of the protocol is handled entirely by Section 5, whose witness sets are used without change. Throughout, the parameters are those of Section 6.1, in particular $\rho = 2^{-R}$ with $R \geq 2$, and

$$\delta^* = \frac{1 - \rho}{2}, \quad 0 < \delta \leq \delta^*.$$

Probabilities are over the challenge space $\mathbb{F} \times \mathbb{F}^\ell \times L^{\times \kappa}$ of Π_{KF} , with $\beta, r = (r_1, \dots, r_\ell)$ and the query points uniform and independent. The set C of committed levels is arbitrary: by Proposition 5.28, the words at the other levels are the exact folds of the previous ones, and every statement below is proved for $\Pi_{\text{FT}}[\ell, \kappa]$ with these virtual words. A set $S \subseteq L$ is *fibre-closed* if $-\xi \in S$ for every $\xi \in S$, that is, if it is a union of fibres (Definition 2.9); a fibre-closed set of size at least $(1 - \delta)M$ consists of at least $(1 - \delta)M/2$ fibres.

7.1 The relation

Definition 7.1 (Evaluation relation).

$$\mathcal{R}_{\text{KF}}^\delta = \{ ((w; z, v), f) : w \in \mathbb{F}^L, f \in \mathcal{L}_n(\mathbb{F}), \Delta_0^{\text{fib}}(w, \text{Enc}(f)) \leq \delta \text{ and } f(z) = v \}.$$

This is the relation $\mathcal{R}_{\text{eval}}^\delta$ of Section 3.3 with $\text{dist} = \Delta_0^{\text{fib}}$ (Definition 5.10); the uniqueness property (3) holds by Lemma 6.3(2). Since $\Delta \leq \Delta_0^{\text{fib}}$ (Lemma 5.11(1)), a witness also satisfies $\Delta(w, \text{Enc}(f)) \leq \delta$; the fibre distance is the natural one because every query of the protocol reads whole fibres, and it is the one used in Section 8, where oracle symbols are fibres. An instance $(w; z, v)$ has a witness if and only if the unique codeword of $\mathcal{C}_0$ within fibre distance δ of w exists and is $\text{Enc}(f)$ with $f(z) = v$.

Lemma 7.2 (Rate condition). $2N < (1 - \delta)M$, that is, $2\rho < 1 - \delta$.

Proof. $1 - \delta \geq 1 - \delta^* = (1 + \rho)/2$, and $(1 + \rho)/2 > 2\rho$ because $\rho \leq 1/4 < 1/3$. $\square$

7.2 The value on an agreement set

Lemma 7.3 (Value from agreement). *Let $w, w_A \in \mathbb{F}^L$, $z \in \mathbb{F}^n$, $v \in \mathbb{F}$ and $h = h[w, w_A; z, v]$ (Definition 6.5). Let $\hat{U}, \hat{A}, \hat{H} \in \mathbb{F}[X]_{<N}$, and let $S \subseteq L$ be a set with $|S| > 2N$ on which $w = \text{ev}_L(\hat{U})$, $w_A = \text{ev}_L(\hat{A})$ and $h = \text{ev}_L(\hat{H})$. Then $v = [X^{N-1}] \hat{U} K_z$; that is, $v = \hat{f}(z)$ for the $\hat{f} \in \mathcal{L}_n(\mathbb{F})$ with $U_{\hat{f}} = \hat{U}$.*

Proof. Let $\Phi = X \hat{U} K_z - \hat{A} - v X^N - X^{N+1} \hat{H}$. For $\xi \in S$, Lemma 6.6(1) and the three agreements give $\Phi(\xi) = \xi w(\xi) K_z(\xi) - w_A(\xi) - v \xi^N - \xi^{N+1} h(\xi) = 0$. Thus Φ has more than $2N$ roots, and $\deg \Phi \leq 2N$ (Lemma 4.12), so $\Phi = 0$ (Lemma 2.2). Conclude with Lemma 4.12; $\hat{f}$ exists by Lemma 4.2(2). $\square$

7.3 The batching lemma

For $w, w_A \in \mathbb{F}^L$, $z \in \mathbb{F}^n$, $v \in \mathbb{F}$ and $\beta \in \mathbb{F}$, write $w_\beta = w + \beta w_A + \beta^2 h$ with $h = h[w, w_A; z, v]$; this is the word w_0 of Definition 6.8.

Lemma 7.4 (Batching). *Let $w, w_A \in \mathbb{F}^L$, $z \in \mathbb{F}^n$, $v \in \mathbb{F}$, and let $Y \subseteq L$ be fibre-closed. Let $\mathcal{G}$ be the set of $\beta \in \mathbb{F}$ for which there are $c \in \mathcal{C}_0$ and a fibre-closed set $T \subseteq Y$ with $|T| \geq (1 - \delta)M$ such that w_β agrees with c on T . If $|\mathcal{G}| > 2M$, then there is $\hat{f} \in \mathcal{L}_n(\mathbb{F})$ with $\hat{f}(z) = v$ such that w agrees with $\text{Enc}(\hat{f})$ on a fibre-closed subset of Y of size at least $(1 - \delta)M$. In particular, for $Y = L$, $((w; z, v), \hat{f}) \in \mathcal{R}_{\text{KF}}^\delta$.*

Proof. Let $h = h[w, w_A; z, v]$.

Correlated agreement. Every $\beta \in \mathcal{G}$ has $\Delta(w_\beta, \mathcal{C}_0) \leq \delta$. Since $|\mathcal{G}| > 2M$, Theorem 2.16 with $t = 2$, $(u_0, u_1, u_2) = (w, w_A, h)$ and $\mathcal{C} = \mathcal{C}_0$ gives codewords $\hat{u} = \text{ev}_L(\hat{U})$, $\hat{a} = \text{ev}_L(\hat{A})$, $\hat{h} = \text{ev}_L(\hat{H})$ with $\hat{U}, \hat{A}, \hat{H} \in \mathbb{F}[X]_{<N}$, and a set of at least $(1 - \delta)M$ points on which $w = \hat{u}$, $w_A = \hat{a}$ and $h = \hat{h}$. Let Agr be the set of all points of L where these three equalities hold; $|\text{Agr}| \geq (1 - \delta)M$.

The value. $|\text{Agr}| \geq (1 - \delta)M > 2N$ (Lemma 7.2), so Lemma 7.3 with $S = \text{Agr}$ gives $\hat{f}(z) = v$, where $U_{\hat{f}} = \hat{U}$.

Bad challenges. For $\xi \in L \setminus \text{Agr}$, the polynomial $d_\xi(Z) = (w - \hat{u})(\xi) + Z(w_A - \hat{a})(\xi) + Z^2(h - \hat{h})(\xi) \in \mathbb{F}[Z]$ is nonzero, of degree at most 2, so it has at most 2 roots (Lemma 2.2). Let Bad be the set of all these roots, over all $\xi \in L \setminus \text{Agr}$; then $|\text{Bad}| \leq 2|L \setminus \text{Agr}| \leq 2\delta M < 2M$. Hence there is $\beta \in \mathcal{G} \setminus \text{Bad}$; fix it, together with c and T as in the definition of $\mathcal{G}$.

The codeword is the combination. Let $\hat{c} = \hat{u} + \beta \hat{a} + \beta^2 \hat{h} \in \mathcal{C}_0$; then $w_\beta = \hat{c}$ on Agr . The codewords c and $\hat{c}$ both agree with w_β on $T \cap \text{Agr}$, of size at least $(1 - \delta)M + (1 - \delta)M - M = (1 - 2\delta)M \geq \rho M = N$. By Lemma 2.12(2), $c = \hat{c}$.

$T \subseteq \text{Agr}$. Let $\xi \in T$. Then $w_\beta(\xi) = c(\xi) = \hat{c}(\xi)$, that is, $d_\xi(\beta) = 0$. If $\xi \notin \text{Agr}$, then β is a root of the nonzero polynomial d_ξ , so $\beta \in \text{Bad}$, which is excluded. So $T \subseteq \text{Agr}$.

Agreement. $\text{Enc}(\hat{f}) = \text{ev}_L(\hat{U}) = \hat{u}$, and $w = \hat{u}$ on T , a fibre-closed subset of Y with $|T| \geq (1 - \delta)M$. For $Y = L$, w and $\text{Enc}(\hat{f})$ agree on the at least $(1 - \delta)M/2$ fibres contained in T , so $\Delta_0^{\text{fb}}(w, \text{Enc}(\hat{f})) \leq \delta$ (Lemma 5.11(2)). $\square$

Remark 7.5 (The parameter Y). For the interactive analysis $Y = L$. The set Y is used in Section 8, where the words are only partially defined (the fibres opened by the adversary) and the lemma is applied to arbitrary completions of them, with Y the union of the defined fibres.

Remark 7.6 (Where $R \geq 2$ is used). Only in the step “the value”: the identity has degree $2N$ as a polynomial, and the agreement set must have more than $2N$ points. With $\rho = 1/2$, the condition $2\rho < 1 - \delta$ fails for every $\delta > 0$ (for $\delta = 1/4$, $(1 - \delta)M = \frac{3}{2}N < 2N$), and the lemma does not conclude; rate $1/2$ would need a different identity or a different argument.

7.4 Direct soundness

Put

$$\varepsilon_{\text{KF}} = \frac{2M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa = \frac{2M + M(1 - 2^{-\ell})}{|\mathbb{F}|} + (1 - \delta)^\kappa < \frac{3M}{|\mathbb{F}|} + (1 - \delta)^\kappa,$$

with $\varepsilon_{\text{fold}}$ as in Section 5.6.

Lemma 7.7 (Witness sets are fibre-closed). *For every $j \in \llbracket 0, \ell \rrbracket$ and $c \in \mathcal{C}_j$, the witness set $\mathcal{W}_j(c)$ of Definition 5.20 is fibre-closed.*

Proof. For $j = 0$ the definition is symmetric in ξ and $-\xi$. For $j \geq 1$, the conditions defining $\mathcal{W}_j(c)$ involve ξ only through ξ^{2^i} with $i \geq 1$, and $(-\xi)^{2^i} = \xi^{2^i}$. $\square$

Theorem 7.8 (Soundness). *Let $(w; z, v)$ be an instance with no witness in $\mathcal{R}_{\text{KF}}^\delta$. Then every prover makes the verifier of Π_{KF} accept with probability at most ε_{KF} .*

Proof. By Lemma 3.4 it suffices to treat a deterministic prover. Its round-1 message w_A is fixed, and so is the set $\mathcal{G}$ of Lemma 7.4 with $Y = L$. Since there is no witness, $|\mathcal{G}| \leq 2M$. For each fixed β , rounds 2 to $\ell+2$ are an execution of $\Pi_{\text{FT}}[\ell, \kappa]$ on $w_0 = w_\beta$ by a deterministic prover; let $\text{Good}_\beta \subseteq \mathbb{F}^\ell$ be the event of Theorem 5.24, so that $\Pr_r[\neg \text{Good}_\beta] \leq \varepsilon_{\text{fold}}$. Let $\mathcal{A} = \{(\beta, r) : \beta \notin \mathcal{G}, r \in \text{Good}_\beta\}$.

By Lemma 5.21(1), a query point is accepted if and only if it lies in $\mathcal{W}_\ell(w_\ell)$. We claim $\text{dens}_\ell(w_\ell) < 1 - \delta$ for $(\beta, r) \in \mathcal{A}$. Otherwise Proposition 5.23 gives $c_0 \in \mathcal{C}_0$ such that w_β agrees with c_0 on $T = \mathcal{W}_\ell(w_\ell)$, which is fibre-closed (Lemma 7.7) and has at least $(1 - \delta)M$ points; so $\beta \in \mathcal{G}$, a contradiction. By Lemma 3.3 with $\Omega' = \mathbb{F} \times \mathbb{F}^\ell$, the verifier accepts with $(\beta, r) \in \mathcal{A}$ with probability at most $(1 - \delta)^\kappa$. Finally

$$\Pr[(\beta, r) \notin \mathcal{A}] \leq \Pr[\beta \in \mathcal{G}] + \max_{\beta} \Pr_r[\neg \text{Good}_\beta] \leq \frac{2M}{|\mathbb{F}|} + \varepsilon_{\text{fold}}. \quad \square$$

7.5 Round-by-round knowledge soundness

Fix an instance $(w; z, v)$. For $i \in \llbracket 0, \ell + 2 \rrbracket$, τ_i denotes a partial transcript of i rounds (Section 3.1). Once τ_1 is fixed, so are w_A , β and the word $w_0 = w_\beta$. For $j \in \llbracket 0, \ell \rrbracket$, the partial transcript τ_{j+1} determines $w_0, \dots, w_{j-1}$ and $r_1, \dots, r_j$, hence the witness sets $\mathcal{W}_j(c)$ of Definition 5.20 for $c \in \mathcal{C}_j$ (for $j = 0$, only w_0), and their densities $\text{dens}_j(c)$.

Definition 7.9 (Doomed set and extractor). • The empty transcript τ_0 is doomed.

- For $j \in \llbracket 0, \ell \rrbracket$, τ_{j+1} is *not doomed* if there is $c \in \mathcal{C}_j$ with $\text{dens}_j(c) \geq 1 - \delta$, and doomed otherwise.
- A full transcript $\tau_{\ell+2}$ is doomed if and only if the verifier rejects it.

The extractor Ext , on every input (τ, m) , reads the oracle w of the instance, and outputs the $\hat{f} \in \mathcal{L}_n(\mathbb{F})$ such that $\text{Enc}(\hat{f})$ is the codeword of $\mathcal{C}_0$ within fibre distance δ^* of w , if it exists, and $\hat{f} = 0$ otherwise.

Lemma 7.10 (The extractor). *Ext is well defined and runs in time polynomial in M . Its output depends only on w , and it is a witness for $(w; z, v)$ whenever a witness exists. If $w \in \mathbb{F}_q^L$, its output has coefficients in $\mathbb{F}_q$.*

Proof. The codeword within fibre distance δ^* is unique by Lemma 5.12, which holds for every radius at most $(1 - \rho)/2$, in particular for δ^* . By Remark 5.14 (at radius δ^*), it is found, and its existence decided, in time polynomial in M ; an inverse NTT (Fact 6.4) gives $U_{\hat{f}}$, whose coefficients are those of $\hat{f}$ (Definition 4.1). If $((w; z, v), f) \in \mathcal{R}_{\text{KF}}^\delta$, then $\text{Enc}(f)$ is within fibre distance $\delta \leq \delta^*$ of w , so it is the decoded codeword and $\hat{f} = f$. The last statement is Lemma 6.3(3). $\square$

Theorem 7.11 (Round-by-round knowledge soundness). *With the doomed set and extractor of Definition 7.9:*

1. (Round 1.) *If Ext does not output a witness, then for every round-1 message w_A , τ_1 is not doomed for at most $2M$ values of β .*

2. (Round $j + 1$, $j \in \llbracket 1, \ell \rrbracket$.) If τ_j is doomed, then for every round- $(j + 1)$ message, τ_{j+1} is not doomed for at most M_j values of r_j .
3. (Round $\ell + 2$.) If $\tau_{\ell+1}$ is doomed, then for every final polynomial P , the verifier accepts with probability at most $(1 - \delta)^\kappa$ over the query points.

Consequently, Π_{KF} is round-by-round knowledge sound for $\mathcal{R}_{\text{KF}}^\delta$ (Definition 3.7) with errors

$$\varepsilon_1 = \min\left(1, \frac{2M}{|\mathbb{F}|}\right), \quad \varepsilon_{j+1} = \min\left(1, \frac{M_j}{|\mathbb{F}|}\right) \ (j \in \llbracket 1, \ell \rrbracket), \quad \varepsilon_{\ell+2} = (1 - \delta)^\kappa.$$

Proof. *Item 1.* Fix w_A , and let $\mathcal{G}$ be the set of Lemma 7.4 with $Y = L$. If τ_1 is not doomed, there is $c \in \mathcal{C}_0$ with $|\mathcal{W}_0(c)| \geq (1 - \delta)M$; by Definition 5.20, $w_\beta = c$ on $T = \mathcal{W}_0(c)$, which is fibre-closed (Lemma 7.7), so $\beta \in \mathcal{G}$. If this happened for more than $2M$ values of β , Lemma 7.4 would give a witness, and then Ext would output a witness (Lemma 7.10), contrary to the hypothesis.

Item 2. Fix the doomed τ_j and the round- $(j + 1)$ message; this fixes w_{j-1} . At most M_j values of r_j are not good for w_{j-1} at level j (Lemma 5.18). Let r_j be good, and suppose τ_{j+1} is not doomed, with $c' \in \mathcal{C}_j$ and $\text{dens}_j(c') \geq 1 - \delta$. Then Lemma 5.22 gives $c \in \mathcal{C}_{j-1}$ with $\text{dens}_{j-1}(c) \geq 1 - \delta$, so τ_j is not doomed, a contradiction.

Item 3. A query point is accepted if and only if it lies in $\mathcal{W}_\ell(w_\ell)$ (Lemma 5.21(1)). Since $w_\ell \in \mathcal{C}_\ell$ and $\tau_{\ell+1}$ is doomed, $\text{dens}_\ell(w_\ell) < 1 - \delta$, and the κ independent uniform points all lie in $\mathcal{W}_\ell(w_\ell)$ with probability less than $(1 - \delta)^\kappa$ (Lemma 3.3).

The conditions of Definition 3.7. Condition 1 holds by definition. Condition 3 holds because a doomed full transcript is rejected by definition. For condition 2, let τ be doomed with $i - 1$ rounds and m_i a message such that the probability over c_i of leaving $\mathcal{D}$ exceeds ε_i . For $i = 1$, item 1 shows that Ext outputs a witness. For $i = j + 1$ with $j \in \llbracket 1, \ell \rrbracket$, and for $i = \ell + 2$, items 2 and 3 show that this cannot happen, so the condition holds vacuously. $\square$

Items 2 and 3 are statements about the folding test alone: from round 2 on, the doomed set is that of $\Pi_{\text{FT}}[\ell, \kappa]$ on the word w_0 , and the evaluation claim is used only in round 1, through the batching lemma.

7.6 Consequences

Corollary 7.12 (Knowledge, soundness and binding). *1. (Knowledge.) If a prover makes the verifier accept $(w; z, v)$ with probability greater than ε_{KF} , then $\hat{f} = \text{Ext}(w)$ satisfies $\Delta_0^{\text{fib}}(w, \text{Enc}(\hat{f})) \leq \delta$ and $\hat{f}(z) = v$.*

2. (Soundness.) If $(w; z, v)$ has no witness, every prover is accepted with probability at most ε_{KF} .

3. (Binding.) Π_{KF} is ε_{KF} -evaluation binding (Definition 3.10).

Proof. (1) By Lemma 3.4, some deterministic prover is accepted with probability greater than $\varepsilon_{\text{KF}} \geq \sum_i \varepsilon_i$. By Lemma 3.8(2) and Theorem 7.11, Ext outputs a witness on some input; its output depends only on w (Lemma 7.10). (2) is (1) in contrapositive form, and also Theorem 7.8. (3) Lemma 3.11, with (3) given by Lemma 6.3(2). $\square$

Remark 7.13 (Base-field commitments). If the commitment is computed over $\mathbb{F}_q$, so that $w \in \mathbb{F}_q^L$, the extracted polynomial has coefficients in $\mathbb{F}_q$ (Lemma 7.10), while the point z , the value v and all challenges lie in $\mathbb{F}$.

7.7 Parameters

The error ε_{KF} has a field term, below $3M/|\mathbb{F}|$, and a query term $(1-\delta)^\kappa$ with δ up to $\delta^* = (1-\rho)/2$. Table 1 gives the smallest κ with $(1-\delta^*)^\kappa \leq 2^{-\lambda}$, and Table 2 the field term for the extensions of degree e of the Goldilocks field ($q = 2^{64} - 2^{32} + 1$, $|\mathbb{F}| = q^e$). For example, $\rho = 1/4$ and $\kappa = 148$ give $\lambda = 100$ bits for the query term, and the quadratic extension keeps the field term below 2^{-100} for $M \leq 2^{26}$; for larger M , or for 128 bits, one takes $e \geq 3$. These are the bounds of the interactive protocol; the non-interactive scheme is analysed in Section 8.

ρ	δ^*	$-\log_2(1-\delta^*)$	κ for $\lambda = 100$	κ for $\lambda = 128$
1/4	3/8	0.678	148	189
1/8	7/16	0.830	121	155
1/16	15/32	0.913	110	141

Table 1: Number of queries κ such that $(1-\delta^*)^\kappa \leq 2^{-\lambda}$.

e	$M = 2^{20}$	$M = 2^{24}$	$M = 2^{26}$	$M = 2^{28}$	$M = 2^{32}$
2	-106.4	-102.4	-100.4	-98.4	-94.4
3	-170.4	-166.4	-164.4	-162.4	-158.4
4	-234.4	-230.4	-228.4	-226.4	-222.4

Table 2: $\log_2(3M/|\mathbb{F}|)$ for $|\mathbb{F}| = q^e$, $q = 2^{64} - 2^{32} + 1$.

Remark 7.14 (Unique decoding only). All results are in the unique-decoding regime $\delta \leq (1-\rho)/2$, where Theorem 2.16 is proved with the error $M/|\mathbb{F}|$ per degree of the curve. Beyond it, correlated agreement holds with larger errors (up to the Johnson bound in [BSCI⁺23]) and the extractor list-decodes; this extension is discussed in Section 12.

8 The Non-Interactive Scheme

We compile Π_{KF} into a non-interactive argument with the BCS transformation for interactive oracle reductions of Chiesa, Di, Hu and Zheng [CDHZ25], and derive post-quantum straightline knowledge soundness in the quantum random oracle model from Theorem 3.15. The hypothesis of that theorem is relaxed round-by-round knowledge soundness (Definition 3.13), in which the commitment and the prover messages may be *partial strings*: the extractor of [CDHZ25] recovers from the adversary only the positions that it opened. The proof is the one of Section 7.5, run on the words in which every undefined entry is replaced by 0, with witness sets restricted to query points that read no undefined entry; the batching lemma is applied with Y the union of the defined fibres of the commitment (Remark 7.5). Every query of Π_{KF} reads whole fibres, so we take fibres as the symbols of all strings, as in [Mkh26, §7.7], whose structure we follow. Throughout, $\delta^* = (1-\rho)/2$; the results of Section 7 hold for every $\delta \in (0, \delta^*]$, and below δ ranges over $[0, 1]$.

8.1 The protocol as an interactive oracle reduction

Fibre strings. For each $j \in \llbracket 0, \ell - 1 \rrbracket$ and $\eta \in L_{j+1}$, fix one of the two square roots $\xi_\eta \in L_j$ of η (Lemma 2.8); the other is $-\xi_\eta$. Let $\Sigma = \mathbb{F}^2$, and for $u \in \mathbb{F}^{L_j}$ let $\text{fib}_j(u) \in \Sigma^{M_{j+1}}$ be the string

$\eta \mapsto (u(\xi_\eta), u(-\xi_\eta))$, indexed by L_{j+1} in a fixed order. The map fib_j is a bijection from $\mathbb{F}^{L_j}$ onto $\Sigma^{M_{j+1}}$, and by Definition 5.10,

$$\Delta(\text{fib}_j(u), \text{fib}_j(c)) = \Delta_j^{\text{fib}}(u, c) \quad (u, c \in \mathbb{F}^{L_j}). \quad (7)$$

For a partial string s over Σ , the *filling* $\bar{s}$ replaces every $\perp$ by $(0, 0)$; since $\bar{s}$ and s agree on $\text{Dom}(s)$, $\Delta(\bar{s}, s') \leq \Delta(s, s')$ for every full string s' .

The relation. Let

$$\mathcal{R}_{\text{KF}} = \{ ((z, v), y, f) : y = \text{fib}_0(\text{Enc}(f)) \text{ and } f(z) = v \},$$

with explicit instance $(z, v) \in \mathbb{F}^n \times \mathbb{F}$, implicit instance $y \in \Sigma^{M/2}$ and witness $f \in \mathcal{L}_n(\mathbb{F})$. For a partial string y and $\delta \in [0, 1]$, $((z, v), y, f) \in (\mathcal{R}_{\text{KF}})_{\leq \delta}$ if and only if $\Delta(y, \text{fib}_0(\text{Enc}(f))) \leq \delta$, with $\perp$ counted as a disagreement, and $f(z) = v$. For a full string $y = \text{fib}_0(w)$, by (7), this says $((w; z, v), f) \in \mathcal{R}_{\text{KF}}^\delta$ (Definition 7.1).

Challenges. Fix an integer $\lambda_{\text{ch}} \geq \log_2 |\mathbb{F}|$ and a bijection $\text{num} : [\![0, |\mathbb{F}| - 1]\!] \rightarrow \mathbb{F}$, and let $\phi : \{0, 1\}^{\lambda_{\text{ch}}} \rightarrow \mathbb{F}$, $\phi(\mathbf{vr}) = \text{num}(\text{int}(\mathbf{vr}) \bmod |\mathbb{F}|)$, where int reads a bit string as a binary integer. Fix a bijection $\text{pos} : \{0, 1\}^{n+R} \rightarrow L$ (recall $M = 2^{n+R}$).

Lemma 8.1 (Sampling). *For every $S \subseteq \mathbb{F}$, $\Pr_{\mathbf{vr} \leftarrow \{0, 1\}^{\lambda_{\text{ch}}}} [\phi(\mathbf{vr}) \in S] \leq |S| (1/|\mathbb{F}| + 2^{-\lambda_{\text{ch}}})$. For $\mathbf{vr}$ uniform in $\{0, 1\}^{\kappa(n+R)}$, cut into κ blocks of $n + R$ bits, the images of the blocks under pos are independent and uniform in L .*

Proof. Each $a \in \mathbb{F}$ has at most $\lceil 2^{\lambda_{\text{ch}}}/|\mathbb{F}| \rceil < 2^{\lambda_{\text{ch}}}/|\mathbb{F}| + 1$ preimages under ϕ ; divide by $2^{\lambda_{\text{ch}}}$ and sum over S . The map $\{0, 1\}^{\kappa(n+R)} \rightarrow L^{\times \kappa}$ induced by pos is a bijection. $\square$

Definition 8.2 (The IOR Π_{KF}^Σ). Π_{KF}^Σ is Π_{KF} (Definition 6.8) with statement $((z, v); y)$, where the commitment is $w = \text{fib}_0^{-1}(y)$, and the following encoding of messages and challenges.

1. Round 1: the message is $\text{fib}_0(w_A) \in \Sigma^{M/2}$, and the challenge $\mathbf{vr}_1 \in \{0, 1\}^{\lambda_{\text{ch}}}$ gives $\beta = \phi(\mathbf{vr}_1)$.
2. Round 2: the message is a single symbol, which the verifier never reads (so that every round has a nonempty string to commit to), and $\mathbf{vr}_2$ gives $r_1 = \phi(\mathbf{vr}_2)$.
3. Round $j + 1$, $j \in [\![2, \ell]\!]$: if $j - 1 \in C$, the message is $\text{fib}_{j-1}(w_{j-1}) \in \Sigma^{M_j}$; otherwise it is a single symbol that the verifier never reads. The challenge $\mathbf{vr}_{j+1}$ gives $r_j = \phi(\mathbf{vr}_{j+1})$.
4. Round $\ell + 2$: the message is the string of the N_ℓ coefficients of P , two per symbol and padded with a zero if $N_\ell = 1$, which the verifier reads entirely; and $\mathbf{vr}_{\ell+2} \in \{0, 1\}^{\kappa(n+R)}$ gives the query points by pos .

On partial strings, the verifier rejects if an entry that it reads is $\perp$; otherwise it runs the verifier of Π_{KF} on the words $w = \text{fib}_0^{-1}(\bar{y})$, $w_A = \text{fib}_0^{-1}(\bar{\Pi}_1)$, $w_{j-1} = \text{fib}_{j-1}^{-1}(\bar{\Pi}_{j+1})$ for $j - 1 \in C$ (the other words are the exact folds, Definition 5.27) and the polynomial read from the filled final message, and outputs $(\top, ())$ if that verifier accepts.

Let $0 = j_0 < j_1 < \dots < j_s = \ell$ be the levels of $C \cup \{0, \ell\}$. For each $t \in [\![1, s]\!]$, the query point ξ reads the string of $w_{j_{t-1}}$ (that is, y and Π_1 for $t = 1$, through the virtual word, Lemma 6.6(2), and $\Pi_{j_{t-1}+2} = \text{fib}_{j_{t-1}}(w_{j_{t-1}})$ for $t \geq 2$) at the $2^{j_t-j_{t-1}-1}$ positions $\eta \in L_{j_{t-1}+1}$ with $\eta^{2^{j_t-j_{t-1}-1}} = \xi^{2^{j_t}}$;

these symbols give $w_{j_{t-1}}$ on the $2^{j_t-j_{t-1}}$ points above $\xi^{2^{j_t}}$, from which the fold check of level j_t is computed (Definition 5.27). For $t \geq 2$ they contain the value $w_{j_{t-1}}(\xi^{2^{j_t-1}})$ compared at level j_{t-1} . For arity 2 this is one symbol per string. These positions are determined by the challenges. So Π_{KF}^Σ is an IOR from $\mathcal{R}_{\text{KF}}$ to $\mathcal{R}_{\text{acc}}$ in the sense of Section 3.4, with $\nu = \ell + 2$ rounds, $\lambda_{\min} = \min(\lambda_{\text{ch}}, \kappa(n + R))$ and longest string $l_{\max} = M/2$. Every string of the prescribed lengths over Σ encodes a message of Π_{KF} , and filled strings are such strings; so the objects of Section 7 ($h, w_0, \hat{w}_j, \mathcal{B}_j, \mathcal{W}_j(c)$) are defined for the fillings.

8.2 Erasures and the knowledge state function

For $j \in \llbracket 0, \ell \rrbracket$, let $\mathcal{Z}_j \subseteq L$ be the set of points ξ such that every symbol read by ξ from y , Π_1 and the strings $\Pi_{j_{t-1}+2}$ with $1 \leq j_{t-1} < j$ is defined (with the notation above; for $j \leq 1$, only y and Π_1). The set $\mathcal{Z}_j$ is determined by the statement and the messages of τ_{j+1} . Then $\mathcal{Z}_0 = \mathcal{Z}_1 \supseteq \mathcal{Z}_2 \supseteq \dots \supseteq \mathcal{Z}_\ell$; every $\mathcal{Z}_j$ is fibre-closed, since it depends on ξ only through ξ^2 ; and $\mathcal{Z}_0 \subseteq Y_y$, where $Y_y = \{\xi \in L : \xi^2 \in \text{Dom}(y)\}$ is the union of the defined fibres of y . With the notation of Section 7.5 applied to the fillings, put, for $c \in \mathcal{C}_j$,

$$\mathcal{W}_j^\perp(c) = \mathcal{W}_j(c) \cap \mathcal{Z}_j, \quad \text{dens}_j^\perp(c) = |\mathcal{W}_j^\perp(c)|/M;$$

by Lemma 7.7, $\mathcal{W}_j^\perp(c)$ is fibre-closed. For $\delta \in (0, \delta^*]$, let $\mathcal{D}_\delta^\perp$ be the set of partial transcripts obtained from Definition 7.9 with $\text{dens}_j^\perp$ in place of dens_j . It depends on the erasures of y and of the messages, not only on their fillings; without erasures, it is the doomed set of Definition 7.9.

Definition 8.3 (Knowledge state function of Π_{KF}^Σ). The algorithm **E**, on every input, reads y and outputs $\text{Ext}(\text{fib}_0^{-1}(\bar{y}))$ (Definition 7.9): the $\hat{f}$ such that $\text{Enc}(\hat{f})$ is the codeword within fibre distance δ^* of $\text{fib}_0^{-1}(\bar{y})$ if it exists, and 0 otherwise. For $\delta \in [0, 1]$, a statement $((z, v), y)$, a transcript tr and $f \in \mathcal{L}_n(\mathbb{F})$, let $\text{KState}_\delta((z, v), y, \text{tr}, f)$ be:

1. 1 if and only if $((z, v), y, f) \in (\mathcal{R}_{\text{KF}})_{\leq \delta}$, if $\text{tr} = \text{tr}_\emptyset$;
2. 1 if and only if the verifier of Π_{KF}^Σ accepts, if tr is a full transcript;
3. for $\text{tr} = \tau_i$ the partial transcript after $i \in \llbracket 1, \ell + 1 \rrbracket$ rounds: 1 if and only if $\tau_i \notin \mathcal{D}_\delta^\perp$, when $\delta \in (0, \delta^*]$; and 1 when $\delta \notin (0, \delta^*]$;
4. $\text{KState}_\delta((z, v), y, \text{tr}', f)$, if $\text{tr} = \text{tr}' \parallel \Pi$ ends with a prover message.

E runs in time polynomial in M (Lemma 7.10) and does not depend on δ . Conditions 1, 2 and 3 of Definition 3.13 hold by items 1, 4 and 2 respectively.

Lemma 8.4 (Round-by-round steps with erasures). *Let $\delta \in (0, \delta^*]$, fix a statement $((z, v), y)$ with y a partial string, and fix a partial transcript and the next prover message (partial strings).*

1. (Round 1.) *If $((z, v), y, \text{E}(y)) \notin (\mathcal{R}_{\text{KF}})_{\leq \delta}$, then for every round-1 message, $\tau_1 \notin \mathcal{D}_\delta^\perp$ for at most $2M$ values of β .*
2. (Round $j + 1$, $j \in \llbracket 1, \ell \rrbracket$.) *If $\tau_j \in \mathcal{D}_\delta^\perp$, then for every round- $(j + 1)$ message, $\tau_{j+1} \notin \mathcal{D}_\delta^\perp$ for at most M_j values of r_j .*
3. (Round $\ell + 2$.) *If $\tau_{\ell+1} \in \mathcal{D}_\delta^\perp$, then for every final message, the verifier of Π_{KF}^Σ accepts with probability at most $(1 - \delta)^\kappa$ over the query points.*

Proof. All objects are those of the filled words $w = \text{fib}_0^{-1}(\bar{y})$, w_A , $w_1, \dots$, which are words of Π_{KF} .

(1) Fix the round-1 message, and apply Lemma 7.4 to w and w_A with $Y = Y_y$, which is fibre-closed; let $\mathcal{G}$ be the set of that lemma. If $\tau_1 \notin \mathcal{D}_\delta^\perp$, there is $c \in \mathcal{C}_0$ with $|\mathcal{W}_0^\perp(c)| \geq (1 - \delta)M$. The set $T = \mathcal{W}_0^\perp(c)$ is fibre-closed and contained in $\mathcal{Z}_0 \subseteq Y_y$, and w_β agrees with c on $\mathcal{W}_0(c) \supseteq T$; so $\beta \in \mathcal{G}$. If there were more than $2M$ such β , Lemma 7.4 would give $\hat{f}$ with $\hat{f}(z) = v$ such that w agrees with $\text{Enc}(\hat{f})$ on a fibre-closed subset of Y_y with at least $(1 - \delta)M$ points, that is, on at least $(1 - \delta)M/2$ fibres over positions where y is defined and equal to $\bar{y}$. Then $\Delta(y, \text{fib}_0(\text{Enc}(\hat{f}))) \leq \delta$ with $\perp$ counted as a disagreement, so $((z, v), y, \hat{f}) \in (\mathcal{R}_{\text{KF}})_{\leq \delta}$. Moreover $\Delta_0^{\text{fib}}(w, \text{Enc}(\hat{f})) = \Delta(\bar{y}, \text{fib}_0(\text{Enc}(\hat{f}))) \leq \delta \leq \delta^*$ by (7), so $E(y) = \hat{f}$ (Lemma 7.10), contradicting the hypothesis.

(2) Fix the round- $(j + 1)$ message; at most M_j values of r_j are not good for w_{j-1} at level j (Lemma 5.18). Let r_j be good, and suppose $\tau_{j+1} \notin \mathcal{D}_\delta^\perp$, witnessed by $c' \in \mathcal{C}_j$ with $\text{dens}_j^\perp(c') \geq 1 - \delta$. Then $\text{dens}_j(c') \geq 1 - \delta$, and Lemma 5.22 gives $c \in \mathcal{C}_{j-1}$ with $\mathcal{W}_j(c') \subseteq \mathcal{W}_{j-1}(c)$. Intersecting with $\mathcal{Z}_j \subseteq \mathcal{Z}_{j-1}$ gives $\mathcal{W}_j^\perp(c') \subseteq \mathcal{W}_{j-1}^\perp(c)$; so $\tau_j \notin \mathcal{D}_\delta^\perp$, a contradiction.

(3) If the verifier accepts, the final message has no $\perp$, and every query point ξ reads no $\perp$, so $\xi \in \mathcal{Z}_\ell$; and ξ passes the checks of Π_{KF} on the fillings, so $\xi \in \mathcal{W}_\ell(w_\ell)$ (Lemma 5.21(1)). Hence every query point lies in $\mathcal{W}_\ell^\perp(w_\ell)$, where $w_\ell = \text{ev}_{L_\ell}(P) \in \mathcal{C}_\ell$. Since $\tau_{\ell+1} \in \mathcal{D}_\delta^\perp$, $\text{dens}_\ell^\perp(w_\ell) < 1 - \delta$, and by Lemmas 3.3 and 8.1 the verifier accepts with probability less than $(1 - \delta)^\kappa$. $\square$

Theorem 8.5 (Relaxed round-by-round knowledge soundness). Π_{KF}^Σ has relaxed round-by-round knowledge soundness errors (Definition 3.13), with KState and E of Definition 8.3,

$$\varepsilon_1^{\text{rr}}(\delta) = 2M \left(\frac{1}{|\mathbb{F}|} + \frac{1}{2\lambda_{\text{ch}}} \right), \quad \varepsilon_{j+1}^{\text{rr}}(\delta) = M_j \left(\frac{1}{|\mathbb{F}|} + \frac{1}{2\lambda_{\text{ch}}} \right) \quad (j \in \llbracket 1, \ell \rrbracket), \quad \varepsilon_{\ell+2}^{\text{rr}}(\delta) = (1 - \delta)^\kappa$$

for $\delta \in (0, \delta^*]$, and $\varepsilon_i^{\text{rr}}(\delta) = 1$ for $\delta \notin (0, \delta^*]$.

Proof. For $\delta \notin (0, \delta^*]$ there is nothing to prove. Fix $\delta \in (0, \delta^*]$, a statement, a round $i \in \llbracket 1, \ell + 2 \rrbracket$, and $\text{tr} = \tau_{i-1} \parallel \Pi_i$, where $\tau_0 = \text{tr}_\emptyset$. By item 4 of Definition 8.3, $\text{KState}_\delta(\cdot, \text{tr}, \cdot) = \text{KState}_\delta(\cdot, \tau_{i-1}, \cdot)$, and by items 2 and 3, $\text{KState}_\delta(\cdot, \text{tr} \parallel \text{vr}_i, f)$ does not depend on f . So the event of Definition 3.13 is contained in the conjunction of $\text{KState}_\delta(\cdot, \tau_{i-1}, E(\cdot)) = 0$, which does not depend on vr_i , and $\text{KState}_\delta(\cdot, \tau_i, \cdot) = 1$. If the first condition fails, the event is empty. Otherwise, for $i = 1$ it says $((z, v), y, E(y)) \notin (\mathcal{R}_{\text{KF}})_{\leq \delta}$, and for $i \geq 2$ it says $\tau_{i-1} \in \mathcal{D}_\delta^\perp$. For $i \leq \ell + 1$, Lemma 8.4(1,2) bounds the number of exceptional values of β or r_j , and Lemma 8.1 converts it into the stated probability. For $i = \ell + 2$, $\text{KState}_\delta(\cdot, \tau_{\ell+2}, \cdot) = 1$ means acceptance, whose probability is at most $(1 - \delta)^\kappa$ by Lemma 8.4(3). $\square$

8.3 Post-quantum knowledge soundness

Corollary 8.6 (Post-quantum knowledge soundness). Let $\delta \in (0, \delta^*]$ and

$$\varepsilon_{\text{rr}}(\delta) = \max \left\{ 2M \left(\frac{1}{|\mathbb{F}|} + \frac{1}{2\lambda_{\text{ch}}} \right), (1 - \delta)^\kappa \right\}.$$

The non-interactive scheme $\text{BCS}[\Pi_{\text{KF}}^\Sigma]$ (Section 3.4) satisfies the conclusion of Theorem 3.15 with $\nu = \ell + 2$, $\lambda_{\min} = \min(\lambda_{\text{ch}}, \kappa(n + R))$, $l_{\max} = M/2$ and this $\varepsilon_{\text{rr}}(\delta)$. In particular, for every quantum adversary that makes at most Q queries to the random oracle and outputs a Merkle root cm , a claim (z, v) and a proof, except with probability $\varepsilon_{\text{ext}}(Q)$, if the verifier accepts then the extractor outputs a partial string $y \in (\Sigma \sqcup \{\perp\})^{M/2}$, a polynomial $f \in \mathcal{L}_n(\mathbb{F})$ and a trapdoor such that: the openings of all positions of $\text{Dom}(y)$ are accepted under cm ; y agrees with $\text{fib}_0(\text{Enc}(f))$ on at least $(1 - \delta)M/2$ fibres; and $f(z) = v$.

Proof. Theorem 8.5 and Theorem 3.15, with $\max_i \varepsilon_i^{\text{rr}}(\delta) = \varepsilon_{\text{rr}}(\delta)$ since $M_j < 2M$; the last sentence unfolds Definition 3.14 for $\mathcal{R}_{\text{KF}}$, with $\perp$ entries counted as disagreements. $\square$

The extracted polynomial is bound to the root: two extracted witnesses for the same root with different polynomials expose a hash collision.

Lemma 8.7 (Uniqueness of relaxed committed witnesses). *Let $\delta \in (0, \delta^*]$ and let cm be a Merkle root. Let $\text{cx} = ((z, v), \text{cm})$ and $\text{cx}' = ((z', v'), \text{cm})$, and suppose that $(\text{cx}, (y, f, \text{td}))$ and $(\text{cx}', (y', f', \text{td}'))$ belong to $\widehat{\text{CR}}[\mathcal{R}_{\text{KF}}, \delta]$. If $f \neq f'$, then td and td' yield two accepting openings, under cm , of one position to different values, hence a collision of the random oracle. In particular, if $z = z'$ and $v \neq v'$, such a collision is obtained.*

Proof. Let $D = \{\eta \in L_1 : y(\eta) = \text{fib}_0(\text{Enc}(f))(\eta)\}$ and $D' = \{\eta \in L_1 : y'(\eta) = \text{fib}_0(\text{Enc}(f'))(\eta)\}$. Then $D \subseteq \text{Dom}(y)$, $D' \subseteq \text{Dom}(y')$ and $|D|, |D'| \geq (1 - \delta)M/2$, so $|D \cap D'| \geq (1 - 2\delta)M/2 \geq \rho M/2 = N/2$. If $y = y'$ on $D \cap D'$, then $\text{Enc}(f)$ and $\text{Enc}(f')$ agree on the at least N points of the fibres over $D \cap D'$, hence are equal (Lemma 2.12(2)), and $f = f'$ (Lemma 6.3(1)). So if $f \neq f'$, some $\eta \in D \cap D'$ has $y(\eta) \neq y'(\eta)$, and the openings of η computed from td and td' are both accepted under cm (Definition 3.14), with the same random oracle since the commitment index is fixed by the scheme; apply the salted form of Lemma 3.12(1) (Section 3.4). If $z = z'$ and $v \neq v'$, then $f(z) = v \neq v' = f'(z)$, so $f \neq f'$. $\square$

Lemma 8.7 is deterministic, as in [Mkh26]; it is the form in which binding follows from Corollary 8.6, since the extractor of Theorem 3.15 is defined for one proof at a time.

8.4 Parameters

Remark 8.8 (Post-quantum parameters). Take $\rho = 1/4$, $\delta = \delta^* = 3/8$, $M \leq 2^{32}$ (so $n \leq 30$ and $\ell \leq 30$), the quartic extension $\mathbb{F} = \mathbb{F}_q[\iota]/(\iota^4 - 7)$ of the Goldilocks field $\mathbb{F}_q$, $q = 2^{64} - 2^{32} + 1$, of size $q^4 > 2^{255.99}$, $\lambda_{\text{ch}} = 320$, hash output length $\lambda_{\text{H}} = 256$, and $\kappa = 248$. The polynomial $\iota^4 - 7$ is irreducible over $\mathbb{F}_q$, because 7 is not a square in $\mathbb{F}_q$ and $q \equiv 1 \pmod{4}$ [LN97, Theorem 3.75]. Then $(5/8)^{248} = 2^{-168.16}$ and $2M(1/|\mathbb{F}| + 2^{-320}) < 2^{-222}$, so $\varepsilon_{\text{rr}}(\delta) = 2^{-168.16}$. The quantities of Theorem 3.15 satisfy:

- $\nu = \ell + 2 \leq 32$, $l_{\text{max}} = M/2 \leq 2^{31}$, and $\lambda_{\text{min}} = \min(320, \kappa(n + R)) = 320$, since $\kappa(n + R) \geq 248 \cdot 3$;
- $q_s \leq \kappa 2^{k-1}(\ell + 1) + \lceil N_\ell/2 \rceil \leq 2^{30}$ for arity $2^k \leq 16$: at most 2^{k-1} symbols of y , of Π_1 and of each of the at most $\ell - 1$ committed words per query point, and the final symbols;
- $q_{\text{V}} \leq \nu + q_s(\lceil \log_2 l_{\text{max}} \rceil + 1) \leq 2^{36}$: one challenge query per round, and one authentication path of at most $\lceil \log_2 l_{\text{max}} \rceil + 1$ hashes per position read;
- $q_{\widehat{\text{CR}}} \leq (M/2) \log_2 M \leq 2^{36}$: one leaf hash and $\log_2(M/2)$ internal hashes per position of y .

For an adversary making $Q = 2^{64}$ queries, Theorem 3.15 gives $\varepsilon_{\text{ext}}(2^{64}) < 2^{-31.8}$: the first term is $2^{-31.84}$, and the other terms are below 2^{-51} . The bound is evaluated exactly in rational arithmetic by `rust/examples/pq_params.rs` in the companion repository, which also checks that 7 is not a square in $\mathbb{F}_q$. The dominant term is $320 Q^2 \varepsilon_{\text{rr}}(\delta)$, so each additional factor 2 in Q costs $2/\log_2(8/5) \approx 3$ further queries. These are the same parameters as for the scheme of [Mkh26], whose dominant error is also the query term; the round-1 error $2M/|\mathbb{F}|$ of the batching lemma is negligible at this field size.

Remark 8.9 (Classical parameters). Against classical adversaries, Theorem 7.8 and Corollary 7.12 give the interactive bound ε_{KF} , and the parameters of Section 7.7 ($\kappa = 148$ for $\rho = 1/4$, quadratic extension, $M \leq 2^{26}$) give about 100 bits for the interactive protocol. For the compiled scheme, Theorem 3.15 applies to classical adversaries as well (Section 3.4), but its Q^2 loss makes these parameters insufficient against 2^{64} queries; the parameters of Section 10 are therefore stated as interactive security levels.

Remark 8.10 (Merkle trees). Each string is committed by one Merkle tree with one leaf per symbol, that is, per fibre: a query opens one leaf and one authentication path per symbol read (one per string for arity 2), and the two values compared by a fold check come from one leaf. Implementations commonly compress the openings: several leaves read by one query share one path when they are adjacent in the tree, the nodes at a fixed depth (a *cap*) are sent once instead of repeated in every path, and a leaf read by several queries is sent once. From such a compressed proof, the full list of individual openings of $\text{BCS}[\Pi_{\text{KF}}^\Sigma]$ is recomputed deterministically by hashing, and the compressed verifier accepts exactly when the recomputed openings are accepted. A prover of the compressed scheme therefore yields a prover of $\text{BCS}[\Pi_{\text{KF}}^\Sigma]$ with the same success probability, using at most q_V further oracle queries; Corollary 8.6 then applies with $Q + q_V$ in place of Q , which for $Q = 2^{64}$ and the parameters of Remark 8.8 changes the bound by a factor below $1 + 2^{-26}$. The commitment y has entries in $\mathbb{F}_q^2 \subseteq \Sigma$ when f has coefficients in $\mathbb{F}_q$, which the implementation uses to halve the size of its leaves (Section 9).

Remark 8.11 (Scope). • The statement concerns one commitment opened at one point z per proof. Several polynomials at one point are handled by the batched protocol of Section 11.2; several points and composition inside a larger Fiat–Shamir transcript are discussed in Section 12.

- Corollary 8.6 applies to the compiler BCS of [CDHZ25, Construction 11.7], with salted Merkle trees and the challenge derivation described in Section 3.4. Unsalted trees are used only to measure the cost of salting; Section 9 describes the variant the implementation uses.
- The bounds also hold against classical adversaries in the random oracle model, with the same (quantum) extractor.

9 Implementation

We implemented Π_{KF} , compiled with salted Merkle trees and a Fiat–Shamir transcript, as a self-contained Rust library, `kronecker-fri`, in the companion repository; it is published on crates.io as the crate `kronecker-fri`. It has about 1,900 lines of library code, including unit tests, and 850 lines of integration tests, examples and benchmarks (formatted with `rustfmt`); its only dependency is the BLAKE3 hash function, with rayon as an optional dependency for multithreading (the parameter check of Remark 8.8 also uses arbitrary-precision integers, as a development dependency). The code follows the paper: every function names the statement it implements, and the test suite (Section 9.4) checks the algebraic statements listed in Appendix B against direct computations. This section records the concrete choices; Section 10 reports measurements.

Index conventions. The paper numbers bits, variables and rounds from 1; the code numbers them from 0. Bit a_k of the paper is bit $k - 1$ of the code, the coordinate z_k is `z[k-1]`, and the challenge r_j is `rs[j-1]`; the oracle w_j , sent in round $j + 2$ for $j \in C$, is the tree of index $j/k - 1$ when the arity is 2^k (Definition 5.27), and the parameter `fold.log` is k . A multilinear polynomial

is given by its coefficient form $(\alpha_a)_{a < N}$, which is also the coefficient vector of U_f (Definition 4.1), or by its table form, converted with the Möbius transform.

9.1 Fields

Base field. Committed polynomials and the commitment word $\text{Enc}(f)$ (whose fibre string is the implicit instance y of Section 8) live over the Goldilocks field $\mathbb{F}_q$, $q = 2^{64} - 2^{32} + 1$ (Example 6.1). We take $L = \langle \omega \rangle$ with $\omega = 7^{(q-1)/M}$; since 7 generates $\mathbb{F}_q^\times$ (the test suite checks $7^{(q-1)/\ell'} \neq 1$ for the six prime factors ℓ' of $q-1 = 2^{32} \cdot 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$), ω has order exactly M . Multiplication reduces a 128-bit product with $2^{64} \equiv 2^{32} - 1$ and $2^{96} \equiv -1 \pmod{q}$; elements are kept in canonical form. This code is shared with the implementation of [Mkh26].

Challenge fields. The code is generic over the challenge field $\mathbb{F} = \mathbb{F}_{q^e}$ with $e \in \{2, 4\}$. The quadratic extension is $\mathbb{F}_q[u]/(u^2 - 7)$, used for the classical parameters of Section 7.7; the quartic extension is $\mathbb{F}_q[\iota]/(\iota^4 - 7)$, used for the post-quantum parameters of Remark 8.8. Both polynomials are irreducible because 7 is not a square in $\mathbb{F}_q$ and $q \equiv 1 \pmod{4}$ (Remark 8.8); the test suite checks $7^{(q-1)/2} = -1$. The quartic extension is implemented as the tower $\mathbb{F}_{q^2}[\iota]/(\iota^2 - u)$ with $u = \iota^2$, so that a product costs three products in $\mathbb{F}_{q^2}$ (Karatsuba) and an inverse costs one inverse in $\mathbb{F}_{q^2}$; elements are serialised in the basis $1, \iota, \iota^2, \iota^3$, which is the basis used by the map num of Section 8.1. The test suite checks $\iota^4 = 7$, associativity, distributivity and inverses on random elements, and compares the tower product with the schoolbook product modulo $\iota^4 - 7$.

9.2 Encoding, opening and folding

Commit. $\text{Enc}(f)$ is computed as in Section 6.2: the Möbius transform if f is given by its table ($\frac{n}{2}N$ subtractions, Lemma 2.6), zero-padding to length M , and an iterative radix-2 NTT over $\mathbb{F}_q$ in natural order (Fact 6.4). The NTT is generic over the field of the coefficients, with twiddle factors in $\mathbb{F}_q$, and is also used for w_A over $\mathbb{F}$.

Domain layout. Position i of a word on L_j is the point ω_j^i , $\omega_j = \omega^{2^j}$. Since $\omega_j^{M_{j+1}} = -1$, the fibre over position k of L_{j+1} is the pair of positions $\{k, k + M_{j+1}\}$, and folding reads the two halves of an array and writes one array of half size, with no permutation. A query index $i \in \llbracket 0, M-1 \rrbracket$ is the point $\xi = \omega^i$; at level $j \in \llbracket 1, \ell \rrbracket$ its fold check reads the fibre over position $i \bmod M_j$ of L_j . More generally, the 2^g points of $L_{j'}$ above position a of $L_{j'+g}$ are the positions $a + tM_{j'+g}$, $t < 2^g$.

Opening polynomials. The prover computes $G = U_f K_z$ as in the proof of Lemma 4.9(3): n successive multiplications by the binomials $X^{2^{k-1}} + z_k$, each a pass of one multiplication and one addition per coefficient. It reads $v = [X^{N-1}]G$, $A = (0, [X^0]G, \dots, [X^{N-2}]G)$ and $H = ([X^N]G, \dots, [X^{2N-2}]G, 0)$ (proof of Lemma 4.10), and computes w_A by one NTT over $\mathbb{F}$.

Virtual word. After β , the prover computes $\text{ev}_L(K_z)$ by the recursion of Lemma 6.7, and then $h(\xi)$ and $w_0(\xi)$ in one pass over L , carrying the powers ξ , ξ^N and $\xi^{-(N+1)}$ multiplicatively; H itself is used only for the coefficients of U_0 . The verifier computes, at each query point, the powers $\xi^2, \dots, \xi^{2^n} = \xi^N$ (n squarings), $K_{z'}(\xi^2)$ from them, and then $K_z(\pm\xi) = (z_1 \pm \xi)K_{z'}(\xi^2)$ (Lemma 4.4(2)), one inversion $\xi^{-(N+1)} = (\xi^N \cdot \xi)^{-1}$ in $\mathbb{F}_q$, and $h(\pm\xi)$ and $w_0(\pm\xi)$ from the four opened values of y and w_A (Proposition 6.11(3)); for arity 2^k , the same computation runs at the 2^{g_0} opened points of the first group, $g_0 = \min(k, \ell)$, sharing the powers; both use the same function for one value of h .

Fold. Prover and verifier use the same function for one fibre, in line form (Lemma 5.3): with $a = w(\xi)$ and $b = w(-\xi)$, $w_e = (a+b)/2$, $w_o = (a-b)(2\xi)^{-1}$ and $\text{fold}_r(w)(\xi^2) = (w_o - w_e) + r(2w_e - w_o)$; this is the count of 7 operations of Proposition 6.11. The prover reads $(2\xi)^{-1}$ from one table of ω^{-t} , $t < M/2$, shared by all levels. The verifier obtains $(2\omega_{j-1}^k)^{-1} = \frac{1}{2}\omega^{M-k2^{j-1}}$ by one exponentiation in $\mathbb{F}_q$, without inversion. With arity 2^k , the verifier folds the 2^g values of $w_{j'}$ above a point of $L_{j'+g}$, $g = \min(k, \ell - j')$, by $2^g - 1$ calls of this function, level by level (2^{g-1} calls at the first level, then half as many at each next level), which computes the check of Definition 5.27. The final polynomial is $P = \text{pfold}_r^{(\ell)}(U_0)$, computed on coefficients (Lemma 5.5), and the verifier evaluates it by Horner's rule.

Multithreading. An optional feature runs the prover on several threads: the NTT, the leaves and layers of the Merkle trees, the product $U_f K_z$, the word $\text{ev}_L(K_z)$, the batched word w_0 , the folds and the final polynomial. Each is split into independent pieces whose results do not depend on the split: every piece starts from its own powers of ω or ξ , and the salts are read from the salt stream at the position of the leaf. The proofs are therefore identical with and without the feature; the continuous integration checks this by comparing digests of proofs of every protocol built with and without it. Without the feature the same loops run serially, and all measurements of Section 10 except Section 10.8 are single-threaded.

9.3 Commitments and transcript

Salted Merkle trees. Leaves are hashed as $H_{K_0}(\text{data} \parallel \gamma)$ with a salt γ of 32 bytes, and internal nodes as $H_{K_1}(\text{left} \parallel \text{right})$, where H_K is BLAKE3 in keyed mode with key K , $K_0 = 0x00^{32}$ and $K_1 = 0x01^{32}$ separate the two domains, and $\lambda_H = 256$. A node, and a salted leaf of one fibre over $\mathbb{F}_{q^2}$, are 64 bytes: one BLAKE3 compression each. The salts of a tree are the consecutive 32-byte blocks of one BLAKE3 output stream, keyed with a 32-byte seed drawn from the operating system's random number generator for every commitment and every proof, and bound to the label of the tree; the salts are what the compiler of [CDHZ25] requires to be random. Setting the salt length to 0 gives unsalted trees, which we use only to measure the cost of salting (Section 10).

Coset leaves. Every tree has leaves of one fibre each, and the leaves are ordered so that the fibres read by one query at one level are contiguous. Let the committed levels be $0 = j_0 < j_1 < \dots < j_s$ and, for the tree of $w_{j'}$, $j' = j_{t-1}$, let $g = j_t - j_{t-1}$ with $j_s = \ell$ (so $g = k$, except possibly for the last group). A fibre of $w_{j'}$ is a position $\eta = a + t M_{j'+g}$ of $L_{j'+1}$, $a < M_{j'+g}$, $t < 2^{g-1}$, and it is stored at leaf $a 2^{g-1} + t$; this leaf holds $(w_{j'}(\omega_{j'}^\eta), w_{j'}(-\omega_{j'}^\eta))$, the symbol $\text{fib}_{j'}(w_{j'})$ at η with the representative $\omega_{j'}^\eta$ (Section 8.1). The 2^{g-1} fibres above position a of $L_{j'+g}$ are then an aligned group of leaves, opened with one authentication path of length $\log_2 M_{j'} - g$. For arity 2 this is the layout of Remark 8.10: one leaf per fibre, in natural order. Leaves of the commitment tree hold two elements of $\mathbb{F}_q$ (16 bytes), and the other leaves two elements of $\mathbb{F}$; the commitment tree and the tree of w_A have the layout of the first group, of size $g_0 = \min(k, \ell)$, so that a commitment fixes g_0 .

Merkle caps and deduplication. Instead of a root, the proof contains the 2^c nodes at depth c of each tree (its *cap*), with c reduced to the depth of the tree minus $\log_2$ of the group size when the tree is smaller; the verifier recomputes the root from the cap, compares it with the commitment for the commitment tree, and authenticates every group against the cap with a path shortened by c . The transcript absorbs the root, so a cap is a representation of the same commitment message, and

Definition 8.2 and Corollary 8.6 are unchanged. The κ query indices can repeat positions at deep levels; the proof opens each distinct group once, in increasing order, and the verifier checks that the list of opened positions is exactly the sorted set of distinct positions derived from the query indices, and that every cap, path and salt has its prescribed length, so that the proof has a unique form. Both are standard optimisations of FRI implementations and change only the encoding of openings.

Fiat–Shamir. The transcript is a BLAKE3 hash chain. It absorbs the parameters $(n, R, \ell, \kappa, k, c)$, the salt length and e , the commitment root, z and v . In round 1 it absorbs the root of w_A and a round salt, and derives β ; in round 2 it absorbs a round salt and derives r_1 ; in round $j + 1$, $j \in \llbracket 2, \ell \rrbracket$, it absorbs the root of w_{j-1} if $j - 1 \in C$, and a round salt, and derives r_j ; in round $\ell + 2$ it absorbs the coefficients of P and a round salt, and derives the query indices. A challenge in $\mathbb{F}$ is the image under the map ϕ of Section 8.1 of $\lambda_{\text{ch}} = 64(e + 1)$ output bits (192 for $e = 2$, 320 for $e = 4$): the bits are read as an integer, whose first e digits in base q are the coordinates. The κ query indices are κ consecutive blocks of $n + R$ output bits (the map pos).

Relation to the compiler of [CDHZ25]. The implementation has the structure of $\text{BCS}[\Pi_{\text{KF}}^\Sigma]$: salted trees, one round salt per round, challenges derived from the roots and salts of all previous rounds, and the maps ϕ and pos . It departs from Definition 8.2 in one inessential point: the rounds whose message is the dummy symbol send no string , so no tree is built for it. Corollary 8.6 is stated for the compiler of [CDHZ25, Construction 11.7], whose structure the implementation follows.

Proof format. A proof consists of the caps of the commitment tree, of the tree of w_A and of the trees of w_j , $j \in C \setminus \{0\}$, the $\ell + 2$ round salts, the N_ℓ coefficients of P , and the openings: for each distinct position of the first group, 2^{g_0-1} leaves of the commitment tree (2^{g_0} elements of $\mathbb{F}_q$) and of the tree of w_A (2^{g_0} elements of $\mathbb{F}$) with their salts and two paths; and for each committed level j' and each distinct position, 2^g elements of $\mathbb{F}$, 2^{g-1} leaf salts and one path. For arity 2, without caps, and with distinct query positions, this is 2 elements of $\mathbb{F}_q$, 2ℓ elements of $\mathbb{F}$, $\ell + 1$ salts and $\ell + 1$ paths per query point. The positions are recomputed by the verifier and are not part of the proof. Section 10 reports the measured sizes.

9.4 Testing

The test suite checks each component against a direct definition, and the protocol end to end.

- *Arithmetic.* Base-field operations against 128-bit integer arithmetic; the extensions as above; the NTT against Horner evaluation, over $\mathbb{F}_{q^4}$.
- *Section 4.* For $n \leq 8$ and both extensions: the coefficients of K_z against the product of binomials of Definition 4.3 (Lemma 4.4(3)); $K_z(\xi)$ against Horner evaluation; the kernel product against schoolbook multiplication; the extraction identity (Theorem 4.5) against $f(z) = \sum_a \alpha_a z^a$; the Möbius transform against the table; and the split identity (4) with $A(0) = 0$ and $[X^{N-1}]H = 0$ at a random point.
- *Section 5.* For $n \leq 7$ and $\ell = n$: the explicit formula (5) against the line form, the consistency of word and polynomial folds (Proposition 5.4), and the coefficient formula of Lemma 5.5, after every fold.
- *Transcript and trees.* Group openings of every size and position, with and without salts and caps, and rejection of a modified leaf or cap node; canonical challenge coordinates.

- *Completeness.* Honest proofs verify for all $1 \leq n \leq 9$ over $\mathbb{F}_{q^2}$ and $1 \leq n \leq 7$ over $\mathbb{F}_{q^4}$, all $1 \leq \ell \leq n$, rates $\rho \in \{1/4, 1/8, 1/16\}$, arities 2, 4 and 8, no cap or caps of 4 and 16 nodes, with and without salts; and the returned value equals $f(z)$ computed from the table form by $\sum_b f(b) \prod_k (b_k z_k + (1 - b_k)(1 - z_k))$, independently of the Möbius transform and of the kernel product.
- *Virtual word.* For $n \leq 6$ and all $\xi \in L$: the virtual word of honest openings equals $H(\xi)$ (Lemma 6.6(3)), and $\text{ev}_L(K_z)$ computed by Lemma 6.7 equals $K_z(\xi)$.
- *Tampering.* Proofs are rejected after changing the claimed value, the point, the commitment, an opened value of y , w_A or w_1 , a coefficient of P , a root, a round salt, a leaf salt, an authentication path or a cap node, or the length of a salt, a path or a cap, for arities 2 to 16, and when a proof for another polynomial is checked against the original commitment.
- *Two cheating provers.* Both commit honestly and claim $v + 1$, and both are rejected by a fold check in every tested configuration (arities 2 to 16, with and without caps). The first sends the honest w_A ; its virtual word is $\text{ev}_L(H) - \text{ev}_L(X^{-1})$, far from $\mathcal{C}_0$ (Remark 6.9(1)). The second sends w_A for $A - X^N$, so that its virtual word is exactly $\text{ev}_L(H)$; now w_A has degree N and the batched word is far from $\mathcal{C}_0$. This is the situation of Remark 4.13: the degree bound N of the proximity test is what prevents A from absorbing the error in v .

Independently, `examples/small_field_checks.rs` checks the extraction identity, the split lemma, the identity (4) at every point of L for honest openings together with the folds, and the root bound $2N$ of Lemma 4.12 for a wrong value, over $\mathbb{F}_{257}$, with $1 \leq n \leq 5$ and 20 random instances each; and `examples/pq_params.rs` evaluates the bound of Remark 8.8 exactly.

10 Experiments

This section measures the implementation of Section 9. The goals are specific: to compare Kronecker-FRI with the kernel-basis scheme of [Mkh26] and with a coefficient-form baseline at identical engineering, to measure the effect of folding arity and Merkle caps, to locate where the prover’s time goes, and to quantify salting and the number of folding rounds. All codes are single-threaded, except in Section 10.8, and use no SIMD and no hash-specific batching, so that the comparison isolates the protocols.

10.1 Setup

Machine. All runs use one core of an Intel Xeon processor at 2.10 GHz (cloud virtual machine, 2 vCPUs, 7 GB RAM), Rust 1.95.0, release profile with fat LTO and one codegen unit, and no `target-cpu` flag. All numbers of this section come from one session on one machine; the absolute timings of [Mkh26] were measured on another machine. Times are medians over 11 runs for $n \leq 18$, 5 for $n = 20$ and 3 for $n = 22$ and for Tables 4, 9 and 10; verification times are medians over 21 runs. The machine is a shared virtual machine, and timings of identical work vary by up to about 10–20% between runs. Except in Section 10.7, the measurements were recorded with an earlier Merkle format of both implementations compared, which separated leaves and nodes by one-byte prefixes, $H(0x00 \parallel \cdot)$ and $H(0x01 \parallel \cdot)$, and derived each salt by its own keyed BLAKE3 call; Section 10.7 measures the format of Section 9.3.

n	Commit (ms)			Open (ms)			Verify (ms)			Proof (KiB)	
	KF-2	KB	coeff.	KF-2	KB	coeff.	KF-2	KB	coeff.	KF-2	KB, coeff.
12	2.6	2.4	2.2	6.9	2.6	2.5	2.5	2.2	2.2	392	387
14	9.2	9.4	9.0	25.7	10.2	10.1	3.2	2.9	3.0	545	531
16	36.1	41.3	41.0	114.0	39.8	40.6	4.3	4.5	4.7	719	693
18	181.6	167.7	173.4	481.7	154.2	157.8	5.2	4.7	4.6	909	873
20	729.5	769.6	740.3	2138.0	710.9	596.6	6.3	5.9	5.7	1117	1072
22	3294.2	3291.9	3355.5	11285.0	3265.2	3020.9	6.7	6.8	7.2	1344	1290

Table 3: Kronecker-FRI with arity 2 (KF-2), KBFold (KB) [Mkh26] and the coefficient-form baseline, single-threaded, $\rho = 1/4$, $\ell = n - 4$, $\kappa = 148$, $\mathbb{F} = \mathbb{F}_{q^2}$, unsalted trees. KBFold and the baseline have proofs of identical shape.

Parameters. Unless stated otherwise: $\rho = 1/4$ ($R = 2$), $\kappa = 148$, the quadratic extension $\mathbb{F} = \mathbb{F}_{q^2}$, and unsalted Merkle trees. We use two configurations of Kronecker-FRI:

- *KF-2*: arity 2, no caps, $\ell = n - 4$ (final polynomial of 16 coefficients), the configuration of the schemes of [Mkh26], for the comparison at identical engineering;
- *KF-8*: arity 8, caps of 128 nodes, $\ell = n - 8$ (final polynomial of 256 coefficients), the configuration we recommend (Sections 10.3 and 10.9).

Both deduplicate openings (Section 9.3). By Theorem 7.8 with $\delta = \delta^* = 3/8$, the query term is $(5/8)^{148} < 2^{-100}$, and the field term is below 2^{-100} for all sizes tested ($M \leq 2^{24}$; Table 2). Polynomials have uniformly random coefficients in $\mathbb{F}_q$, and z is uniformly random in $\mathbb{F}^n$.

Schemes compared. We compare three schemes that share the field arithmetic, the NTT, the BLAKE3 Merkle trees with fibre leaves and the transcript, and differ only in how an evaluation is proved.

- *Kronecker-FRI*: Π_{KF} as implemented in Section 9.
- *KBFold*: the evaluation protocol of [Mkh26], from its reference implementation: the table f is encoded in the Boolean-kernel basis, and a sumcheck of ℓ rounds runs alongside the kernel folds.
- *Coefficient-form baseline*: the second mode of the same implementation [Mkh26, §8], which encodes the coefficient form of f and folds with the classical FRI fold, alongside the same sumcheck. This is the coefficient-form folding of Reed–Solomon instantiations of BaseFold [ZCF24]; it is *not* the BaseFold code base.

The implementation of [Mkh26] uses unsalted trees, arity 2, no caps and no deduplication, so the comparison uses unsalted trees for Kronecker-FRI too; salting is measured separately in Section 10.6. All three schemes are run with the same ρ , ℓ and κ , which give the same query term; for the two sumcheck-based schemes the field term is $\sum_j (M_j + 2)/|\mathbb{F}|$ [Mkh26], and for Kronecker-FRI it is $\varepsilon_{\text{KF}} - (1 - \delta)^\kappa < 3M/|\mathbb{F}|$, both below 2^{-100} here.

10.2 Comparison at identical engineering

Table 3 compares the three schemes in the configuration of [Mkh26], for $n = 12, \dots, 22$.

arity	$\ell = 16$				$\ell = 12$			
	no cap		cap 2^7		no cap		cap 2^7	
	KiB	ms	KiB	ms	KiB	ms	KiB	ms
2	1117	6.3	681	5.4	1008	5.5	648	4.5
4	613	4.1	388	3.1	570	4.0	371	3.1
8	478	4.2	315	3.5	440	4.0	299	3.4
16	424	5.3	308	3.9	402	5.0	295	4.6

Table 4: Proof size (KiB) and verification time (ms) of Kronecker-FRI at $n = 20$, $\rho = 1/4$, $\kappa = 148$, $\mathbb{F} = \mathbb{F}_{q^2}$, unsalted, for arities 2^k , $k \leq 4$ (Definition 5.27), without caps and with caps of 128 nodes per tree. The soundness error is the same in every cell (Proposition 5.28).

Commit. The three schemes perform the same work up to the transform that precedes the NTT: the Boolean-kernel transform for KBFold, the Möbius transform for the baseline, and nothing for Kronecker-FRI, which we run from the coefficient form. From a table, Kronecker-FRI also performs the Möbius transform, which takes 12 ms at $n = 20$ and 57 ms at $n = 22$ in the same session (the transform column of the baseline), below 2% of the commitment. For $n \geq 14$ the commitment times agree within 15%, in both directions; at $n = 12$ the timings, of 2 to 3 ms, are dominated by noise.

Verify. Verification times agree within the variability of the machine, between 2.2 and 7.2 ms: the three verifiers are dominated by the authentication paths. Kronecker-FRI replaces the sumcheck checks by the computation of $K_z(\pm\xi)$ and $h(\pm\xi)$ at each query point, $O(n)$ field operations per point (Proposition 6.11(3)), which is not visible at this resolution.

Proof size. Kronecker-FRI opens one more tree, that of w_A , at every query point, while the ℓ sumcheck messages of the other schemes are negligible. With deduplication, which saves 56 to 59 KiB here, Kronecker-FRI proofs are within 1.4–4.2% of those of KBFold (5 to 54 KiB larger); without it, the difference is the κ openings of w_A , $\kappa(32(\log_2 M - 1) + 32)$ bytes minus the sumcheck messages (101 KiB at $n = 20$).

Open. The opening of Kronecker-FRI with arity 2 takes 2.5 to 3.5 times the opening time of KBFold, since the prover commits to one more word of length M over $\mathbb{F}$; Section 10.5 measures each step, and higher arity reduces it (Section 10.4).

10.3 Folding arity and Merkle caps

Table 4 varies the arity and the cap at $n = 20$; by Proposition 5.28, every cell has the soundness error of Theorem 7.8. Arity 2^k divides the number of trees of folded words by k , and the paths opened per query with them, while each opened group grows to 2^{k-1} leaves; caps of 2^7 nodes remove the 7 top levels of every path, which the 148 queries share. Together they reduce the proof by a factor of 3.4 to 3.6 with respect to arity 2 without caps: from 1117 to 315 KiB at $\ell = 16$ and from 1008 to 299 KiB at $\ell = 12$. With caps, arities 8 and 16 give the smallest proofs, within 3% of each other, and arity 8 verifies faster than arity 16, which folds 15 values per committed word and query; arity 4 verifies fastest, with proofs 23–24% larger. Opening time is lower for larger arities (about 2.2 s for arity 2, 1.4–1.8 s for arities 8 and 16), since fewer trees are built; commitment time does not depend on it. These optimisations are generic: they apply to every FRI-type test, including

n	Commit (ms)		Open (ms)		Verify (ms)		Proof (KiB)	
	unsalted	salted	unsalted	salted	unsalted	salted	unsalted	salted
12	2.3	3.1	5.0	6.7	1.6	1.7	95	136
14	8.9	11.8	21.0	26.3	1.9	2.0	130	183
16	40.5	49.2	91.6	126.4	2.4	2.5	184	247
18	160.2	222.8	387.8	488.5	3.0	2.5	248	324
20	730.3	910.1	1689.8	2044.3	3.4	3.6	299	390
22	3175.5	4094.0	9282.5	10894.0	4.5	4.2	372	474

Table 5: Kronecker-FRI in the recommended configuration KF-8 (arity 8, caps of 128 nodes, $\ell = n - 8$), $\rho = 1/4$, $\kappa = 148$, $\mathbb{F} = \mathbb{F}_{q^2}$, without and with 32-byte salts, measured in the same session as Table 3.

n	$U_f K_z$	NTT of w_A	tree of w_A	h, w_0	folds, trees	total open
16	12.2	16.8	31.8	15.8	10.6	91.6
18	63.9	79.6	128.5	66.9	43.4	387.8
20	327.0	360.8	565.1	324.7	214.9	1689.8
22	1893.0	2664.8	2314.8	1263.6	906.7	9282.5

Table 6: Components of the KF-8 opening (ms), measured separately with the field, NTT and hash code of the prover (trees in natural leaf order, of the same cost), and the total opening time of Table 5.

those of KBFold and of the baseline, whose implementation does not use them; the comparison at identical engineering is that of Table 3.

10.4 Scaling of the recommended configuration

Table 5 reports KF-8 for $n = 12, \dots, 22$. Proofs are 95 to 372 KiB unsalted, 3.6 to 4.2 times smaller than those of KF-2, and grow by 35 to 73 KiB per two variables. Verification takes 1.6 to 4.5 ms, 1.5 to 1.8 times faster than KF-2. Opening is 1.2 to 1.4 times faster than KF-2, since the folded words are committed at one level in three, and takes 1.9 to 2.8 times the opening time of KBFold with arity 2. Commitment is unchanged.

10.5 Where the prover’s time goes

Table 6 splits the KF-8 opening into its steps; the components add up to the measured total within the variability of the machine (-5% to $+6\%$). The last column, the folds and the trees of the committed levels, is the work of the folding test itself; with arity 8 it is between 9.8% and 12.7% of the opening. The other four columns are the commitment to the opening polynomial:

- the product $U_f K_z$, with $2nN$ multiplications in $\mathbb{F}$ (Lemma 4.9(3));
- the NTT of A over $\mathbb{F}$, which costs $e = 2$ base-field NTTs of size M (Fact 6.4);
- the Merkle tree of w_A , with $M/2$ leaves of two elements of $\mathbb{F}$;
- the words $\text{ev}_L(K_z)$, h and w_0 , with $O(M)$ operations in $\mathbb{F}$ (Proposition 6.11(2)).

Each costs between 0.3 and 0.85 of a commitment, and together 1.9 to 2.6 commitments. In exchange, the verifier runs no sumcheck and the soundness argument has a single algebraic step

n	format	commit (ms)	open (ms)	verify (ms)	proof (KiB)
16	prefixes, salt per leaf	69.9	155.0	3.57	249.8
	prefixes, salt stream	49.9	133.4	3.58	247.9
	keys, salt stream	38.7	111.9	3.09	247.9
18	prefixes, salt per leaf	288.1	671.6	4.27	325.0
	prefixes, salt stream	221.9	589.4	4.33	326.5
	keys, salt stream	163.6	496.1	3.63	325.6
20	prefixes, salt per leaf	1268.3	3248.1	5.39	391.7
	prefixes, salt stream	1050.4	2945.0	5.38	390.7
	keys, salt stream	835.3	2576.8	4.11	389.5

Table 7: Kronecker-FRI in the configuration KF-8 with 32-byte salts, $\rho = 1/4$, $\kappa = 148$, $\mathbb{F} = \mathbb{F}_{q^2}$, for three Merkle formats: one-byte prefixes and one keyed BLAKE3 call per salt (the format of the other tables of this section), one-byte prefixes and one salt stream per tree, and the format of Section 9.3 (keys and one salt stream). One core of an Intel Xeon at 2.80 GHz, a different machine from the other tables; the three versions interleaved in one session, means of two medians. Proof sizes differ only through the queried positions. Source: `bench/merkle_format.csv`.

(Lemma 7.4). The asymptotic cost of the opening is $O(M \log M)$; the product, the NTT and the tree of w_A are independent passes that parallelise like the commitment (Section 10.8).

10.6 Salting

Salting every leaf with 32 bytes (Table 5) costs 21–39% in commit time and 17–38% in open time (in the recorded version the salts were derived by one keyed BLAKE3 call per leaf; Section 10.7 measures the current derivation), and adds 32 bytes per opened leaf and per round to the proof, 41 to 102 KiB; it does not change the verification time beyond the variability of the machine. The post-quantum statement (Corollary 8.6) requires salts; Table 10 measures the salted parameter sets. The commitment is unaffected by the choice of $\mathbb{F}$, since $\text{Enc}(f)$ is computed over $\mathbb{F}_q$.

10.7 Merkle hashing

Table 7 compares the Merkle formats on one machine and in one session. Reading the salts of a tree from one BLAKE3 output stream instead of one keyed call per leaf, and separating the domains by keys, so that a node and a salted leaf are 64 bytes (one compression), reduce the commit time by 34–45%, the open time by 21–28% and the verification time by 13–24%; the proofs keep their shape and size.

10.8 Multithreading

Table 8 runs the prover on the two vCPUs of the machine of Table 7. The commitment is 1.3 to 1.6 times faster and the opening 1.4 to 1.5 times faster, the larger gains at $n = 20$, where the passes are long enough to amortise the synchronisation; the verifier is unchanged, since it runs on one thread. Every pass of the prover is split into independent pieces, so more cores reduce the prover time further.

n	prover	commit (ms)	open (ms)	verify (ms)
16	one thread	40.1	108.4	3.03
	two threads	30.2	78.9	3.04
20	one thread	920.2	2544.6	4.19
	two threads	580.5	1709.2	4.32

Table 8: Kronecker-FRI in the configuration KF-8 with 32-byte salts, $\rho = 1/4$, $\kappa = 148$, $\mathbb{F} = \mathbb{F}_{q^2}$, with the prover on one thread and on two threads (Section 9.2, multithreading); the proofs are identical. Intel Xeon at 2.80 GHz with 2 vCPUs, the machine of Table 7; the two builds interleaved in one session, means of two medians. Source: `bench/parallel.csv`.

ℓ	N_ℓ	open (ms)	verify (ms)	proof (KiB)
8	4096	1808	7.7	320
10	1024	1781	4.4	309
12	256	1799	3.6	300
14	64	1791	2.7	311
16	16	1772	3.4	314
18	4	1763	3.5	312
20	1	1749	3.6	313

Table 9: Effect of the number of folding rounds ℓ (Remark 6.12) for arity 8 and caps of 128 nodes, $n = 20$, $\rho = 1/4$, $\kappa = 148$, $\mathbb{F} = \mathbb{F}_{q^2}$, unsalted.

10.9 The number of folding rounds

Table 9 varies ℓ at $n = 20$ with arity 8 and caps. Stopping earlier removes committed levels but adds $16N_\ell$ bytes for P and $O(N_\ell)$ operations per query for its evaluation; with caps the levels are cheap, so proof size is nearly flat, between 300 and 320 KiB, with its minimum at $\ell = 12$. Verification is fastest at $\ell = 14$ (2.7 ms) and slowest at $\ell = 8$, where the κ evaluations of P take over. Opening time does not depend on ℓ beyond the variability of the machine, since the first folds dominate it. We use $\ell = n - 8$, the smallest proofs at $n = 20$.

10.10 Summary

At identical engineering, Kronecker-FRI commits and verifies as fast as KBFold and as the coefficient-form baseline, with proofs within 5% of theirs, and its opening pays for the commitment to A . With arity 8 and Merkle caps, which Proposition 5.28 covers without changing the soundness error, proofs are 95 to 372 KiB and verification 1.6 to 4.5 ms for $n = 12, \dots, 22$, unsalted, with interactive soundness error below 2^{-99} , and opening is 1.2 to 1.4 times faster than with arity 2. The protocol has no sumcheck: its verifier is an FRI verifier with one local computation per query, and its soundness reduces to the folding test and one application of correlated agreement for curves of degree 2. Section 11 gives parameter sets and compares with other schemes.

11 Practical Use, Parameters and Comparison

This section places Kronecker-FRI in its intended setting, extends it to the opening of several polynomials at one point, recommends parameter sets, and compares it with existing multilinear polynomial commitment schemes.

11.1 Where the scheme fits

Sumcheck-based SNARKs. Multilinear polynomial commitments are the cryptographic component of SNARKs built from the sumcheck protocol [LFKN92], such as Spartan [Set20] and HyperPlonk [CBBZ23]. In these systems the prover commits to multilinear polynomials, typically given by their tables on B_n , and the sumcheck reduces the statement to claims $f_i(z) = v_i$ at one random point $z \in \mathbb{F}^n$ chosen by the verifier; claims at several points are reduced to one point by a further sumcheck [CBBZ23]. This is exactly the setting of Π_{KF} and of its batched version below: polynomials committed over the base field $\mathbb{F}_q$ (Lemma 6.3(3)), and one evaluation point in the extension $\mathbb{F}$. A table is converted to the coefficient form of f by the Möbius transform before the NTT; this costs less than $1/(3 \cdot 2^R)$ of the commitment’s arithmetic (Section 6.2), and below 2% of the commitment time in the measurements of Section 10.2.

Reuse of univariate FRI. Kronecker-FRI is a univariate low-degree test in disguise: the prover commits to univariate polynomials U_f and A of degree less than N , the verifier forms the batched word $w + \beta w_A + \beta^2 h$ pointwise, and the rest is FRI with the kernel fold (or, equivalently, the classical fold, Remark 5.6). The virtual word h is a pointwise quotient of the same kind as those used in FRI-based STARKs [BSCR⁺19]. A code base that already has an FRI prover and verifier over a smooth multiplicative domain therefore obtains a multilinear commitment by adding the product $U_f K_z$ on the prover side and the $O(n)$ computation of $h(\pm\xi)$ per query on the verifier side, with no sumcheck and no new proximity test.

Fields. The scheme needs a smooth domain of order $M = 2^{n+R}$ in $\mathbb{F}_q^\times$, that is, $2^{n+R} \mid q - 1$ (Lemma 2.8). Besides Goldilocks ($2^{32} \mid q - 1$), this holds for the 31-bit primes BabyBear, $q = 15 \cdot 2^{27} + 1$, and KoalaBear, $q = 127 \cdot 2^{24} + 1$, for $n + R \leq 27$ and $n + R \leq 24$ respectively; the extension degree e must then make the field term $3M/q^e$ negligible, which requires $e \geq 5$ for 100 bits at $M = 2^{24}$. For the Mersenne prime $2^{31} - 1$, circle-domain analogues of FRI [HLP24] play this role, and binary fields are served by the schemes of Section 11.4. The implementation uses Goldilocks.

11.2 Opening several polynomials at one point

A SNARK prover typically opens several committed polynomials at the same point. Opening them one by one costs one run of Π_{KF} each; the standard random linear combination reduces this to one run. We state and prove it for the interactive protocol.

Definition 11.1 (Batched evaluation protocol Π_{KF}^m). Let $m \geq 2$. The instance is

$$(w^{(1)}, \dots, w^{(m)}; z, v_1, \dots, v_m),$$

with oracles $w^{(i)} \in \mathbb{F}^L$, a point $z \in \mathbb{F}^n$ and values $v_i \in \mathbb{F}$; honestly $w^{(i)} = \text{Enc}(f_i)$ and $v_i = f_i(z)$. The protocol has $\ell + 3$ rounds. In round 1 the prover sends nothing and the verifier sends $\gamma \in \mathbb{F}$; put

$$w_\gamma = \sum_{i=1}^m \gamma^{i-1} w^{(i)}, \quad v_\gamma = \sum_{i=1}^m \gamma^{i-1} v_i.$$

Rounds 2 to $\ell + 3$ are rounds 1 to $\ell + 2$ of Π_{KF} on the instance $(w_\gamma; z, v_\gamma)$, where the verifier computes each value $w_\gamma(\pm\xi)$ from the $2m$ queries $w^{(i)}(\pm\xi)$. The honest prover runs the honest prover of Π_{KF} for $f_\gamma = \sum_i \gamma^{i-1} f_i$.

Completeness follows from Theorem 6.10, since $w_\gamma = \text{Enc}(f_\gamma)$ and $v_\gamma = f_\gamma(z)$ by linearity of Enc and of evaluation. The relation asks that the commitments be *jointly* close to codewords, on one common set of fibres:

$$\mathcal{R}_{\text{KF}}^{\delta, m} = \left\{ ((w^{(i)})_i; z, (v_i)_i, (f_i)_i) : \exists S \subseteq L \text{ fibre-closed with } |S| \geq (1 - \delta)M \right. \\ \left. \text{such that } w^{(i)} = \text{Enc}(f_i) \text{ on } S \text{ and } f_i(z) = v_i \text{ for all } i \right\}.$$

A witness satisfies $\Delta_0^{\text{fb}}(w^{(i)}, \text{Enc}(f_i)) \leq \delta$ for every i , so it is unique when it exists (Lemma 6.3(2)); for $m = 1$ the relation is $\mathcal{R}_{\text{KF}}^\delta$.

Theorem 11.2 (Soundness of batched openings). *Let $0 < \delta \leq \delta^*$. If $((w^{(i)})_i; z, (v_i)_i)$ has no witness in $\mathcal{R}_{\text{KF}}^{\delta, m}$, then every prover makes the verifier of Π_{KF}^m accept with probability at most*

$$\frac{(m-1)M}{|\mathbb{F}|} + \varepsilon_{\text{KF}}.$$

Proof. By Lemma 3.4 it suffices to treat a deterministic prover. Its messages from round 2 on are functions of γ and of the later challenges, so for each fixed γ rounds 2 to $\ell + 3$ are an execution of Π_{KF} on $(w_\gamma; z, v_\gamma)$ by a deterministic prover. Let Γ be the set of $\gamma \in \mathbb{F}$ such that $(w_\gamma; z, v_\gamma)$ has a witness in $\mathcal{R}_{\text{KF}}^\delta$ (Definition 7.1). For $\gamma \notin \Gamma$, Theorem 7.8 bounds the acceptance probability by ε_{KF} ; it remains to show $|\Gamma| \leq (m-1)M$.

Suppose $|\Gamma| > (m-1)M$. For $\gamma \in \Gamma$, let g_γ be the witness; then $\Delta(w_\gamma, \text{Enc}(g_\gamma)) \leq \Delta_0^{\text{fb}}(w_\gamma, \text{Enc}(g_\gamma)) \leq \delta$ (Lemma 5.11(1)), so $\Delta(w_\gamma, \mathcal{C}_0) \leq \delta$.

Correlated agreement. The words w_γ form the curve generated by $(w^{(1)}, \dots, w^{(m)})$, of degree $t = m-1 \geq 1$, and more than tM of its members are within distance δ of $\mathcal{C}_0$. By Theorem 2.16, there are codewords $\hat{c}_1, \dots, \hat{c}_m \in \mathcal{C}_0$ and a set of at least $(1-\delta)M$ points on which $w^{(i)} = \hat{c}_i$ for every i . Let Agr be the set of all points where these m equalities hold, and $\hat{f}_i \in \mathcal{L}_n(\mathbb{F})$ with $\text{Enc}(\hat{f}_i) = \hat{c}_i$ (Lemma 6.3(1)).

The combination is the decoded codeword. Let $\gamma \in \Gamma$. The codeword $\sum_i \gamma^{i-1} \hat{c}_i = \text{Enc}(\sum_i \gamma^{i-1} \hat{f}_i)$ agrees with w_γ on Agr, so it is within distance δ of w_γ , and so is $\text{Enc}(g_\gamma)$. By Lemma 2.12(3) they are equal, and by Lemma 6.3(1), $g_\gamma = \sum_i \gamma^{i-1} \hat{f}_i$.

The values. Hence $\sum_{i=1}^m \gamma^{i-1} (\hat{f}_i(z) - v_i) = g_\gamma(z) - v_\gamma = 0$ for every $\gamma \in \Gamma$. The polynomial $\sum_{i=1}^m Z^{i-1} (\hat{f}_i(z) - v_i) \in \mathbb{F}[Z]$ has degree at most $m-1$ and more than $(m-1)M \geq m-1$ roots, so it is zero (Lemma 2.2), and $\hat{f}_i(z) = v_i$ for all i .

A common set of fibres. For $\xi \in L \setminus \text{Agr}$, the polynomial $d_\xi(Z) = \sum_{i=1}^m Z^{i-1} (w^{(i)} - \hat{c}_i)(\xi) \in \mathbb{F}[Z]$ is nonzero, of degree at most $m-1$; let Bad be the set of all its roots, over all such ξ . Then $|\text{Bad}| \leq (m-1)|L \setminus \text{Agr}| \leq (m-1)\delta M < |\Gamma|$, so there is $\gamma \in \Gamma \setminus \text{Bad}$. For this γ , w_γ and $\text{Enc}(g_\gamma) = \sum_i \gamma^{i-1} \hat{c}_i$ agree exactly on Agr: at $\xi \notin \text{Agr}$ they differ by $d_\xi(\gamma) \neq 0$. Since $\Delta_0^{\text{fb}}(w_\gamma, \text{Enc}(g_\gamma)) \leq \delta$, they agree on a fibre-closed set S with $|S| \geq (1-\delta)M$ (Lemma 5.11(2)), and $S \subseteq \text{Agr}$. So every $w^{(i)}$ agrees with $\text{Enc}(\hat{f}_i)$ on S , and $(\hat{f}_i)_i$ is a witness, a contradiction. $\square$

Corollary 11.3 (Knowledge and binding for batched openings). *Let Ext^m output, for each i , the $\hat{f}_i \in \mathcal{L}_n(\mathbb{F})$ such that $\text{Enc}(\hat{f}_i)$ is the codeword of $\mathcal{C}_0$ within fibre distance δ^* of $w^{(i)}$, or 0 if there is none; it runs in time polynomial in mM (Remark 5.14). If a prover makes the verifier of Π_{KF}^m accept with probability greater than $(m-1)M/|\mathbb{F}| + \varepsilon_{\text{KF}}$, then $(\hat{f}_i)_i$ is a witness. In particular Π_{KF}^m is evaluation binding for each of the m commitments, with the same error.*

Proof. By Theorem 11.2, a witness $(f_i)_i$ exists. Each $\text{Enc}(f_i)$ is within fibre distance $\delta \leq \delta^*$ of $w^{(i)}$, so it is the codeword found by Ext^m (Lemma 5.12), and $\hat{f}_i = f_i$. Binding follows as in Lemma 3.11, since the witness is determined by the commitments. $\square$

target	$\mathbb{F}$	ρ	κ	field term	commit (s)	open (s)	verify (ms)	proof (KiB)
100 bits	$\mathbb{F}_{q^2}$	1/4	148	$2^{-104.4}$	0.96	2.18	3.6	390
100 bits	$\mathbb{F}_{q^2}$	1/8	121	$2^{-103.4}$	1.83	4.45	3.2	346
128 bits	$\mathbb{F}_{q^4}$	1/4	189	$2^{-232.4}$	0.96	3.96	9.0	586
post-quantum	$\mathbb{F}_{q^4}$	1/4	248	$2^{-232.4}$	0.92	3.95	9.5	755

Table 10: Recommended parameter sets, with measurements at $n = 20$ in the configuration KF-8 (arity 8, caps of 128 nodes, $\ell = n - 8$), salted trees, on the machine of Section 10. The field term is the bound $3M/|\mathbb{F}|$ of Section 7.7; the query term $(1 - \delta^*)^\kappa$ is below 2^{-100} in the first two rows, 2^{-128} in the third and 2^{-168} in the fourth. The last row is a post-quantum level in the sense of Corollary 8.6, not a bit-security level comparable to the others.

Remark 11.4 (Costs and scope of batching). Compared with one run of Π_{KF} , the batched protocol has one more round (the challenge γ); the prover computes f_γ (mN operations) and w_γ (mM operations), and the verifier makes $2m$ instead of 2 queries to the commitments per query point, that is, m authentication paths instead of one in the compiled scheme; all other costs are those of a single opening. The batched protocol is stated for the interactive setting. Its relation is fibre-based, like that of Section 8, so the erasure analysis of Section 8.2 applies to it in the same way; its post-quantum statement and its implementation are part of the roadmap (Section 12).

11.3 Recommended parameters

Table 10 gives four parameter sets, with $\delta = \delta^* = (1 - \rho)/2$ and κ from Table 1.

- 100 *bits*, $\rho = 1/4$ is the default: both terms of ε_{KF} are below 2^{-100} , so $\varepsilon_{\text{KF}} < 2^{-99}$, for $M \leq 2^{26}$, that is, $n \leq 24$.
- 100 *bits*, $\rho = 1/8$ trades a prover about 2 times slower for proofs 11% smaller and a slightly faster verifier; it is preferable when proofs are stored or sent many times.
- 128 *bits* needs a larger field, since the field term over $\mathbb{F}_{q^2}$ is only about 2^{-104} at this size; with $\mathbb{F}_{q^4}$ it is negligible, and $\kappa = 189$.
- *Post-quantum* is the set of Remark 8.8, for which Corollary 8.6 gives $\varepsilon_{\text{ext}}(2^{64}) < 2^{-31.8}$ against quantum adversaries making 2^{64} oracle queries, for one opening, for the compiler of [CDHZ25] (Remark 8.11).

The first three rows are levels of the interactive protocol (Theorem 7.8). For the compiled scheme, the loss of the Fiat–Shamir transformation must be accounted for separately (Remark 8.9). We choose arity 8 with caps of 128 nodes, whose proofs are within 3% of the smallest and which verifies faster than arity 16 (Table 4), and $\ell = n - 8$, which gives the smallest proofs at $n = 20$ (Table 9); none of these choices changes the soundness error (Proposition 5.28).

11.4 Comparison with existing schemes

Table 11 lists representative multilinear polynomial commitments. Timings are compared only for schemes run in the same environment (Section 10), since published timings use different machines, fields, codes and security targets.

scheme	setup, assumption	proof	opening prover	evaluation via
Gemini [BCHO22]	SRS, pairings	$O(\log N)$	$O(N)$ group ops	univariate KZG
Zeromorph [KT24]	SRS, pairings	$O(\log N)$	$O(N)$ group ops	univariate KZG
Mercury [EG25]	SRS, pairings	$O(1)$	$O(N)$ field, $O(N)$ scalar mults	univariate KZG
Samaritan [GPS25]	SRS, pairings	$O(1)$	$O(N)$	univariate KZG
Brakedown [GLS ⁺ 23]	transparent, hashes	$O_\lambda(\sqrt{N})$	$O(N)$ field ops	linear-time code
BaseFold [ZCF24]	transparent, hashes	$O_\lambda(\log^2 N)$	$O(N \log N)$	sumcheck + folding
DeepFold [GLH ⁺ 25]	transparent, hashes	$O_\lambda(\log^2 N)$	$O(N \log N)$	sumcheck + folding
FRI-Binius [DP24]	transparent, hashes	polylog	$O(N \log N)$	sumcheck + folding
WHIR [ACFY24]	transparent, hashes	polylog	$\tilde{O}(N)$	sumcheck + constr. RS
KBFold [Mkh26]	transparent, hashes	$O_\lambda(\log^2 N)$	$O(N \log N)$	sumcheck + kernel folding
Kronecker-FRI	transparent, hashes	$O_\lambda(\log^2 N)$	$O(N \log N)$	extraction + FRI

Table 11: Multilinear polynomial commitment schemes; SRS is a universal structured reference string (trusted setup). Proof sizes count group elements for the pairing-based schemes, and field elements and hashes for the hash-based ones, with O_λ hiding the dependence on the security parameter through the number of queries. The entries for other schemes are as stated in the cited papers.

Pairing-based schemes. Gemini, Zeromorph, Mercury and Samaritan reduce a multilinear evaluation to univariate KZG commitments, and reach logarithmic or constant proof size. Mercury is the closest in spirit to Kronecker-FRI: it also expresses $f(z)$ as a coefficient of a product with a tensor-structured polynomial (Section 4.6). These schemes need a universal structured reference string and pairing-friendly curves, and are not post-quantum. Kronecker-FRI is transparent, uses only a hash function, and has the post-quantum guarantee of Corollary 8.6, with proofs of a few hundred kilobytes (390 KiB at $n = 20$ and 100 bits, salted).

Hash-based schemes. Among hash-based schemes, Brakedown has a linear-time prover and proofs of size $O(\sqrt{N})$; the folding-based schemes have polylogarithmic proofs. The closest comparison is with the schemes that combine a sumcheck with Reed–Solomon folding, which we measured at identical engineering in Section 10.2: Kronecker-FRI commits and verifies at the same speed as KBFold and the coefficient-form (BaseFold-style) baseline, with proofs within 5% of theirs, and its opening additionally commits to A . With arity 8 and caps, Kronecker-FRI proofs are 95 to 372 KiB for $n = 12, \dots, 22$ and verification takes 1.6 to 4.5 ms, unsalted, with interactive soundness error below 2^{-99} (Table 5); salting adds 41 to 102 KiB; these optimisations apply to the folding phase of the sumcheck-based schemes as well. DeepFold and WHIR reduce proof size and verification time substantially below these schemes by working in the list-decoding regime, with fewer queries; WHIR reports 63 KiB proofs and $360 \mu\text{s}$ verification for degree 2^{22} at 100 bits [ACFY24]; the main theorem of [ACFY24] (their Theorem 1) assumes a conjecture on correlated agreement (their Conjecture 1), and the authors note that parameters provable without it are less efficient. Part of the difference is the number of queries: our analysis, like that of KBFold, is confined to the unique-decoding regime (Remark 7.14), which forces $\kappa = 148$ at $\rho = 1/4$; the other part is that WHIR’s proximity test is itself more query-efficient than FRI [ACFY24].

Positioning. Kronecker-FRI is designed around a simple verifier and a short soundness argument.

- The verifier is an FRI verifier plus an $O(n)$ local computation per query; there is no sumcheck, and no round of the protocol depends on the multilinear structure except the first.

- The soundness proof has one algebraic step, the batching lemma, on top of the folding test (Section 7); the folding test itself is independent of the evaluation claim.
- The scheme is built from components that already exist in FRI-based systems (Section 11.1).
- Standard FRI optimisations, higher folding arity and Merkle caps, apply without changing the soundness error (Proposition 5.28), and give proofs of 299 KiB unsalted and 390 KiB salted, and verification in 3.4 to 3.6 ms, at $n = 20$ (Table 5).
- Its interactive soundness is fully proved in the unique-decoding regime, without conjectures, from the correlated-agreement theorem of [BSCI⁺23]; its post-quantum guarantee is proved for one opening from the quoted theorem of [CDHZ25] (Corollary 8.6), for their compiler.

The natural next step is a list-decoding analysis of the batching lemma, which would reduce κ and bring proof sizes closer to those of DeepFold and WHIR.

12 Conclusion and Open Problems

Kronecker-FRI proves a multilinear evaluation $f(z) = v$ by reading v as the coefficient of X^{N-1} in $U_f K_z$, where U_f is the Kronecker encoding of f and $K_z = \prod_k (X^{2^{k-1}} + z_k)$. The prover commits to one opening polynomial; the verifier derives a second one as a virtual word, batches the three words, and runs an FRI-style folding test. We proved the scheme complete, sound, round-by-round knowledge sound and evaluation binding in the unique-decoding regime, with the correlated-agreement theorem of [BSCI⁺23] as the only external input about codes, and we derived post-quantum knowledge soundness of the compiled scheme from the theorem of [CDHZ25]. The results of Sections 4 to 7 and the batched openings are formalised in Lean 4, with correlated agreement as the only axiom. At identical engineering, it commits and verifies at the same cost as a sumcheck-based Reed–Solomon scheme; with arity 8 and Merkle caps, which the analysis covers with the same error, proofs are 95 to 372 KiB unsalted for 2^{12} to 2^{22} coefficients at an interactive 100-bit level (Section 10).

Future work. The following directions extend the present results.

- *List decoding.* Extend the batching lemma (Lemma 7.4) and the folding test (Theorem 5.24) to proximity parameters beyond $(1 - \rho)/2$, using correlated agreement up to the Johnson bound [BSCI⁺23]. The identity lemma needs $2N < (1 - \delta)M$, which still holds for $\rho \leq 1/4$ and every $\delta < 1 - 2\rho$; the main work is to replace unique decoding by lists. This would reduce κ : at $\rho = 1/4$ and 100 bits, from 148 to about 100 queries near the Johnson bound $\delta \rightarrow 1/2$, at the price of larger field terms; larger reductions would need proximity-gap results beyond the Johnson bound.
- *Rate 1/2.* The batching lemma needs $\rho < 1/3$ (Remark 7.6). Is there an identity of degree less than $(1 - \delta)M$ at rate $1/2$?
- *Several openings.* Extend Corollary 8.6 to batched openings (Theorem 11.2), which needs correlated agreement for fibre strings (Remark 11.4), to several openings of one commitment, and to several points, for which Corollary 4.6 is a starting point.
- *Conformance of the implementation.* Check the transcript and the salts of the implementation against [CDHZ25, Construction 11.7], so that Corollary 8.6 applies to the code, and implement the batched protocol.

- *Opening cost.* The opening pays for the product $U_f K_z$, one NTT, one Merkle tree and the virtual word (Table 6). All of them run on several threads with identical proofs, 1.4 to 1.5 times faster on two cores (Section 10.8); SIMD arithmetic would reduce them further, as for any FRI-based scheme, and whether the product and the NTT can be merged is open.
- *Recursion.* The verifier is an FRI verifier plus $O(n)$ operations per query, which makes it a natural candidate for recursive verification; measure its cost inside a recursive circuit.
- *Formal verification of Section 8.* The results of Sections 4 to 7 and Section 11.2 are formalised in Lean 4 (Appendix B); the post-quantum analysis would need a formalisation of the theorem of [CDHZ25].
- *Zero knowledge.* Adapt standard masking techniques for FRI-based schemes to the opening polynomial A and the folded words.

Code availability. The Rust implementation of Section 9, the benchmarks of Sections 10 and 11 and the numerical checks of Section 9.4 are released as open source together with this paper, and the library is published on crates.io as the crate `kronecker-fri`.

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A Notation

Symbol	Meaning	Where
<i>Integers and bits</i>		
$\mathbb{N}$, $\llbracket a, c \rrbracket$	non-negative integers; integers i with $a \leq i \leq c$	§2.1
n , $N = 2^n$	number of variables; number of coefficients	§2.1
a_k , B_n	bit k of an index a (least significant first); Boolean hypercube, identified with $\llbracket 0, N - 1 \rrbracket$	§2.1
$\text{wt}(a)$, $\bar{a}$, $b \preceq a$	Hamming weight; complement $N - 1 - a$; submask	§2.1
<i>Fields and polynomials</i>		

Symbol	Meaning	Where
$\mathbb{F}_q, \mathbb{F}$	base field of odd characteristic; challenge field $\mathbb{F} \supseteq \mathbb{F}_q$	§2.2, §6.1
e	degree of $\mathbb{F}$ over $\mathbb{F}_q$ (implementation)	§9.1
$\mathbb{F}[X]_{<d}, [X^a]P$	polynomials of degree $< d$; coefficient of X^a	§2.2
P_e, P_o	even and odd parts, $P(X) = P_e(X^2) + XP_o(X^2)$	Lemma 2.4
<i>Multilinear polynomials</i>		
$\mathcal{L}_n(\mathbb{A})$	multilinear polynomials in n variables over a ring $\mathbb{A}$	§2.3
x^a, z^a	monomial $\prod_{k:a_k=1} x_k$; its value at z	Definition 2.5
$\alpha = (\alpha_a)_a, (f(b))_b$	coefficient form; table form	Definition 2.5 and Lemma 2.6
<i>Kernel polynomial and identities</i>		
U_f	Kronecker encoding $\sum_a \alpha_a X^a = f(X, X^2, \dots, X^{2^{n-1}})$	Definition 4.1
K_z	kernel polynomial $\prod_{k=1}^n (X^{2^{k-1}} + z_k)$	Definition 4.3
$A_{U,z}, H_{U,z}$	opening polynomials: $XUK_z = A + vX^N + X^{N+1}H$	Lemma 4.10 and Definition 4.11
Φ	$XUK_z - A - vX^N - X^{N+1}H$, of degree $\leq 2N$	Lemma 4.12
<i>Domains and codes</i>		
$L, M = 2^\mu$	smooth domain and its order	Definition 2.7
$L_j = L^{2^j}, M_j, N_j$	domain, order and degree bound at level j	§5
fibre $\{\xi, -\xi\}$	the two square roots of $\eta = \xi^2$	Definition 2.9
$\text{ev}_L, \text{RS}_{\mathbb{F}}[L, d], \rho, R$	evaluation map; Reed–Solomon code; rate $d/M = 2^{-R}$	Definition 2.10, §6.1
$\mathcal{C}_j, P_c$	code $\text{RS}_{\mathbb{F}}[L_j, N_j]$; polynomial of a codeword	§5, Lemma 2.12
$\Delta, \Delta_j^{\text{fib}}$	relative Hamming distance; fibre distance	Definitions 2.10 and 5.10
w_e, w_o	even and odd parts of a word	(1)
δ, δ^*	proximity parameter; $(1 - \rho)/2$	§5, §7
$\text{dec}_j(w)$	decoded codeword at fibre distance $\leq \delta$	Definition 5.13
<i>Folding</i>		
$\text{fold}_r, \text{pfold}_r$	kernel fold of a word and of a polynomial	Definition 5.1
$\text{pfold}_r^{(j)}, \chi_t(r)$	iterated fold; its coefficients	Lemma 5.5
cfold_θ	classical FRI fold	Remark 5.6
$\Pi_{\text{FT}}[\ell, \kappa]$	folding proximity test	Definition 5.7
$\hat{w}_j, \mathcal{B}_j$	folded word; points where the level- j check fails	§5.5
$\mathcal{W}_j(c), \text{dens}_j(c)$	witness set of a codeword; its density	Definition 5.20
$\varepsilon_{\text{fold}}$	$\sum_{j=1}^\ell M_j/ \mathbb{F} $	§5.6
<i>The scheme</i>		
$\text{Enc}(f)$	commitment $\text{ev}_L(U_f)$	Definition 6.2
ℓ, κ	number of folding rounds; number of queries	§6.1
$h[w, w_A; z, v]$	virtual word	Definition 6.5
$w_A, w_0 = w_\beta$	opening oracle; batched word $w + \beta w_A + \beta^2 h$	Definition 6.8, §7.3
$\beta, r_j, \xi^{(i)}$	batching challenge; folding challenges; query points	Definition 6.8
P	final polynomial, N_ℓ coefficients	Definition 6.8
$\Pi_{\text{KF}}, \Pi_{\text{KF}}^m$	evaluation protocol; batched protocol	Definitions 6.8 and 11.1
$\mathcal{R}_{\text{KF}}^\delta, \mathcal{R}_{\text{KF}}^{\delta, m}$	evaluation relations	Definition 7.1, §11.2

Symbol	Meaning	Where
ε_{KF}	soundness error $(2M + M(1 - 2^{-\ell}))/ \mathbb{F} + (1 - \delta)^\kappa$	§7.4
$\mathcal{G}$	challenges β admitting a fibre-closed agreement set	Lemma 7.4
$\mathcal{D}$, Ext	doomed transcripts; extractor	Definition 7.9
<i>Non-interactive scheme</i>		
$\Sigma = \mathbb{F}^2$, fib_j	alphabet of fibre symbols; fibre string of a word	§8.1
$\perp$, $\bar{s}$, $\text{Dom}(s)$	undefined entry; filling; defined positions	§3.4, §8.1
ϕ , pos, λ_{ch}	challenge maps; challenge length	§8.1
$\mathcal{Z}_j$, $\mathcal{W}_j^\perp(c)$	points reading no $\perp$; witness sets with erasures	§8.2
KState, E	knowledge state function; its extractor	Definition 8.3
λ_H , Q	hash output length; number of oracle queries	§3.4

B Verification Status

This appendix records, for each result, how it is established. *Proved* means proved in this paper; *from [Mkh26]* means that the statement and its proof are reproduced from the companion paper; *adapted* means that the proof follows one of [Mkh26] with changes; *quoted* means that the result is cited and not reproved.

Lean 4. The results of Sections 4 to 7 and of Section 11.2 are formalised in Lean 4 with Mathlib, in the directory `lean/` of the companion repository, without `sorry`. The formalisation has a single axiom, Theorem 2.16 (`KBFold.bciks_curves`); the case $t = 1$ used by the folding test is derived from it. A continuous-integration job builds the project, checks that no `sorry` occurs and that `bciks_curves` is the only axiom, and prints the axioms on which every main theorem depends. The column *Lean* gives the name of the main Lean statement; the file `lean/README.md` gives the complete correspondence. The formalisation models a deterministic prover by its messages as functions of the challenges, with the hypotheses that a message depends only on the earlier challenges and that the final polynomial has fewer than N_ℓ coefficients; in the generic model of Definition 3.1, used for Theorem 7.11, neither hypothesis is needed. Operation counts and the base-field statements are established in the paper and checked by the test suite. The modules shared with [Mkh26] (fibre distance, decoding, good challenges, probability lemmas and the framework of Section 3) are those of its Lean project.

Tested refers to the Rust test suite and examples of Section 9.4, which check the statement on random instances; tests are evidence, not proofs.

Result	Status	Lean	Tested
Facts 2.3, 2.13 and 6.4	quoted [LN97, WB86, Gao03, CT65]	–	NTT against Horner evaluation
Theorem 2.16 (correlated agreement)	quoted [BSCI ⁺ 23]	axiom <code>bciks_curves</code>	–
Theorem 3.15 (BCS for IORs)	quoted [CDHZ25]	–	bound of Remark 8.8 evaluated exactly

Result	Status	Lean	Tested
Lemmas 2.1, 2.2, 2.4, 2.6, 2.8, 2.11, 2.12 and 2.15	proved; from [Mkh26]	from [Mkh26]	Möbius transform
Lemma 2.14	proved; from [Mkh26]	–	–
Lemmas 3.2 to 3.4, 3.8 and 3.11	proved; from [Mkh26]	from [Mkh26]	–
Lemma 3.12	proved; from [Mkh26]	–	Merkle openings
Lemma 4.2	proved	kronEnc_eq_subst, kronEncLin_bijective	–
Lemma 4.4	proved	natDegree_kernel, kernel_recursion, coeff_kernel_bits	coefficient rule
Theorem 4.5	proved	extraction	$n \leq 8$, two fields; $\mathbb{F}_{257}$
Corollaries 4.6 and 4.7, Proposition 4.8, and Lemma 4.14	proved	extraction_comb, cube_sum_half, bilinear, kernel_reflect	–
Lemma 4.9	proved	– (operation counts)	kernel product
Lemma 4.10	proved	split_iff, split_unique	split identity
Lemma 4.12, Remark 4.13	proved	identity_lemma, without_X_absorbs	root bound; two cheating provers
Lemmas 5.2 and 5.3 and Proposition 5.4	proved; from [Mkh26]	from [Mkh26]	$n \leq 7$
Lemma 5.5	proved	coeff_pfoldUp	$n \leq 7$
Lemma 5.9(1)	proved	honestFT_accepts	completeness tests
Lemmas 5.11, 5.12, 5.15, 5.16, 5.18 and 5.19	proved; from [Mkh26]	from [Mkh26]	–
Lemmas 5.21 and 5.22 and Proposition 5.23	proved; adapted from [Mkh26]	queryOK_iff, rbr_step, chain_final	–
Theorem 5.24	proved; adapted from [Mkh26]	fpt_sound	–
Proposition 5.28	proved	acceptsC_iff, fptC_sound	arities 2 to 16, with and without caps
Lemma 6.3(1)–(2)	proved	encode_injective, encode_fib_unique	–
Lemma 6.3(3)	proved	– (base field)	–
Lemma 6.6, Remark 6.9(1)	proved	virt_honest, virt_wrong_value, inv_agree_le	virtual word
Lemma 6.7 and Proposition 6.11	proved	– (operation counts)	$ev_L(K_z)$
Theorem 6.10	proved	honestKF_prob_one	$n \leq 9$, all ℓ , three rates, arities 2, 4, 8
Lemmas 7.2 and 7.3	proved	rate_condition, value_of_agreement	–
Lemma 7.4	proved	batching	–
Lemma 7.7	proved	negPt_mem_Wset	–
Theorem 7.8	proved	soundness, soundnessC	tampering tests
Lemma 7.10	proved	extKF_witness (running time not modelled)	–
Theorem 7.11	proved	rbr_knowledge	–

Result	Status	Lean	Tested
Corollary 7.12	proved	<code>knowledge</code> , <code>binding</code>	–
Lemmas 8.1, 8.4 and 8.7, Theorem 8.5, and Corollary 8.6	proved (using Theorem 3.15)	–	–
Remark 8.8 (parameters)	computed	–	exact rational evaluation; non-squareness of 7 not implemented
Theorem 11.2 and Corollary 11.3	proved	<code>batch_soundness</code> , <code>batch_knowledge</code> , <code>batch_binding</code>	